Model Analysis

Table of contents

1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df():
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
```

```
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    cols_to_check = target_col + features_X
    mask = np.isfinite(df[cols_to_check]).all(axis=1)
    df = df[mask]


    cols = list(dict.fromkeys(target_col + features_X  +
                              feats_to_balance + feats_grouping +
                              feats_reference+ feats_postproc))
    df = df[cols]
    #df['campaign'] = "ALL"

    return df
```

```
def import_data_IR():
    if target_col[0] == "i_NH3":
        #file_path = "csv/innova/switched/df_signal_decomposition_NH3_current_db2.csv"
        file_path = "csv/innova/notswitched/df_signal_decomposition_current.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv/innova/switched/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/notswitched/features_decomposition_list.json", "r") as f:
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
```

```python
        with open("csv/innova/switched/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")


    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)
    #Removing of rabbit 4
    df["column"] = df.apply(assign_colonna, axis = 1)
    df['height'] = df.apply(assign_altezza, axis = 1)




    return df,dict_features
```

```python
def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")
```

Function print :

1.2 Metadata

Nombre de lignes df	8168
Nombre de lignes df clean	8168
Nombre de features	5
% de NaN	0.0
Taille dataset entraînement	5717
Taille dataset test	2450
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI

feats_to_balance :

feats_grouping : height

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

Grouping : height

2 Model(s) analysis

2.1 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv

- r_train: train_cv/test_cv

Colors:

[Orange] r_test < 0.95 (under performance of the test)

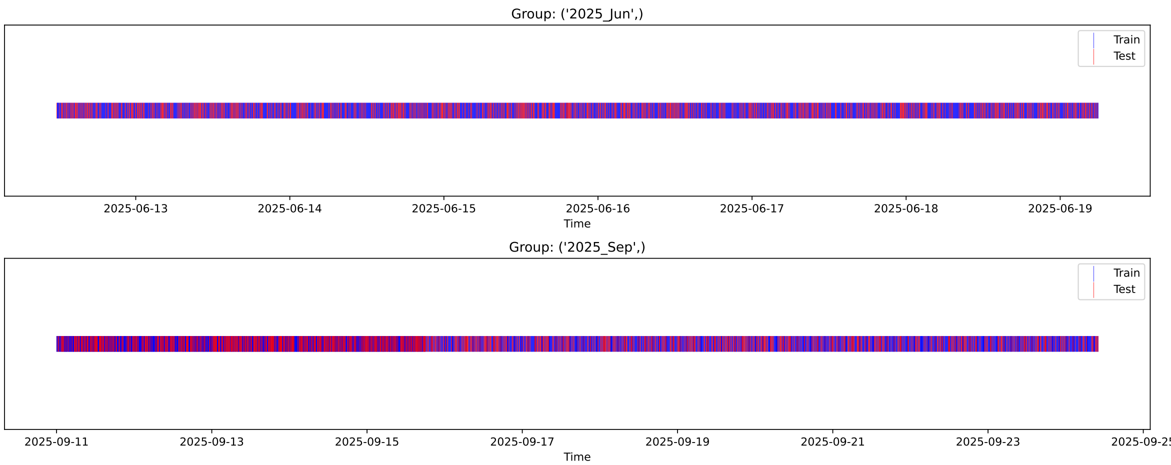
[Green] r_test > 1.05 (over performance of the test)

[Red] r_train not in [0.95, 1.05] (gap between train/test)

2.2 Repartition of train and test

2.2.1 External Split: X_train vs X_test

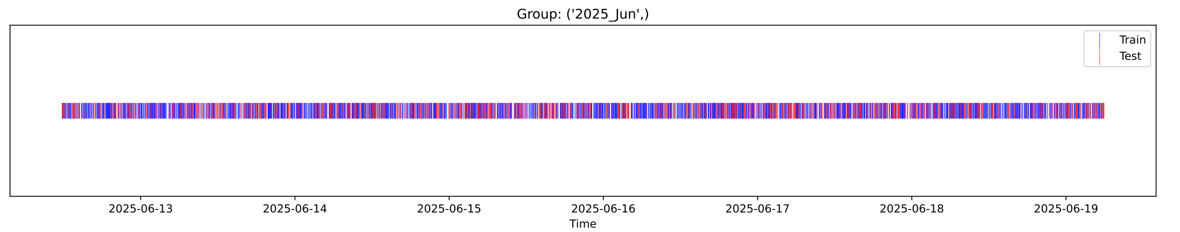
External Split: Train: 5717 samples Test: 2451 samples

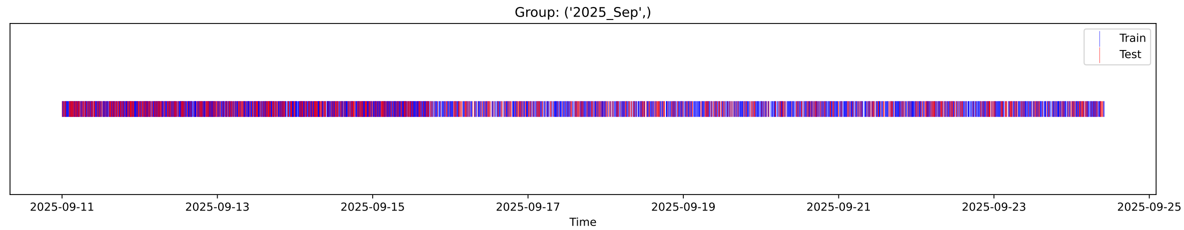


2.2.2 Internal CV Folds

Fold 1/5

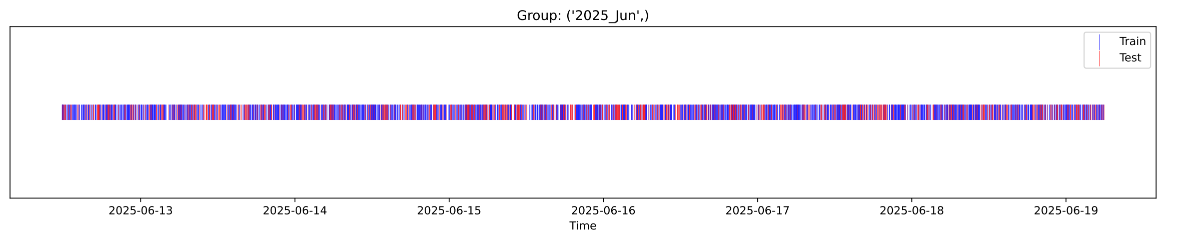
Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h

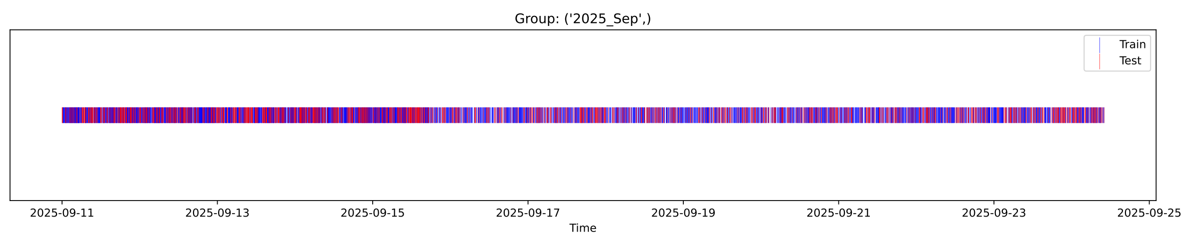




Fold 2/5

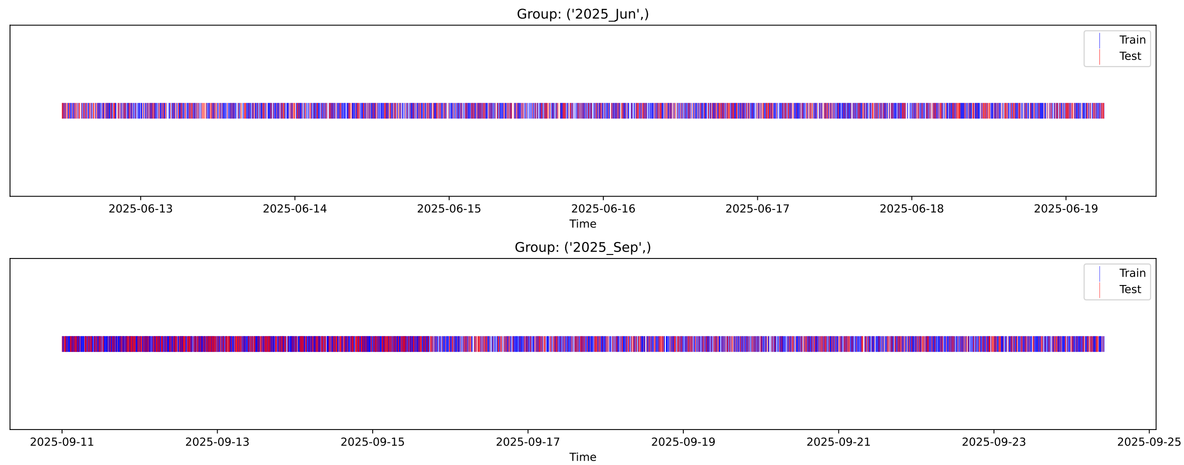
Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h





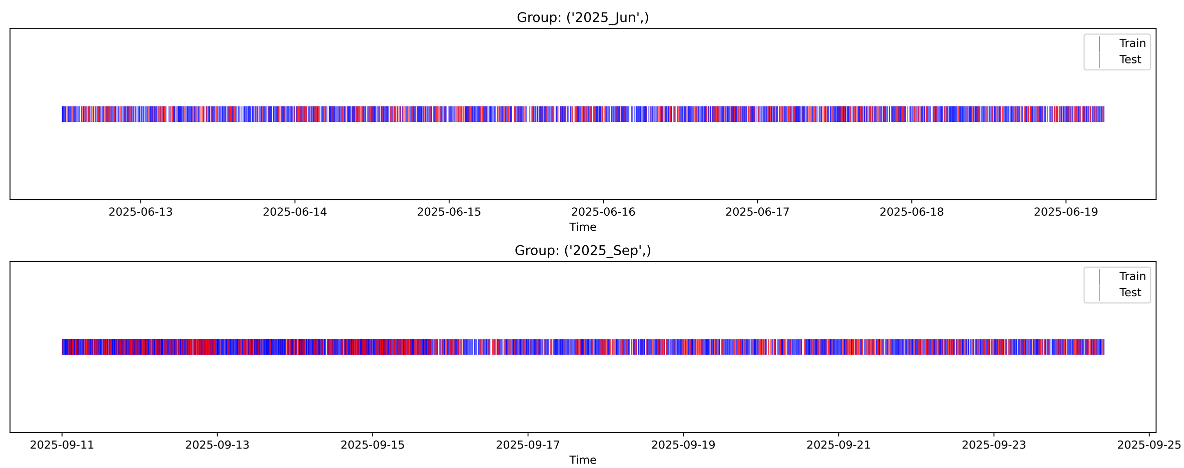
Fold 3/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



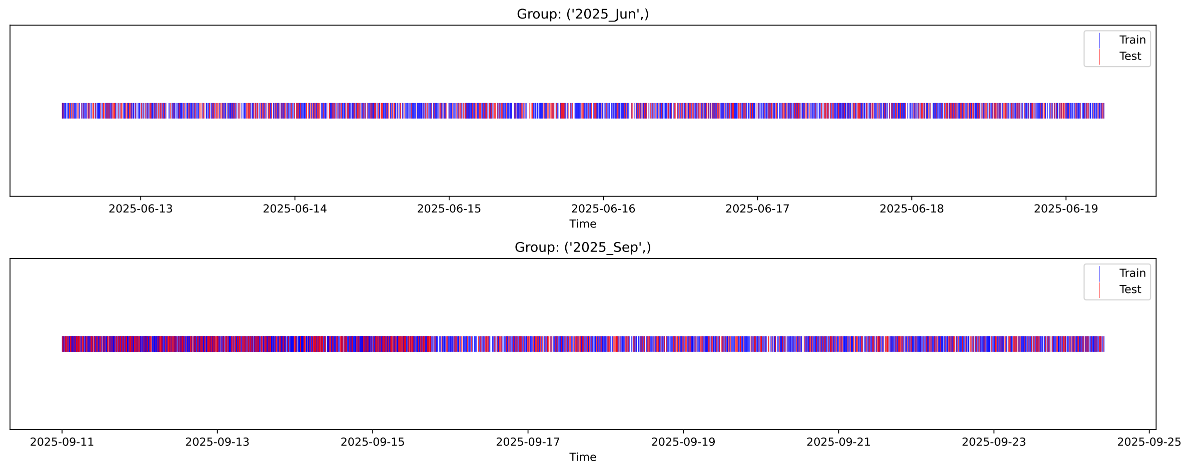
Fold 4/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



2.3 Model : Linear

2.3.1 Gridsearch — Group: 155

Len(group) : 4082, **Len(training)** : 2858, **Len(test)** : 1224

Distribution y_train vs y_test: y_train: mean=2.469, std=0.854, min=1.041, max=13.845 y_test: mean=2.472, std=0.811, min=1.068, max=7.277

Fold 0 — y_test: mean=2.47, std=0.81, y_train: mean=2.47, std=0.87

Fold 1 — y_test: mean=2.48, std=0.87, y_train: mean=2.46, std=0.85

Fold 2 — y_test: mean=2.48, std=0.84, y_train: mean=2.46, std=0.86

Fold 3 — y_test: mean=2.46, std=0.81, y_train: mean=2.47, std=0.87

Fold 4 — y_test: mean=2.52, std=0.97, y_train: mean=2.45, std=0.80

train scores CV: [0.40437693 0.42152043 0.41039248 0.40144919 0.47407666]

test scores CV: [0.47213342 0.42219944 0.45346593 0.47697468 0.3372002]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.4224	0.4324	0.0513	0.4495	0.4497	0.0173	1.04	0.9768

Coefficients:

coef_r_CO2 = 0.0145

coef_r_NH3 = -0.0250

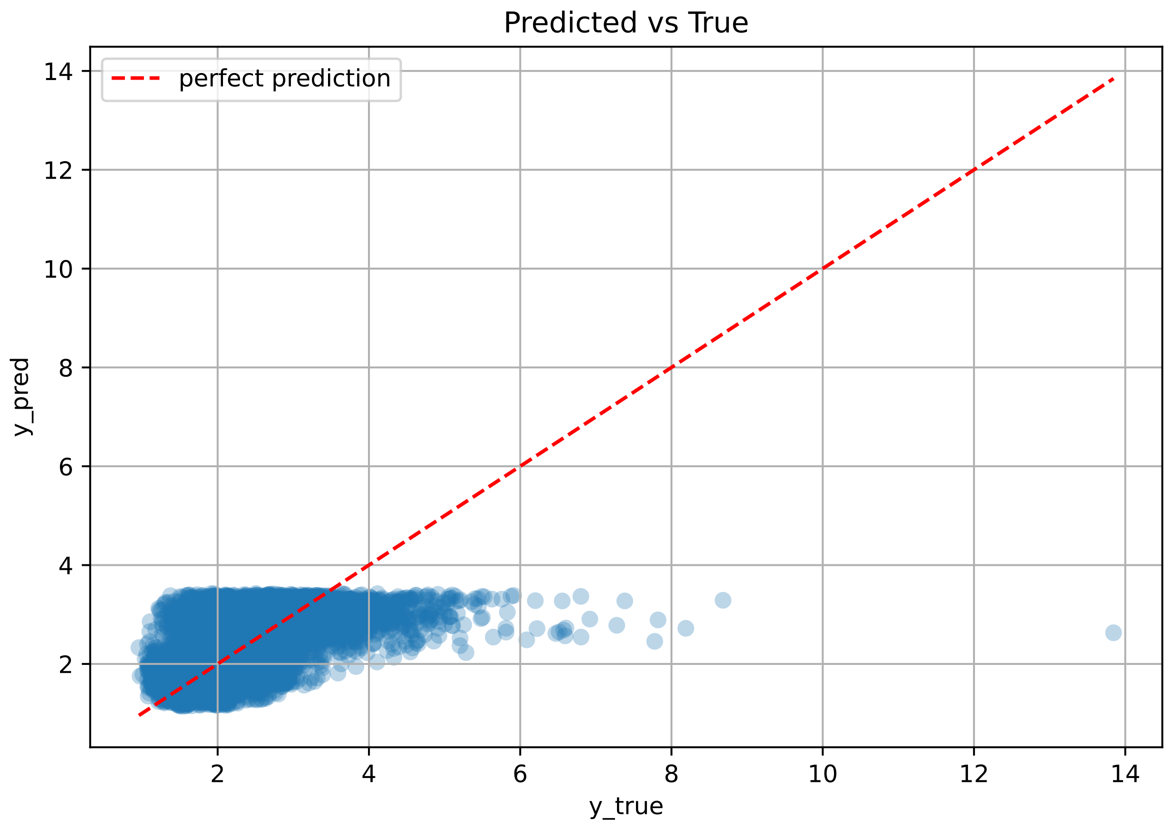
coef_r_THI = 0.6825

coef_r_temperature = -0.7279

coef_r_umidity = -0.6609

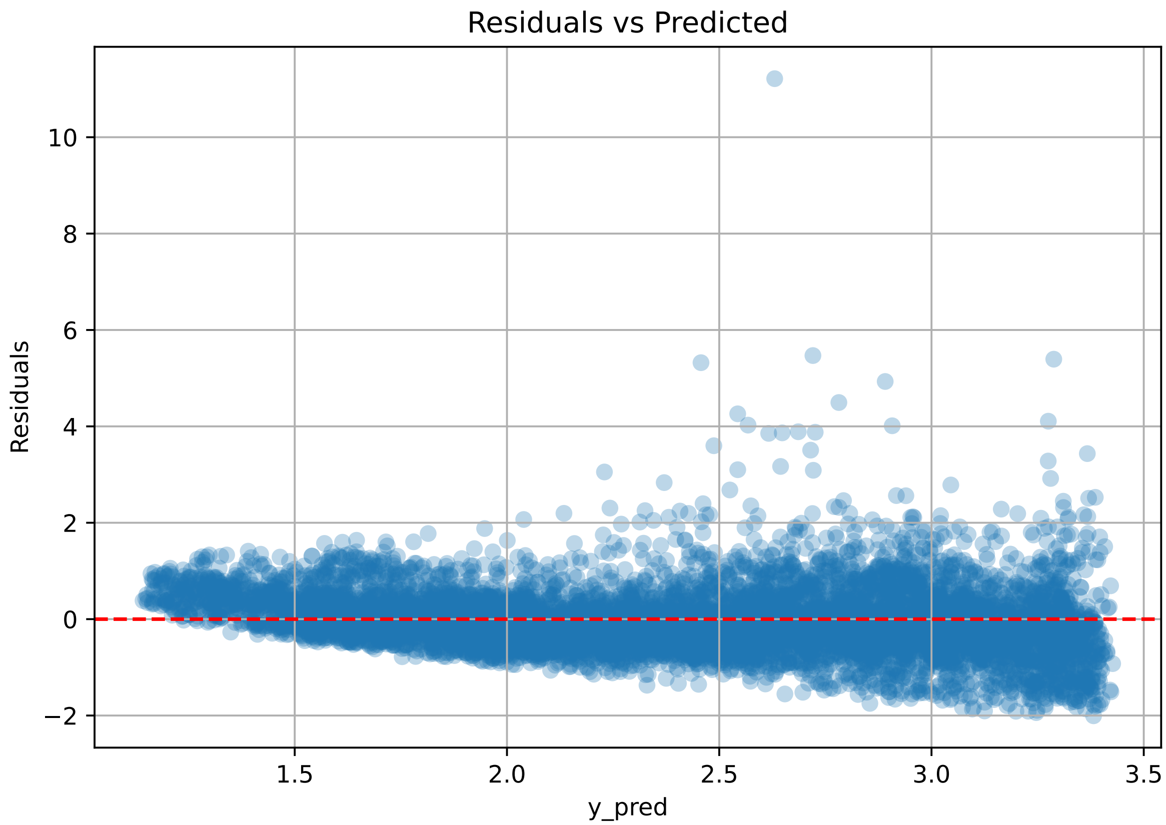
intercept = 2.4690

2.3.2 Scatter



Predicted vs True
perfect prediction
y_pred
y_true
14
12
10
8
6
4
2
2
4
6
8
10
12
14

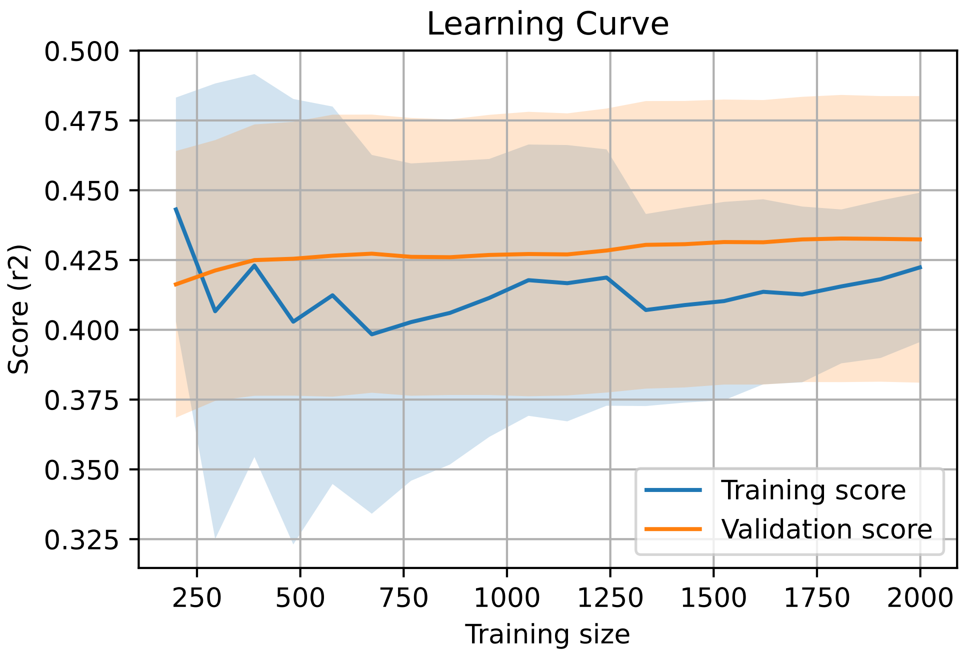
2.3.3 Scatter residuals



2.3.4 Learning curve — Group: 155

Len(group) : 4082, **Len(training)** : 2858,

Len(training_real) : 2001, **Len(test_of_training)** : 857, **Len(test)** : 1224



2.3.5 Gridsearch — Group: 270

Len(group) : 4086, **Len(training)** : 2861, **Len(test)** : 1225

Distribution y_train vs y_test: y_train: mean=2.228, std=0.729, min=0.959, max=7.822 y_test: mean=2.232, std=0.719, min=0.973, max=6.804

Fold 0 — y_test: mean=2.26, std=0.75, y_train: mean=2.22, std=0.72

Fold 1 — y_test: mean=2.23, std=0.73, y_train: mean=2.23, std=0.73

Fold 2 — y_test: mean=2.21, std=0.72, y_train: mean=2.23, std=0.73

Fold 3 — y_test: mean=2.24, std=0.70, y_train: mean=2.22, std=0.74

Fold 4 — y_test: mean=2.27, std=0.72, y_train: mean=2.21, std=0.73

train scores CV: [0.32915933 0.34769981 0.33648741 0.31499108 0.30717472]

test scores CV: [0.30873323 0.26233521 0.29007871 0.34463252 0.35651321]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.3271	0.3125	0.0346	0.3634	0.3641	0.0517	1.1654	1.0469

Coefficients:

coef_r_CO2 = -0.0880

coef_r_NH3 = 0.0921

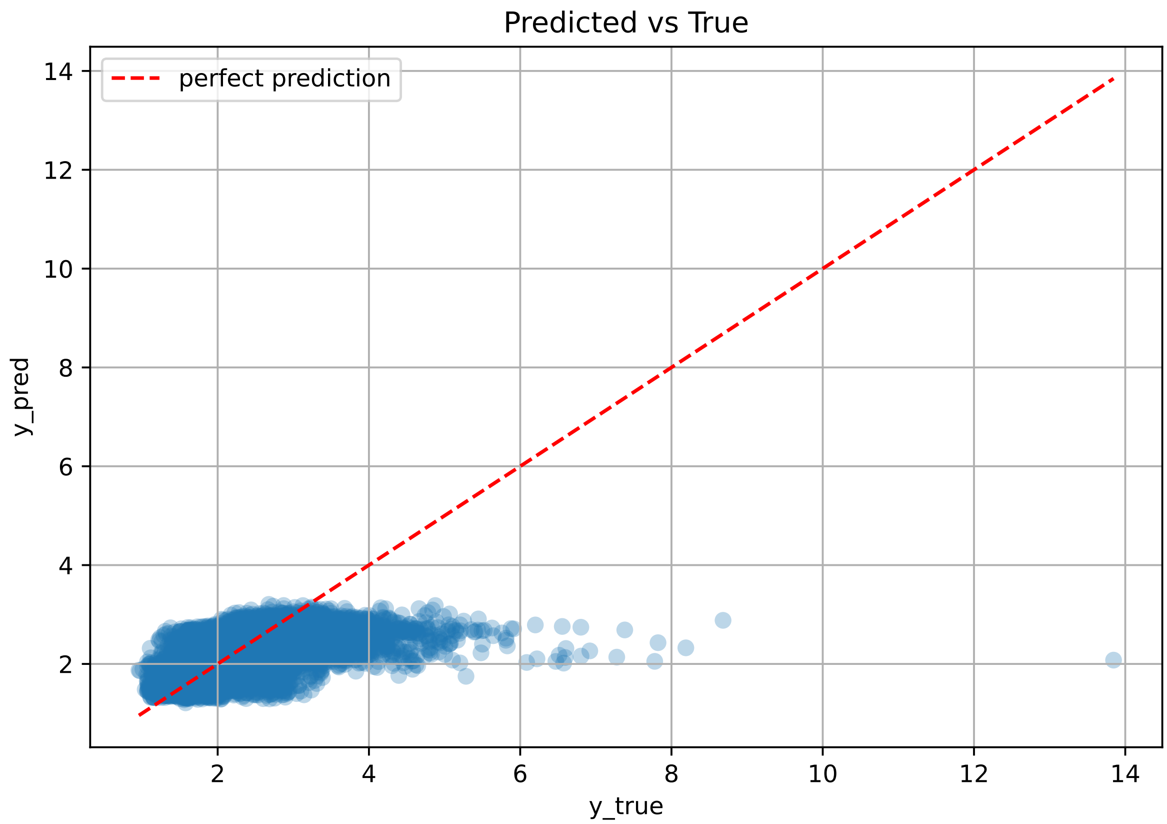
coef_r_THI = 0.4274

coef_r_temperature = -0.7123

coef_r_umidity = -0.7150

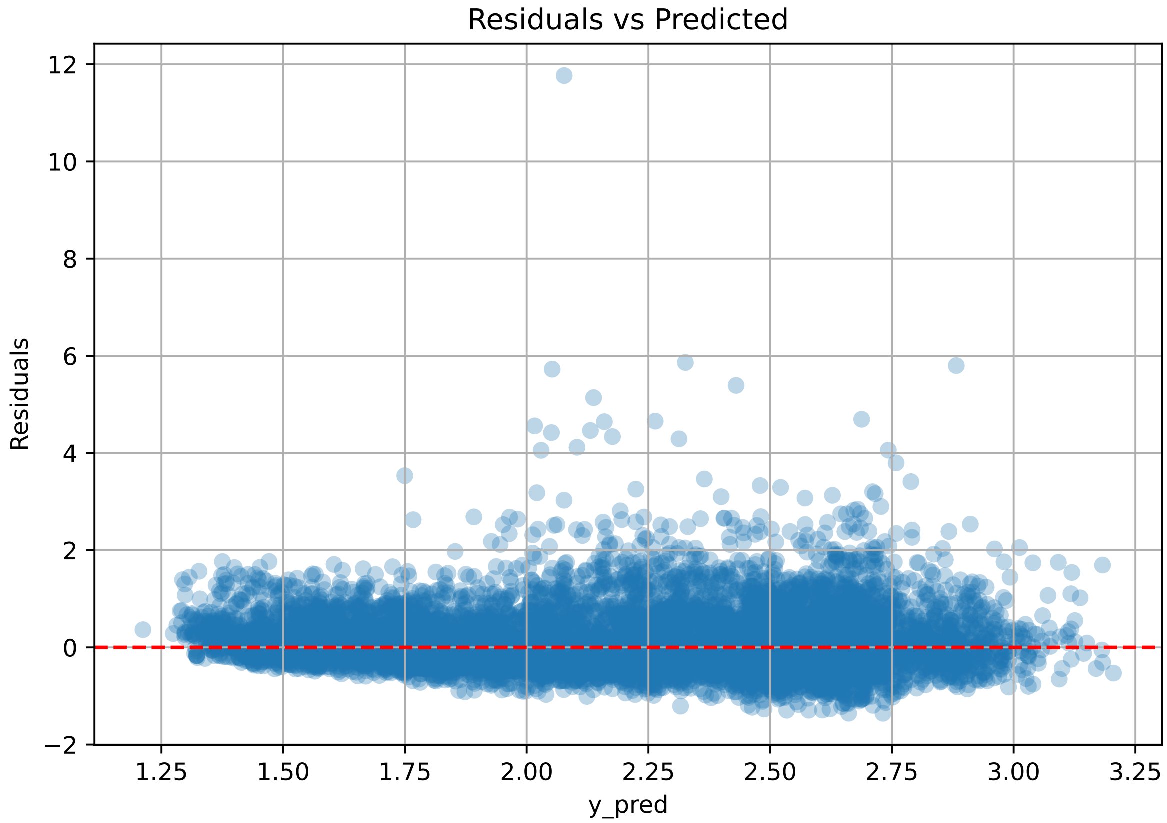
intercept = 2.2282

2.3.6 Scatter



Predicted vs True
perfect prediction
y_pred
y_true
2
4
6
8
10
12
14

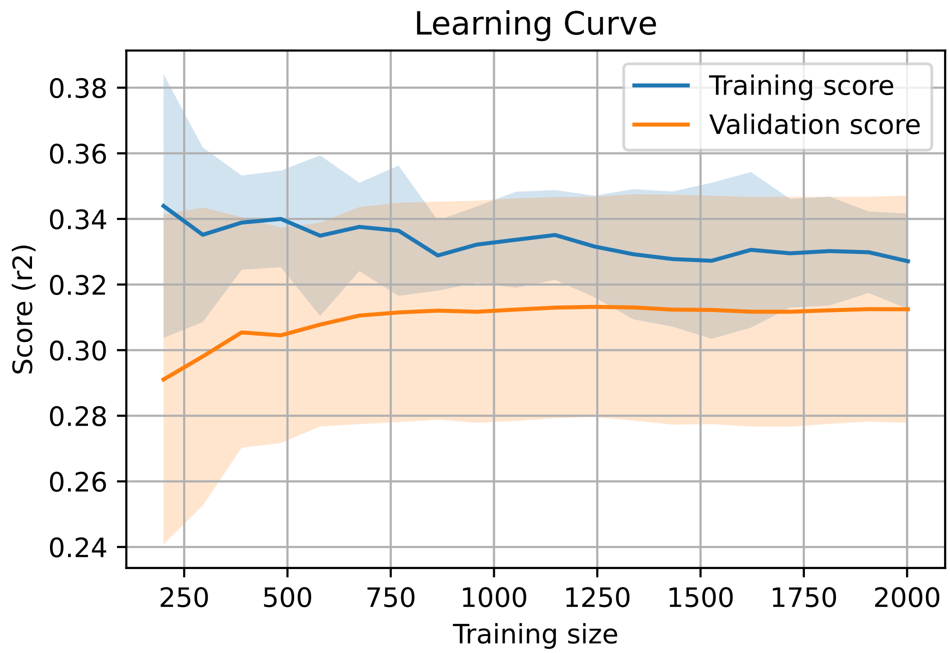
2.3.7 Scatter residuals



2.3.8 Learning curve — Group: 270

Len(group) : 4086, **Len(training)** : 2861,

Len(training_real) : 2003, **Len(test_of_training)** : 858, **Len(test)** : 1225



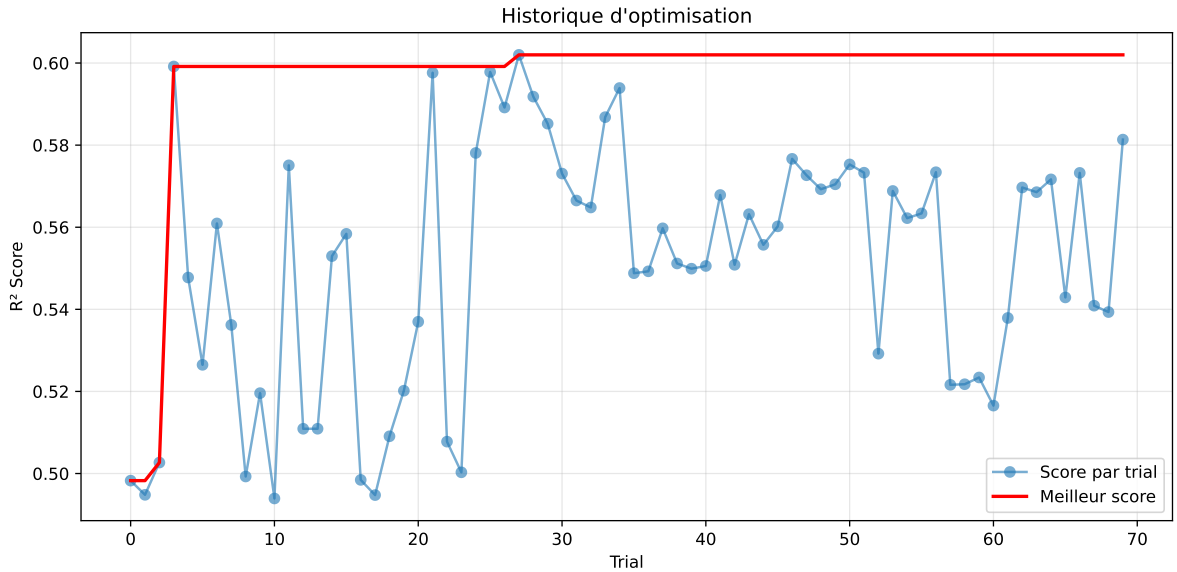
2.4 Model : LGBM

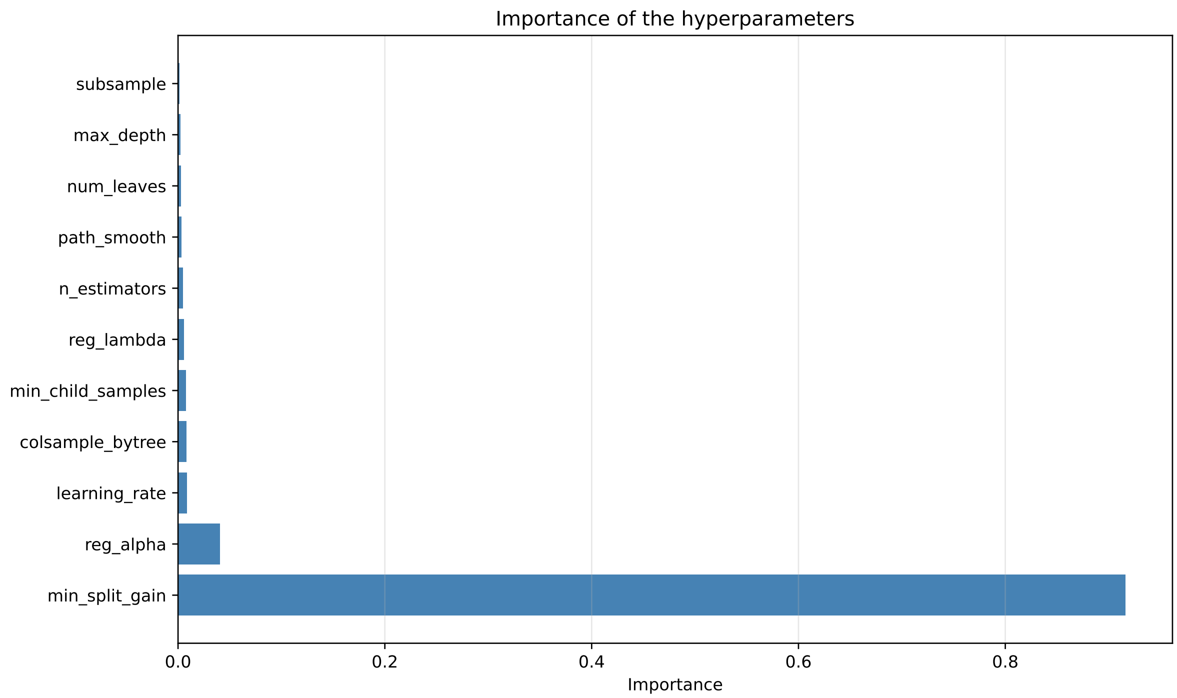
2.4.1 Gridsearch — Group: 155

Len(group) : 4082, **Len(training)** : 2858, **Len(test)** : 1224

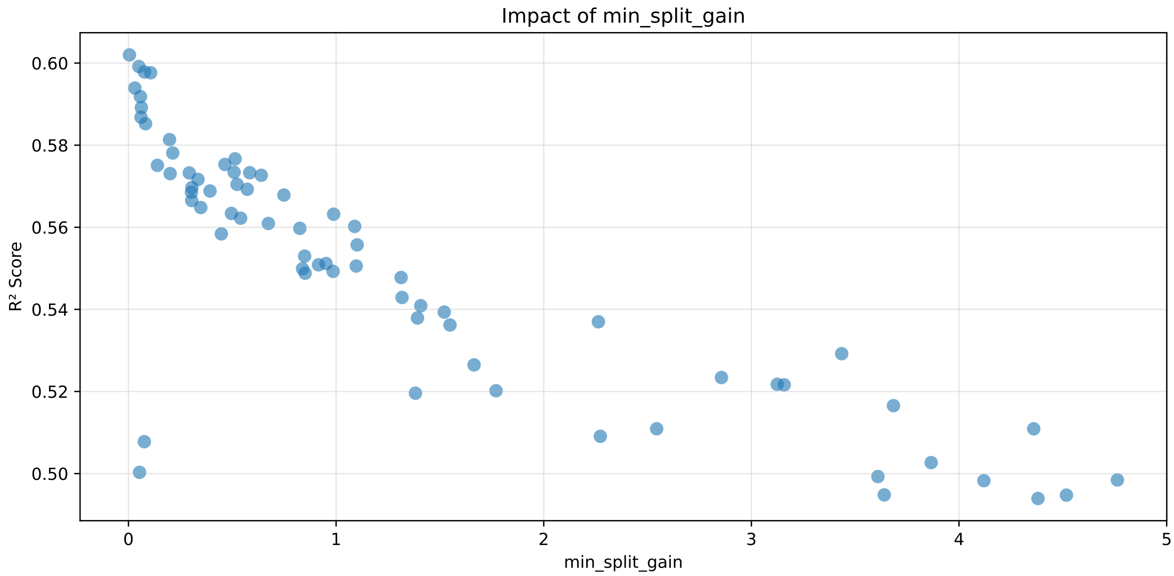
Distribution y_train vs y_test: y_train: mean=2.469, std=0.854, min=1.041, max=13.845 y_test: mean=2.472, std=0.811, min=1.068, max=7.277

==================== TOP Importance FEATURES ==================== feature importance r_umidity 95 r_NH3 76 r_CO2 47 r_temperature 30 r_THI 13





Paramètres avec >5% d'importance : ['min_split_gain']



Pas assez de paramètres avec >5% d'importance pour une heatmap 2D.

rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.526	0.5027	0.5398	0.0592	0.5027	0.5632	0.0605	1.1203	1.0465
2	0.5222	0.4993	0.5354	0.0586	0.4993	0.5636	0.0643	1.1287	1.046
3	0.5224	0.4984	0.5356	0.0572	0.4984	0.5621	0.0637	1.1278	1.048
4	0.5222	0.4983	0.536	0.0588	0.4983	0.566	0.0677	1.1359	1.048
5	0.5166	0.4948	0.5299	0.0583	0.4948	0.5544	0.0596	1.1204	1.0441
6	0.5166	0.4947	0.5291	0.0574	0.4947	0.5527	0.0579	1.117	1.0441
7	0.5173	0.4939	0.5279	0.0571	0.4939	0.5554	0.0614	1.1244	1.0474
8	0.5694	0.537	0.5829	0.0596	0.4396	0.6069	0.0699	1.1302	1.0605
9	0.5385	0.5109	0.5512	0.0584	0.4282	0.5804	0.0695	1.136	1.054
10	0.5381	0.5091	0.5477	0.0591	0.4219	0.5722	0.0631	1.124	1.0571
---	---	---	---	---	---	---	---	---	---
70	0.7629	0.602	0.6639	0.0642	0.1194	0.6859	0.0839	1.1394	1.2672

Hyperparameters:

Rank 1: {'reg___n_estimators': 700, 'reg___max_depth': 3, 'reg___learning_rate': 0.0569718099136699, 'reg___num_leaves': 52, 'reg___subsample': 0.6545935475021628, 'reg___colsample_bytree': 0.8217923721216703, 'reg___min_child_samples': 45, 'reg___reg_alpha': 6.149091008823988, 'reg___reg_lambda': 31.462936785736805, 'reg___min_split_gain': 3.8658719311240675, 'reg___path_smooth': 4.853979898657569}

Rank 2: {'reg___n_estimators': 300, 'reg___max_depth': 7, 'reg___learning_rate': 0.016882837921747856, 'reg___num_leaves': 35, 'reg___subsample': 0.6649773749038699, 'reg__colsample_bytree': 0.8018957420029273, 'reg__min_child_samples': 36, 'reg__reg_alpha': 8.457930385409233, 'reg__reg_lambda': 36.7751108277607, 'reg__min_split_gain': 3.609365249613034, 'reg__path_smooth': 2.725241486485059}

Rank 3: {'reg__n_estimators': 400, 'reg__max_depth': 5, 'reg__learning_rate': 0.02543768399293609, 'reg__num_leaves': 50, 'reg__subsample': 0.5861382154979525, 'reg__colsample_bytree': 0.5613211483895724, 'reg__min_child_samples': 17, 'reg__reg_alpha': 6.051200109321586, 'reg__reg_lambda': 5.563179980852034, 'reg__min_split_gain': 4.762616302593406, 'reg__path_smooth': 0.22520327163503606}

Rank 4: {'reg__n_estimators': 400, 'reg__max_depth': 4, 'reg__learning_rate': 0.019316984334629913, 'reg__num_leaves': 25, 'reg__subsample': 0.7461095730587116, 'reg__colsample_bytree': 0.7272330635427171, 'reg__min_child_samples': 29, 'reg__reg_alpha': 8.938025205237167, 'reg__reg_lambda': 10.338656142873148, 'reg__min_split_gain': 4.119769713165049, 'reg__path_smooth': 2.1986318174659694}

Rank 5: {'reg__n_estimators': 800, 'reg__max_depth': 5, 'reg__learning_rate': 0.015120366535856418, 'reg__num_leaves': 39, 'reg__subsample': 0.6687362717335177, 'reg__colsample_bytree': 0.5476101731989418, 'reg__min_child_samples': 31, 'reg__reg_alpha': 8.906845137400307, 'reg__reg_lambda': 40.3393410682075, 'reg__min_split_gain': 3.6398081369566397, 'reg__path_smooth': 1.4240996133028794}

Rank 6: {'reg__n_estimators': 200, 'reg__max_depth': 6, 'reg__learning_rate': 0.02701302802000631, 'reg__num_leaves': 61, 'reg__subsample': 0.8149816502674865, 'reg__colsample_bytree': 0.5536017649415724, 'reg__min_child_samples': 44, 'reg__reg_alpha': 6.670841225230605, 'reg__reg_lambda': 24.922234375121775, 'reg__min_split_gain': 4.517327116208692, 'reg__path_smooth': 1.671134250531932}

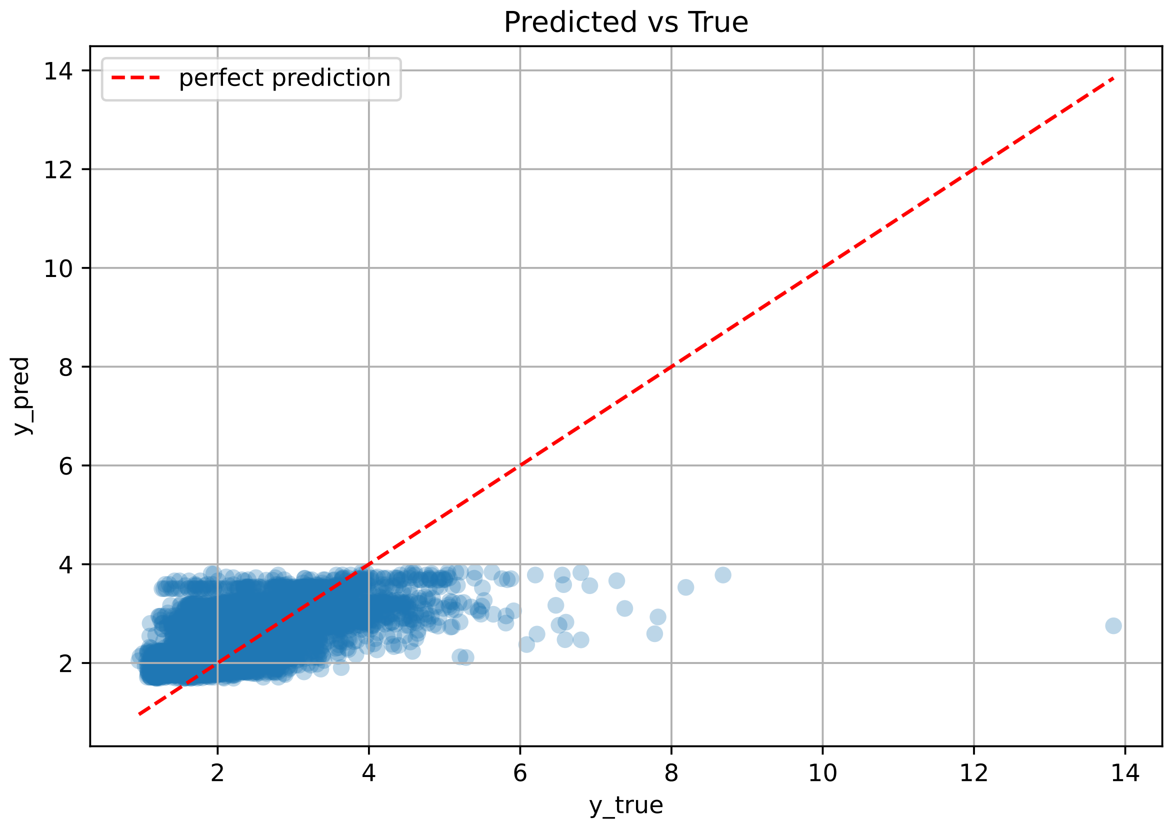
Rank 7: {'reg__n_estimators': 500, 'reg__max_depth': 6, 'reg__learning_rate': 0.08051267412577423, 'reg__num_leaves': 18, 'reg__subsample': 0.8550363461358509, 'reg__colsample_bytree': 0.5747470774798634, 'reg__min_child_samples': 29, 'reg__reg_alpha': 6.889078066777906, 'reg__reg_lambda': 23.22230738271756, 'reg__min_split_gain': 4.380291457590371, 'reg__path_smooth': 3.4749796884649697}

Rank 8: {'reg__n_estimators': 600, 'reg__max_depth': 3, 'reg__learning_rate': 0.07335447747384058, 'reg__num_leaves': 48, 'reg__subsample': 0.7656352626016194, 'reg__colsample_bytree': 0.744730680416382, 'reg__min_child_samples': 20, 'reg__reg_alpha': 0.9844671757480439, 'reg__reg_lambda': 19.542021623066848, 'reg__min_split_gain': 2.263121253042409, 'reg__path_smooth': 2.6990580954501695}

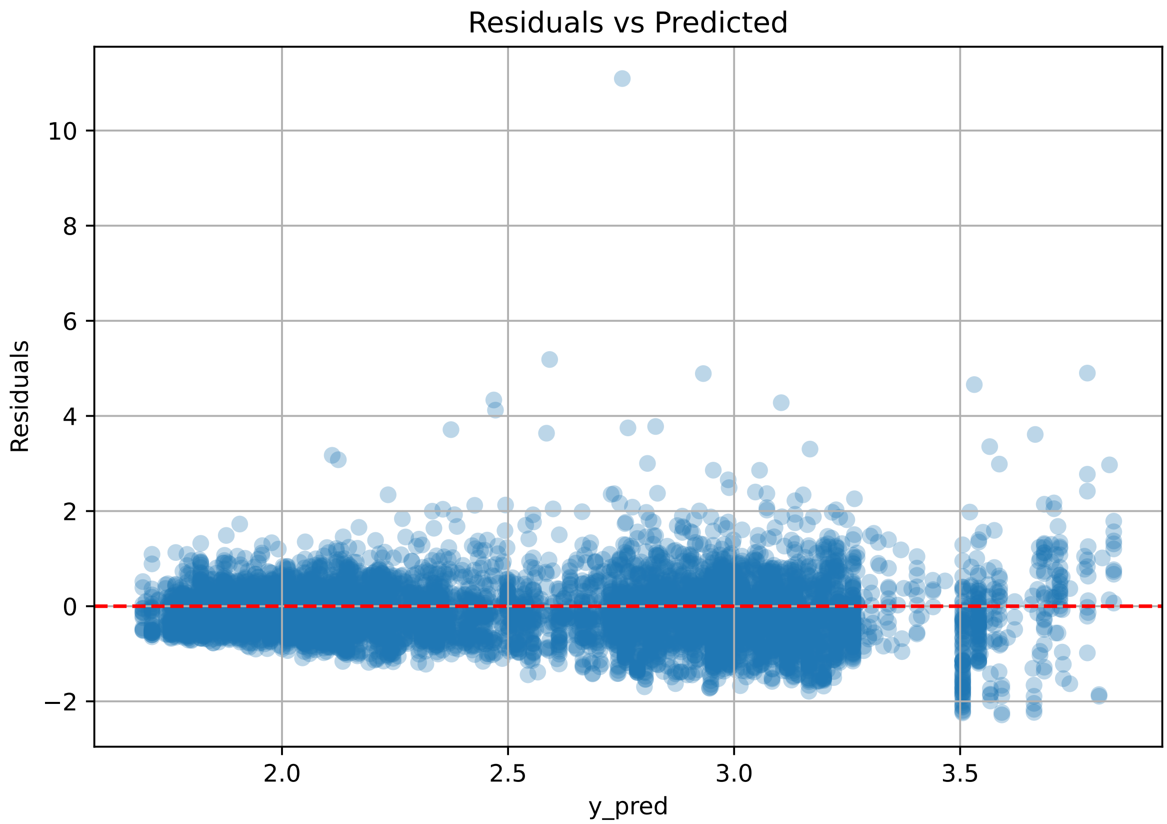
Rank 9: {'reg__n_estimators': 800, 'reg__max_depth': 8, 'reg__learning_rate': 0.016148829847724817, 'reg__num_leaves': 33, 'reg__subsample': 0.8384168188238373, 'reg__colsample_bytree': 0.8472358482089478, 'reg__min_child_samples': 34, 'reg__reg_alpha': 2.191433466321986, 'reg__reg_lambda': 23.32437987191379, 'reg__min_split_gain': 4.359695999121214, 'reg__path_smooth': 1.920283863140075}

Rank 10: {’reg__n_estimators’: 300, ’reg__max_depth’: 5, ’reg__learning_rate’: 0.0179674634112215, ’reg__num_leaves’: 53, ’reg__subsample’: 0.7097214367795934, ’reg__colsample_bytree’: 0.5702788348521327, ’reg__min_child_samples’: 48, ’reg__reg_alpha’: 9.619552762056877, ’reg__reg_lambda’: 8.304665775432902, ’reg__min_split_gain’: 2.272665831965755, ’reg__path_smooth’: 4.0012700253307125}

2.4.2 Scatter



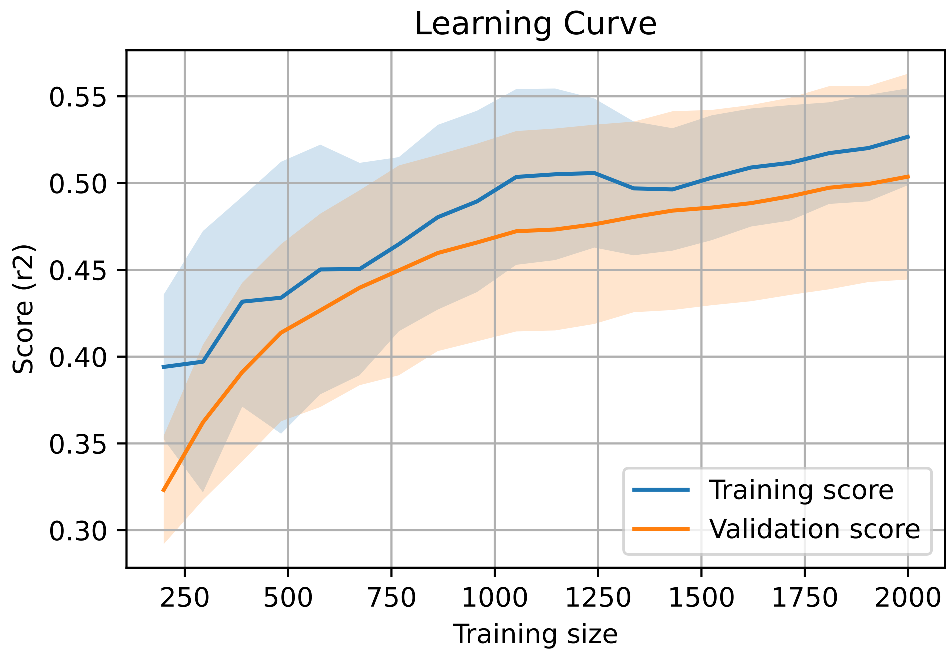
2.4.3 Scatter residuals



2.4.4 Learning curve — Group: 155

Len(group) : 4082, **Len(training)** : 2858,

Len(training_real) : 2001, **Len(test_of_training)** : 857, **Len(test)** : 1224

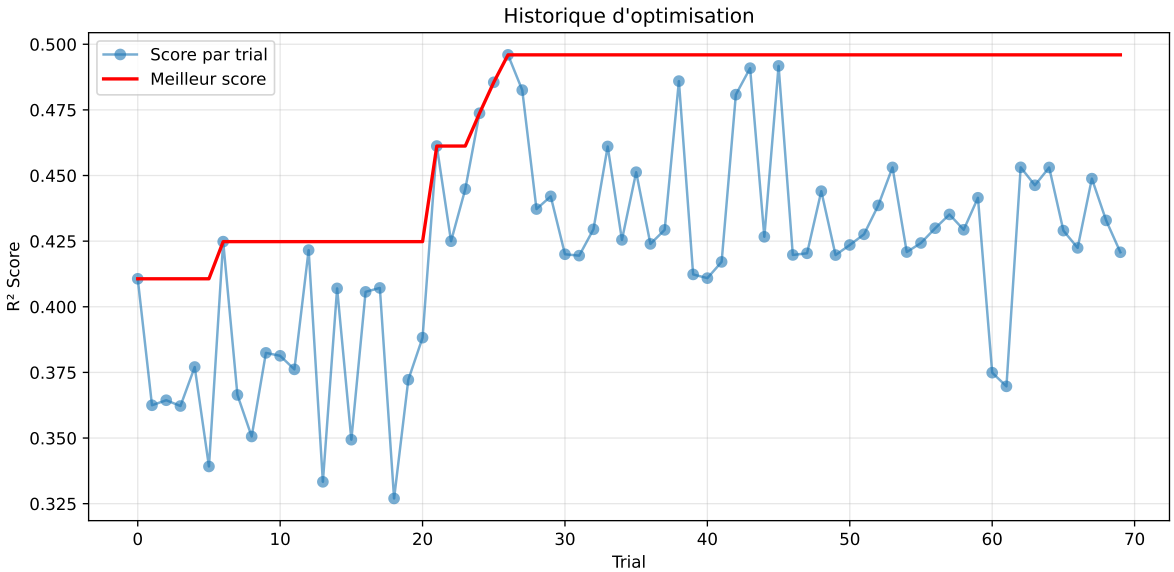


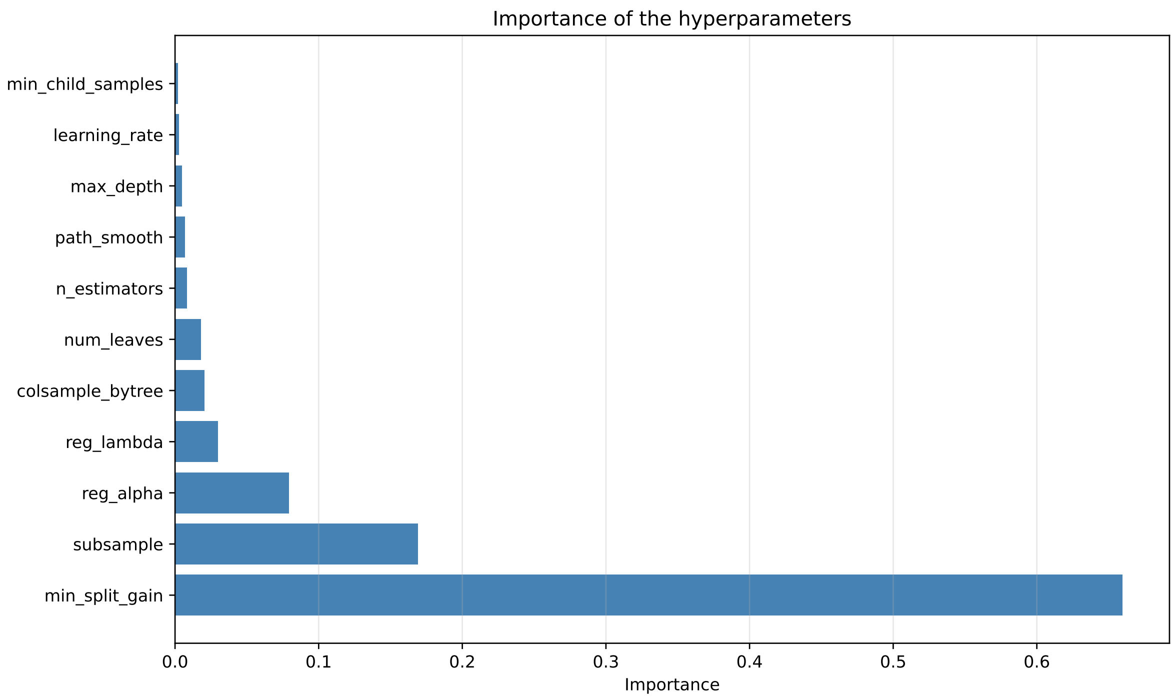
2.4.5 Gridsearch — Group: 270

Len(group) : 4086, **Len(training)** : 2861, **Len(test)** : 1225

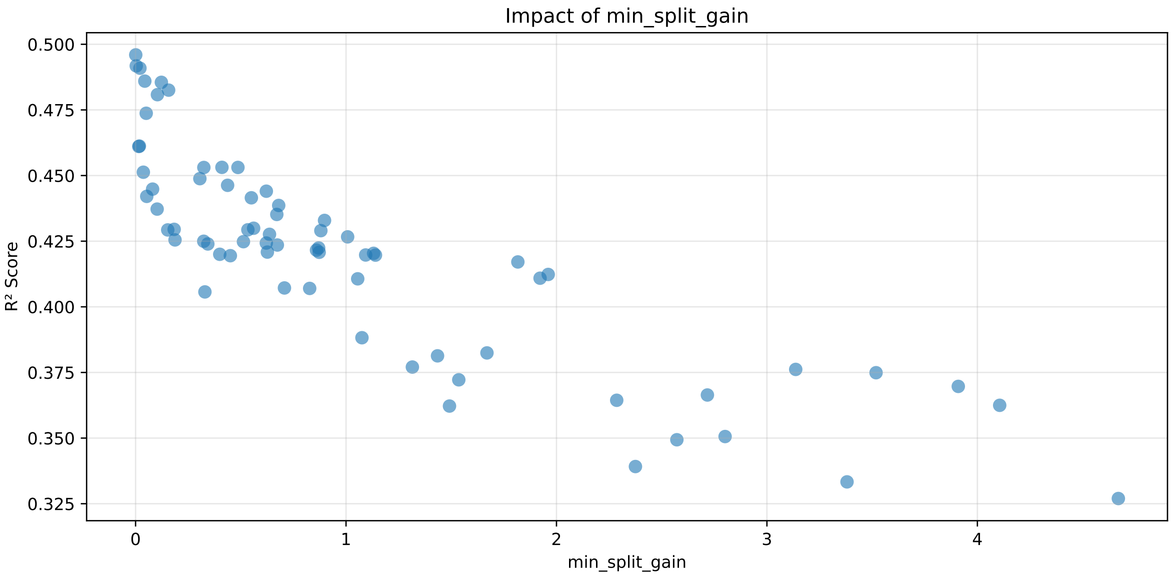
Distribution y_train vs y_test: y_train: mean=2.228, std=0.729, min=0.959, max=7.822 y_test: mean=2.232, std=0.719, min=0.973, max=6.804

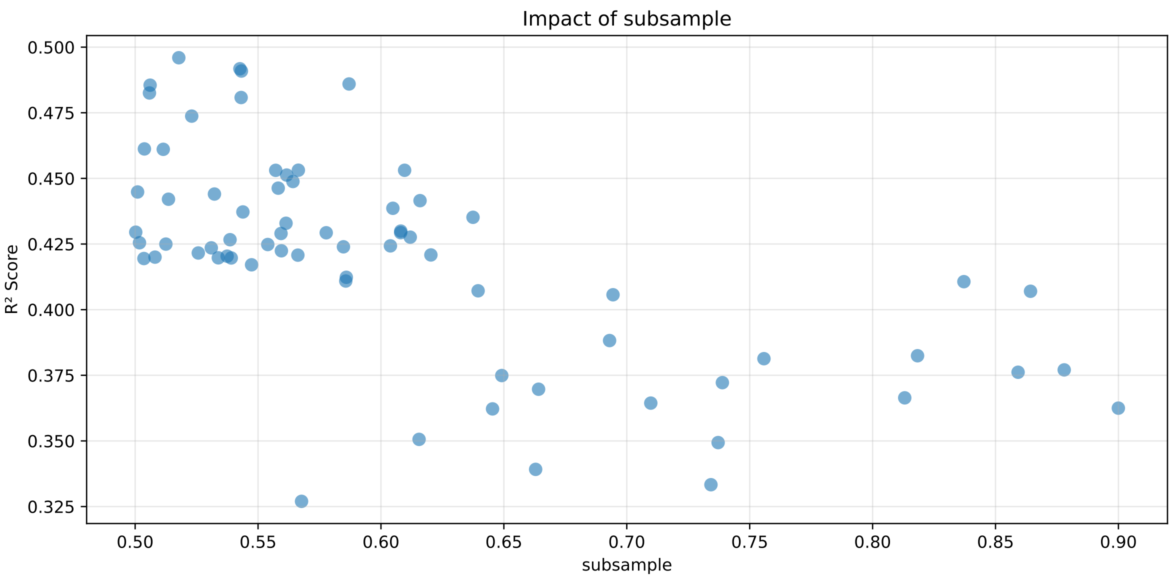
==================== TOP Importance FEATURES ==================== feature importance r_umidity 323 r_THI 108 r_temperature 107 r_NH3 104 r_CO2 53

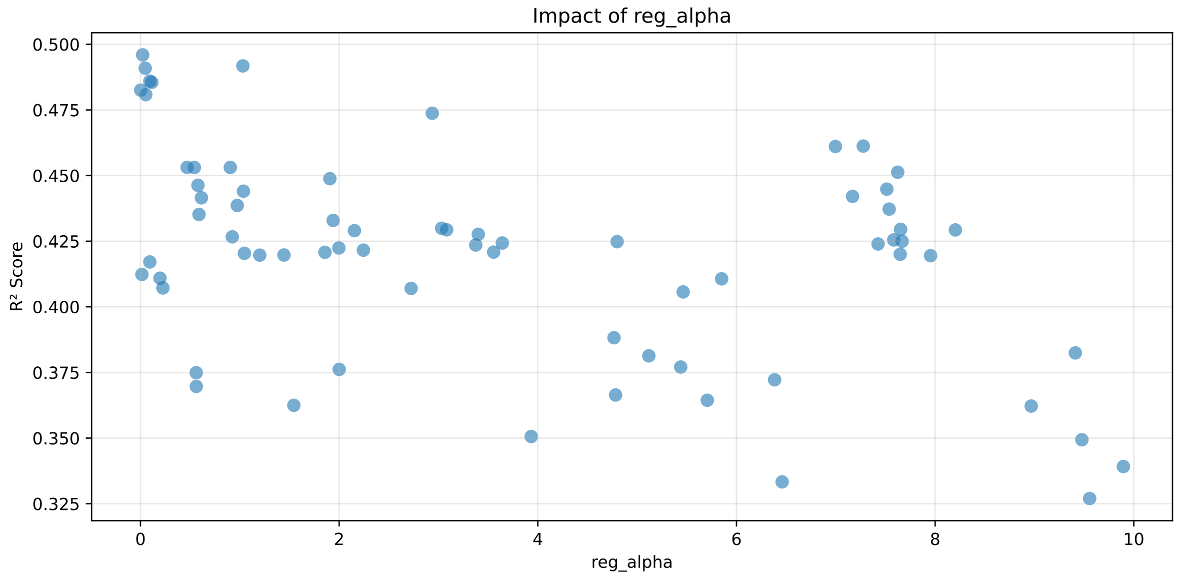




Paramètres avec >5% d'importance : ['min_split_gain', 'subsample', 'reg_alpha']

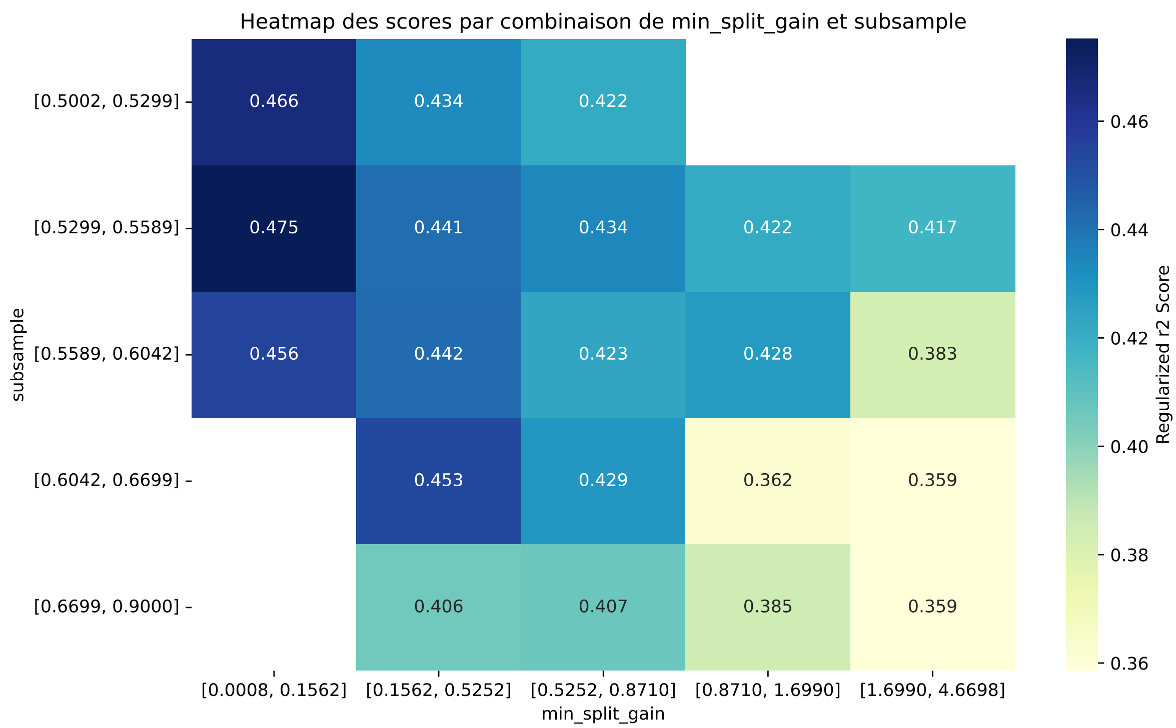

Impact of min_split_gain
R² Score
0.500
0.475
0.450
0.425
0.400
0.375
0.350
0.325
0
1
2
3
4
min_split_gain


Impact of subsample
R² Score
0.500
0.475
0.450
0.425
0.400
0.375
0.350
0.325
0.50
0.55
0.60
0.65
0.70
0.75
0.80
0.85
0.90
subsample



C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a futu



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.3674	0.327	0.3713	0.0264	0.2056	0.3969	0.0699	1.2137	1.1237
2	0.398	0.3494	0.3973	0.0282	0.2034	0.4198	0.0705	1.2017	1.1393
3	0.4158	0.3625	0.4089	0.029	0.2025	0.443	0.0805	1.2221	1.1471
4	0.4228	0.3664	0.4161	0.0292	0.1973	0.4437	0.0773	1.211	1.1539
5	0.4274	0.3697	0.4201	0.0267	0.1964	0.444	0.0743	1.2011	1.1562
6	0.3794	0.3333	0.3796	0.0308	0.1952	0.4132	0.0799	1.2398	1.1382
7	0.3878	0.3392	0.3857	0.0321	0.1931	0.4194	0.0803	1.2366	1.1435
8	0.4356	0.3749	0.4245	0.0283	0.1929	0.4486	0.0737	1.1966	1.1618
9	0.4059	0.3506	0.395	0.0298	0.1847	0.423	0.0725	1.2067	1.1578
10	0.4495	0.3824	0.4337	0.0308	0.1812	0.4616	0.0791	1.2069	1.1754
---	---	---	---	---	---	---	---	---	---
70	0.7946	0.496	0.5545	0.0216	-0.4	0.5954	0.0995	1.2006	1.6022

Hyperparameters:

Rank 1: {’reg__n_estimators’: 600, ’reg__max_depth’: 8, ’reg__learning_rate’: 0.013221648136254736, ’reg__num_leaves’: 43, ’reg__subsample’: 0.567692945174141, ’reg__colsample_bytree’: 0.5578874103098892, ’reg__min_child_samples’: 45, ’reg__reg_alpha’: 9.555473666579674, ’reg__reg_lambda’: 1.9556167888654385, ’reg__min_split_gain’: 4.669836309682736, ’reg__path_smooth’: 0.8676806306499735}

Rank 2: {’reg__n_estimators’: 200, ’reg__max_depth’: 3, ’reg__learning_rate’: 0.0552024331426968, ’reg__num_leaves’: 57, ’reg__subsample’: 0.737158772120165, ’reg__colsample_bytree’: 0.7841799361297062, ’reg__min_child_samples’: 34, ’reg__reg_alpha’: 9.477280533754815, ’reg__reg_lambda’: 33.17546915143546, ’reg__min_split_gain’: 2.572048772241832, ’reg__path_smooth’: 3.5649871258541417}

Rank 3: {’reg__n_estimators’: 600, ’reg__max_depth’: 5, ’reg__learning_rate’: 0.09219689519608633, ’reg__num_leaves’: 40, ’reg__subsample’: 0.8999703086063169, ’reg__colsample_bytree’: 0.8941525115035482, ’reg__min_child_samples’: 38, ’reg__reg_alpha’: 1.5438794861055005, ’reg__reg_lambda’: 22.979928280703767, ’reg__min_split_gain’: 4.105974660162584, ’reg__path_smooth’: 4.470254820580772}

Rank 4: {’reg__n_estimators’: 300, ’reg__max_depth’: 4, ’reg__learning_rate’: 0.013747802983629685, ’reg__num_leaves’: 35, ’reg__subsample’: 0.8130695926842988, ’reg__colsample_bytree’: 0.7144205556567388, ’reg__min_child_samples’: 26, ’reg__reg_alpha’: 4.7829640333734265, ’reg__reg_lambda’: 34.59151272001192, ’reg__min_split_gain’: 2.7171248837480833, ’reg__path_smooth’: 0.268395511314517}

Rank 5: {’reg__n_estimators’: 300, ’reg__max_depth’: 6, ’reg__learning_rate’: 0.04705818012540393, ’reg__num_leaves’: 34, ’reg__subsample’: 0.6641152656363156, 'reg___colsample_bytree': 0.8628381391452083, 'reg___min_child_samples': 34, 'reg___reg_alpha': 0.5616413654639034, 'reg___reg_lambda': 24.455511429033525, 'reg___min_split_gain': 3.909243731870333, 'reg___path_smooth': 0.8403622915934827}

Rank 6: {'reg___n_estimators': 700, 'reg___max_depth': 4, 'reg___learning_rate': 0.06422801315401955, 'reg___num_leaves': 46, 'reg___subsample': 0.7342355726144958, 'reg___colsample_bytree': 0.6625313115785015, 'reg___min_child_samples': 13, 'reg___reg_alpha': 6.459524123011766, 'reg___reg_lambda': 49.81699904614757, 'reg___min_split_gain': 3.3805654066963378, 'reg___path_smooth': 2.6599959366019506}

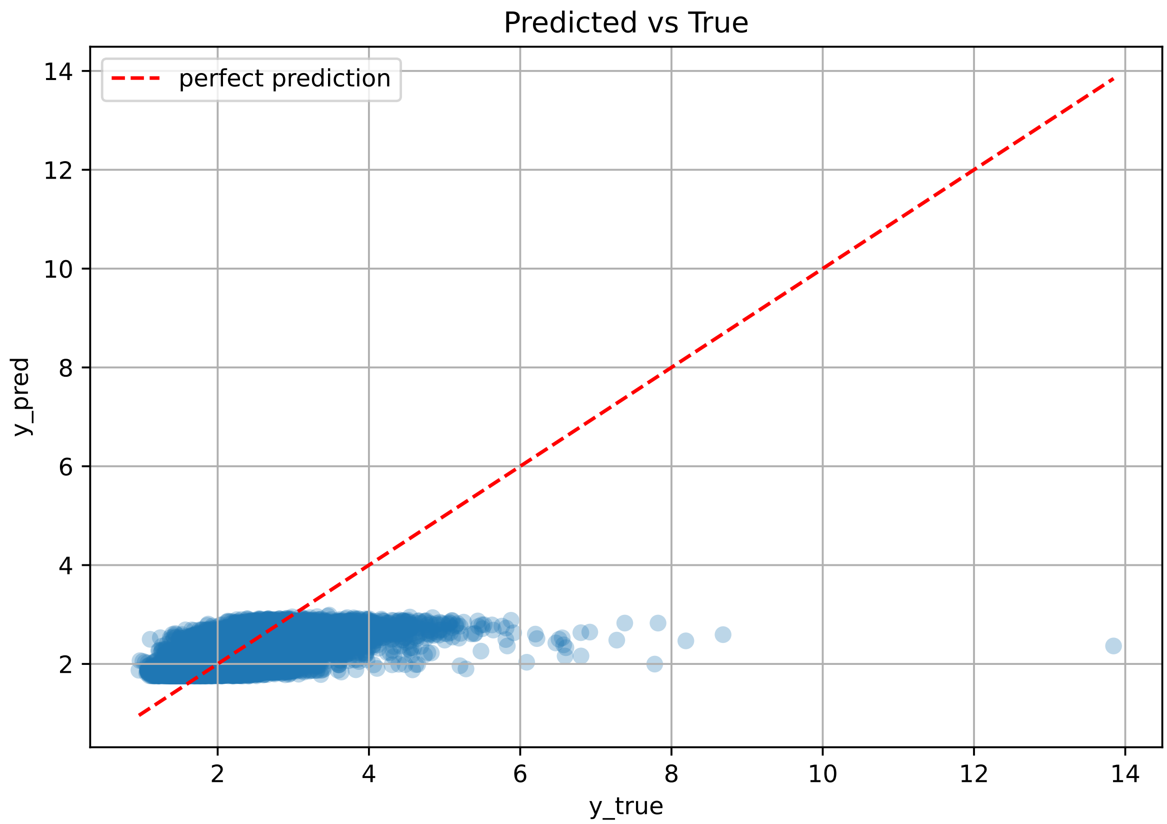
Rank 7: {'reg___n_estimators': 700, 'reg___max_depth': 7, 'reg___learning_rate': 0.06887126124211797, 'reg___num_leaves': 18, 'reg___subsample': 0.6629208055457368, 'reg___colsample_bytree': 0.5612817649929875, 'reg___min_child_samples': 13, 'reg___reg_alpha': 9.894750199979116, 'reg___reg_lambda': 46.52275108274889, 'reg___min_split_gain': 2.3751535948066604, 'reg___path_smooth': 2.6477511624880146}

Rank 8: {'reg___n_estimators': 300, 'reg___max_depth': 6, 'reg___learning_rate': 0.04935733581913025, 'reg___num_leaves': 34, 'reg___subsample': 0.6491498097522376, 'reg___colsample_bytree': 0.8649892137846198, 'reg___min_child_samples': 32, 'reg___reg_alpha': 0.5611392826157781, 'reg___reg_lambda': 24.89830312322897, 'reg___min_split_gain': 3.5185291514456454, 'reg___path_smooth': 0.8428471693194668}

Rank 9: {'reg___n_estimators': 800, 'reg___max_depth': 5, 'reg___learning_rate': 0.07739675274225073, 'reg___num_leaves': 19, 'reg___subsample': 0.6155045808137617, 'reg___colsample_bytree': 0.5994003932406398, 'reg___min_child_samples': 40, 'reg___reg_alpha': 3.933133181358839, 'reg___reg_lambda': 42.84339232713499, 'reg___min_split_gain': 2.801211682745432, 'reg___path_smooth': 0.42678003851219615}

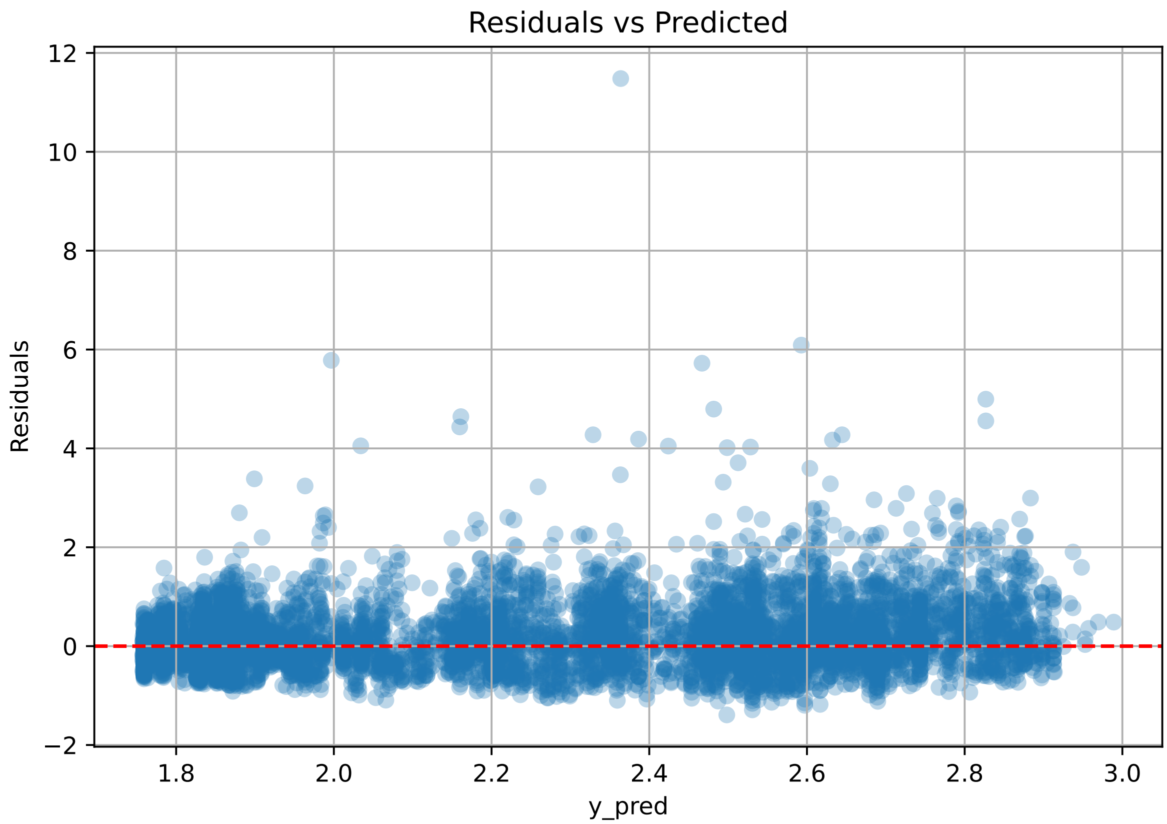
Rank 10: {'reg___n_estimators': 300, 'reg___max_depth': 8, 'reg___learning_rate': 0.023122442009506614, 'reg___num_leaves': 49, 'reg___subsample': 0.8182825192226798, 'reg___colsample_bytree': 0.7843879981453838, 'reg___min_child_samples': 18, 'reg___reg_alpha': 9.41145293097038, 'reg___reg_lambda': 2.2501861215338814, 'reg___min_split_gain': 1.6696743852829425, 'reg___path_smooth': 0.6385739190243978}

2.4.6 Scatter



Predicted vs True
perfect prediction
y_pred
y_true
2
4
6
8
10
12
14

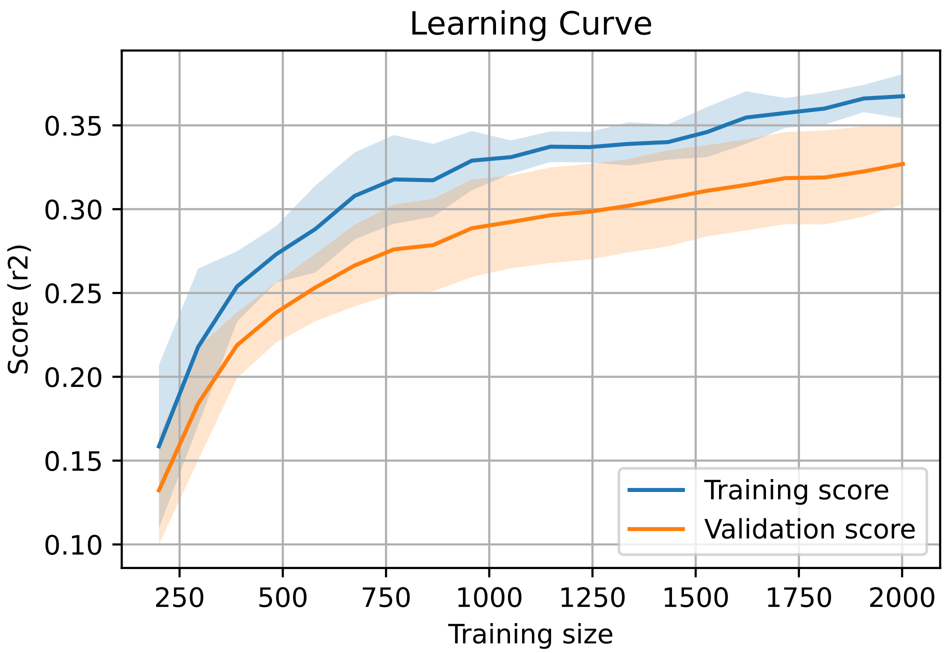
2.4.7 Scatter residuals



2.4.8 Learning curve — Group: 270

Len(group) : 4086, **Len(training)** : 2861,

Len(training_real) : 2003, **Len(test_of_training)** : 858, **Len(test)** : 1225



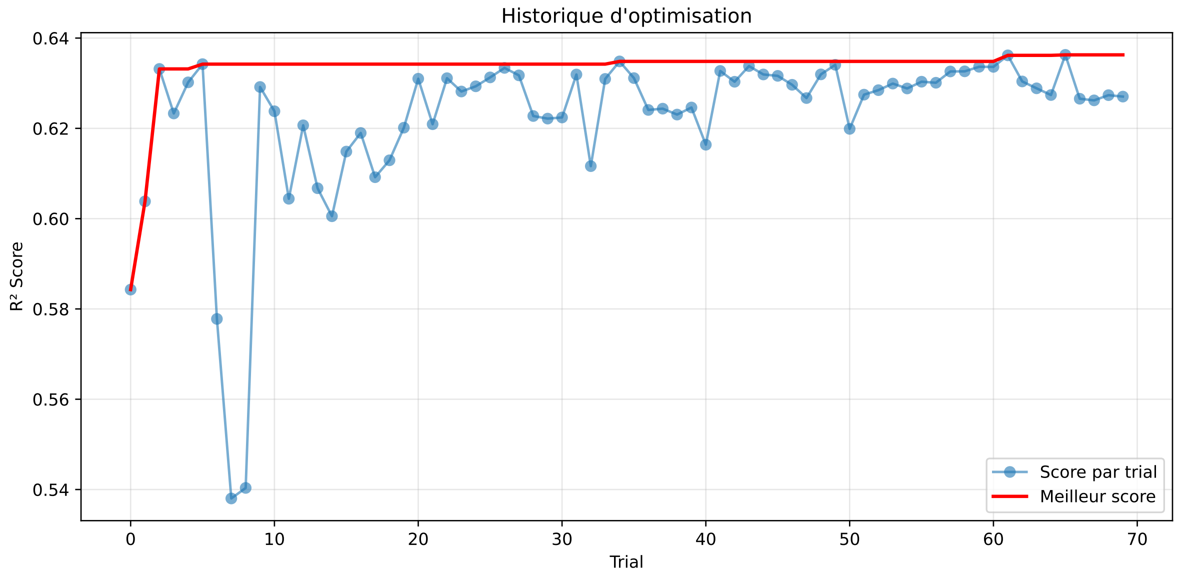
2.5 Model : CatBoost

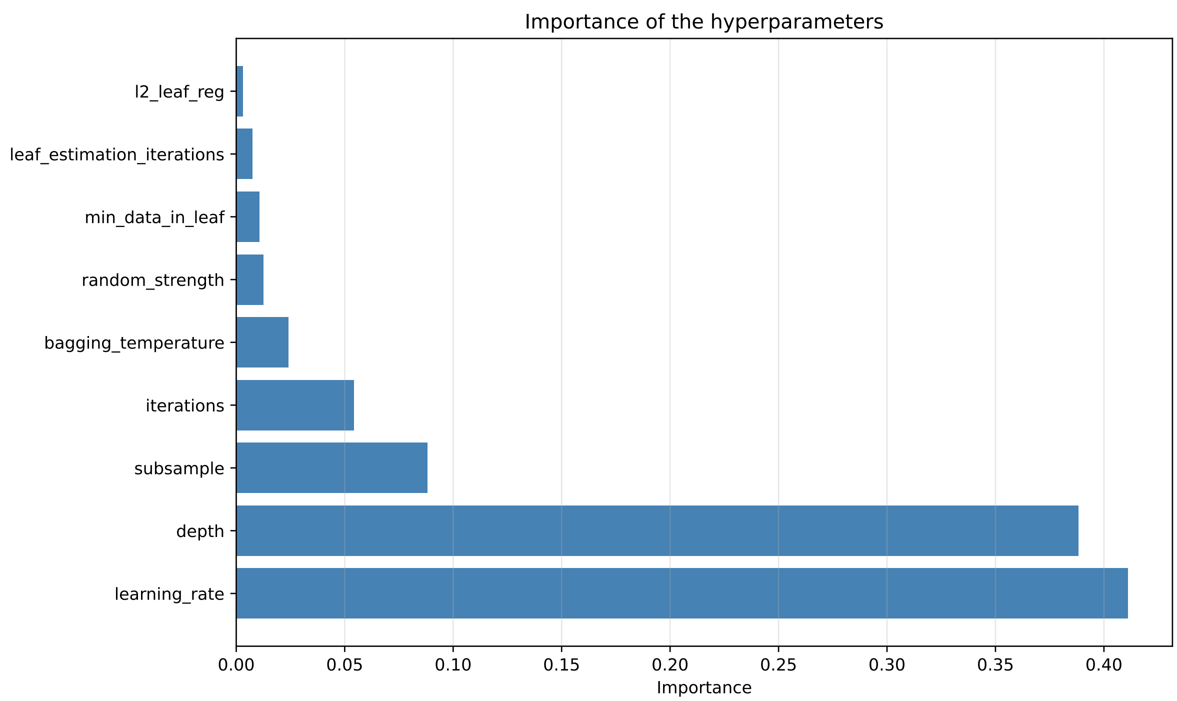
2.5.1 Gridsearch — Group: 155

Len(group) : 4082, **Len(training)** : 2858, **Len(test)** : 1224

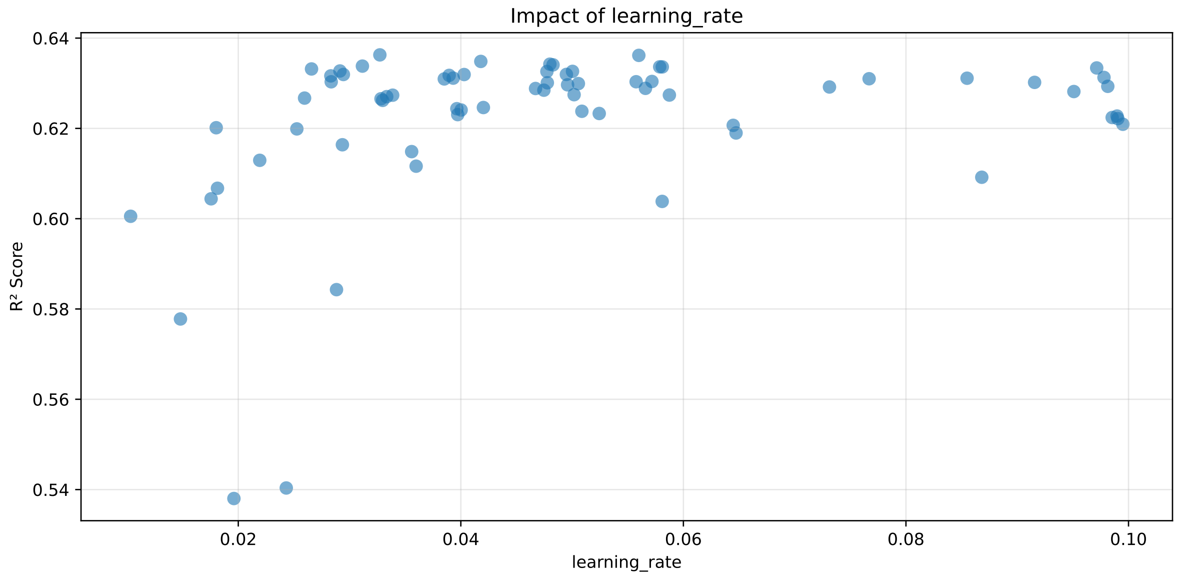
Distribution y_train vs y_test: y_train: mean=2.469, std=0.854, min=1.041, max=13.845 y_test: mean=2.472, std=0.811, min=1.068, max=7.277

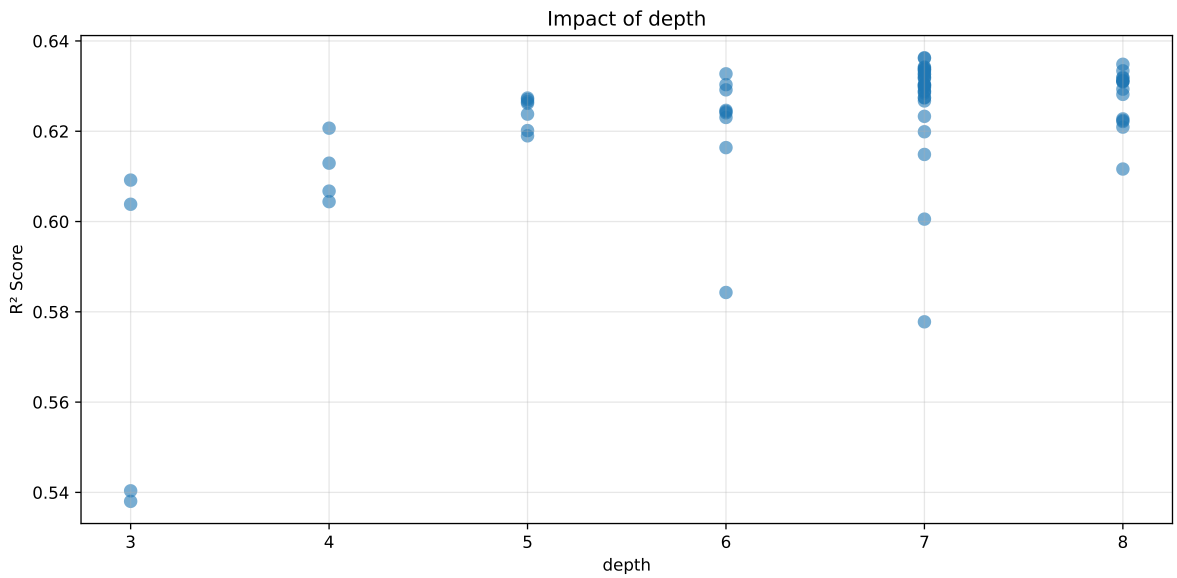
==================== TOP Importance FEATURES ==================== feature importance r_umidity 47.585895 r_NH3 25.478205 r_temperature 13.858462 r_CO2 8.108778 r_THI 4.968660

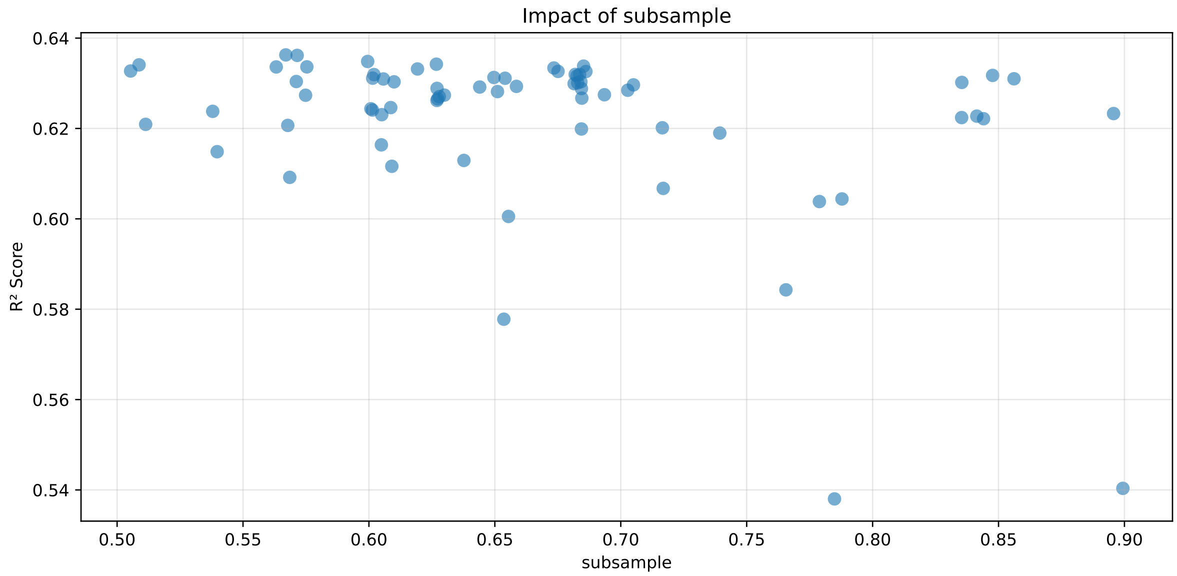


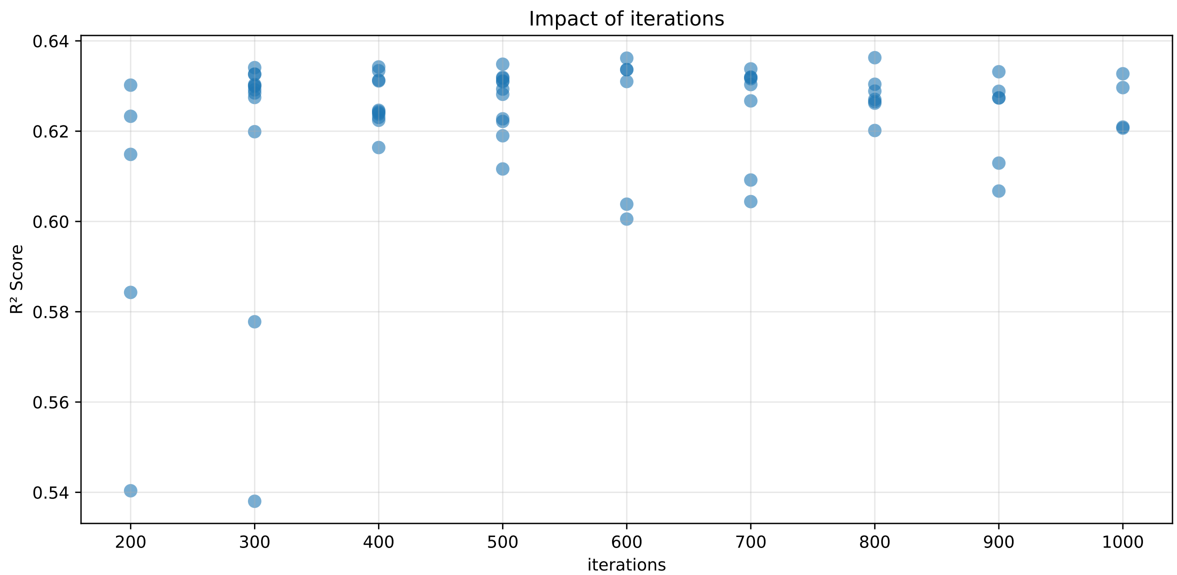


Paramètres avec >5% d'importance : ['learning_rate', 'depth', 'subsample', 'iterations']


Impact of learning_rate
R² Score
0.64
0.62
0.60
0.58
0.56
0.54
0.02
0.04
0.06
0.08
0.10
learning_rate


Impact of depth
R² Score
0.64
0.62
0.60
0.58
0.56
0.54
3
4
5
6
7
8
depth





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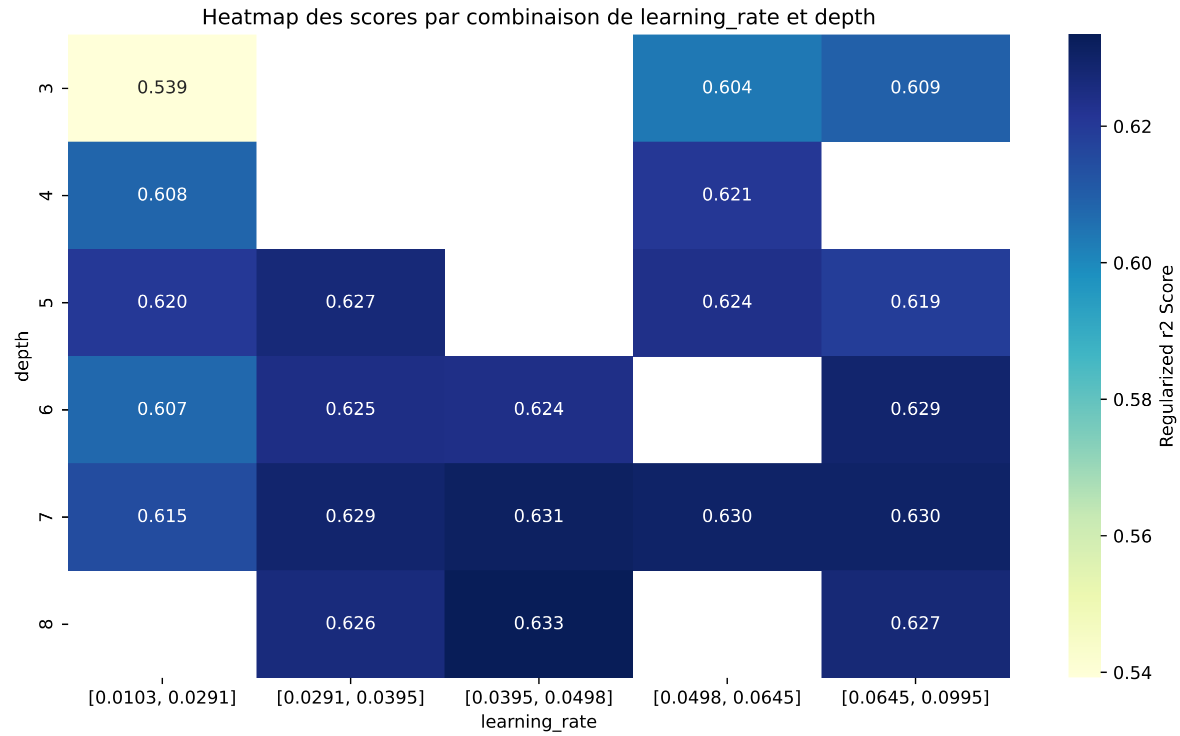
The default value of observed=False is deprecated and will change to observed=True in a futu



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.5554	0.5404	0.5761	0.0578	0.5404	0.5806	0.0402	1.0744	1.0279
2	0.5493	0.538	0.5743	0.0588	0.538	0.5788	0.0408	1.0758	1.0209
3	0.6202	0.5843	0.6307	0.062	0.4764	0.6386	0.0543	1.0929	1.0615
4	0.6142	0.5778	0.6277	0.0616	0.4685	0.638	0.0602	1.1041	1.063
5	0.6528	0.6005	0.6516	0.0623	0.4439	0.6579	0.0574	1.0956	1.087
6	0.6958	0.6149	0.6679	0.0599	0.372	0.6736	0.0588	1.0956	1.1317
7	0.6993	0.6164	0.6673	0.0589	0.3677	0.6747	0.0583	1.0946	1.1345
8	0.6875	0.6067	0.6567	0.0569	0.3645	0.6659	0.0592	1.0975	1.1331
9	0.6984	0.6044	0.6527	0.055	0.3225	0.657	0.0526	1.087	1.1555
10	0.7231	0.6199	0.6739	0.0596	0.3103	0.6825	0.0626	1.1011	1.1665
---	---	---	---	---	---	---	---	---	---
70	0.9867	0.6209	0.6778	0.0606	-0.4763	0.6874	0.0665	1.1071	1.589

Hyperparameters:

Rank 1: {'reg___iterations': 200, 'reg___depth': 3, 'reg___learning_rate': 0.024312991681750922, 'reg___l2_leaf_reg': 10.115691825411377, 'reg___bagging_temperature': 0.2551736968607432, 'reg___random_strength': 2.691891710598817, 'reg___subsample': 0.8993774246179816, 'reg___min_data_in_leaf': 40, 'reg___leaf_estimation_iterations': 8}

Rank 2: {’reg__iterations’: 300, ’reg__depth’: 3, ’reg__learning_rate’: 0.019607742293889, ’reg__l2_leaf_reg’: 38.729910296019376, ’reg__bagging_temperature’: 1.997657253411963, ’reg__random_strength’: 4.079818220839949, ’reg__subsample’: 0.7848708940522413, ’reg__min_data_in_leaf’: 28, ’reg__leaf_estimation_iterations’: 8}

Rank 3: {’reg__iterations’: 200, ’reg__depth’: 6, ’reg__learning_rate’: 0.02882734594546585, ’reg__l2_leaf_reg’: 25.579425816540184, ’reg__bagging_temperature’: 2.032936180642379, ’reg__random_strength’: 3.5481440800417015, ’reg__subsample’: 0.7655773936784336, ’reg__min_data_in_leaf’: 35, ’reg__leaf_estimation_iterations’: 4}

Rank 4: {’reg__iterations’: 300, ’reg__depth’: 7, ’reg__learning_rate’: 0.014805120178567924, ’reg__l2_leaf_reg’: 32.42229526706602, ’reg__bagging_temperature’: 4.218898210031042, ’reg__random_strength’: 3.3588714970062092, ’reg__subsample’: 0.6535715763178709, ’reg__min_data_in_leaf’: 21, ’reg__leaf_estimation_iterations’: 4}

Rank 5: {’reg__iterations’: 600, ’reg__depth’: 7, ’reg__learning_rate’: 0.010327024329068873, ’reg__l2_leaf_reg’: 39.28068457133074, ’reg__bagging_temperature’: 2.2625379140665776, ’reg__random_strength’: 3.8561879088876285, ’reg__subsample’: 0.6554167716857197, ’reg__min_data_in_leaf’: 15, ’reg__leaf_estimation_iterations’: 10}

Rank 6: {’reg__iterations’: 200, ’reg__depth’: 7, ’reg__learning_rate’: 0.035581147607270824, ’reg__l2_leaf_reg’: 24.95262648975925, ’reg__bagging_temperature’: 4.834255640401825, ’reg__random_strength’: 2.020418187207338, ’reg__subsample’: 0.5397326107104115, ’reg__min_data_in_leaf’: 45, ’reg__leaf_estimation_iterations’: 9}

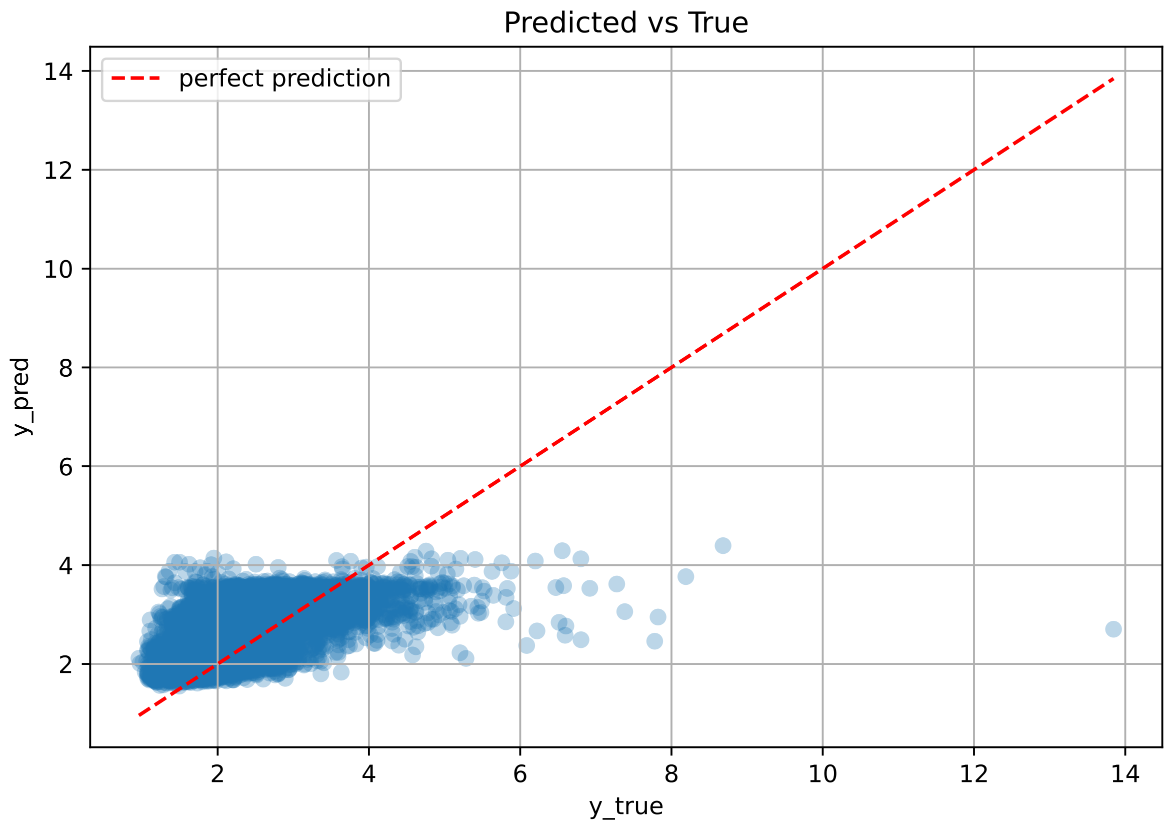
Rank 7: {’reg__iterations’: 400, ’reg__depth’: 6, ’reg__learning_rate’: 0.02935689494913306, ’reg__l2_leaf_reg’: 32.75976844568392, ’reg__bagging_temperature’: 3.769406908258169, ’reg__random_strength’: 4.910167562073605, ’reg__subsample’: 0.6049462761883266, ’reg__min_data_in_leaf’: 33, ’reg__leaf_estimation_iterations’: 6}

Rank 8: {’reg__iterations’: 900, ’reg__depth’: 4, ’reg__learning_rate’: 0.018130979762835467, ’reg__l2_leaf_reg’: 39.603889619332975, ’reg__bagging_temperature’: 0.8702038003770912, ’reg__random_strength’: 1.491472724902877, ’reg__subsample’: 0.7168595362424677, ’reg__min_data_in_leaf’: 48, ’reg__leaf_estimation_iterations’: 4}

Rank 9: {’reg__iterations’: 700, ’reg__depth’: 4, ’reg__learning_rate’: 0.017559652924231235, ’reg__l2_leaf_reg’: 11.652394760550424, ’reg__bagging_temperature’: 1.53377897208625, ’reg__random_strength’: 2.1296565243236945, ’reg__subsample’: 0.7878413766143956, ’reg__min_data_in_leaf’: 42, ’reg__leaf_estimation_iterations’: 9}

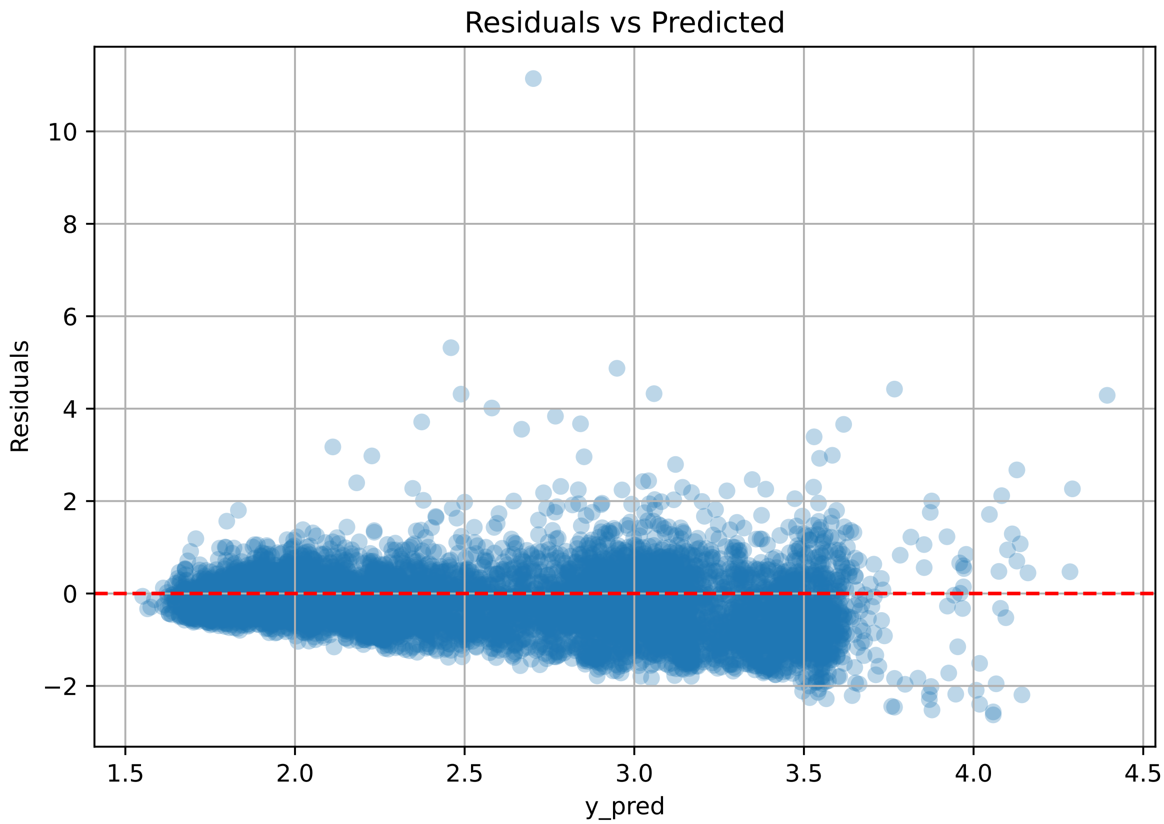
Rank 10: {’reg__iterations’: 300, ’reg__depth’: 7, ’reg__learning_rate’: 0.025263725434064684, ’reg__l2_leaf_reg’: 43.02567800107932, ’reg__bagging_temperature’: 1.8175037582611533, ’reg__random_strength’: 0.6232222362088541, ’reg__subsample’: 0.6843873477482194, ’reg__min_data_in_leaf’: 37, ’reg__leaf_estimation_iterations’: 7}

2.5.2 Scatter



Predicted vs True
perfect prediction
y_pred
y_true
2
4
6
8
10
12
14

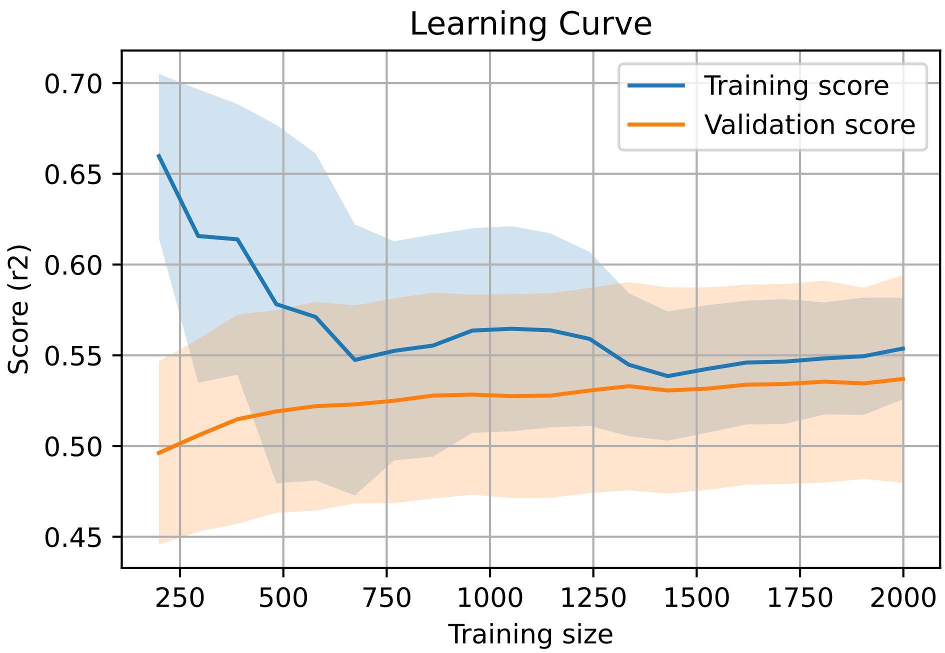
2.5.3 Scatter residuals



2.5.4 Learning curve — Group: 155

Len(group) : 4082, **Len(training)** : 2858,

Len(training_real) : 2001, **Len(test_of_training)** : 857, **Len(test)** : 1224

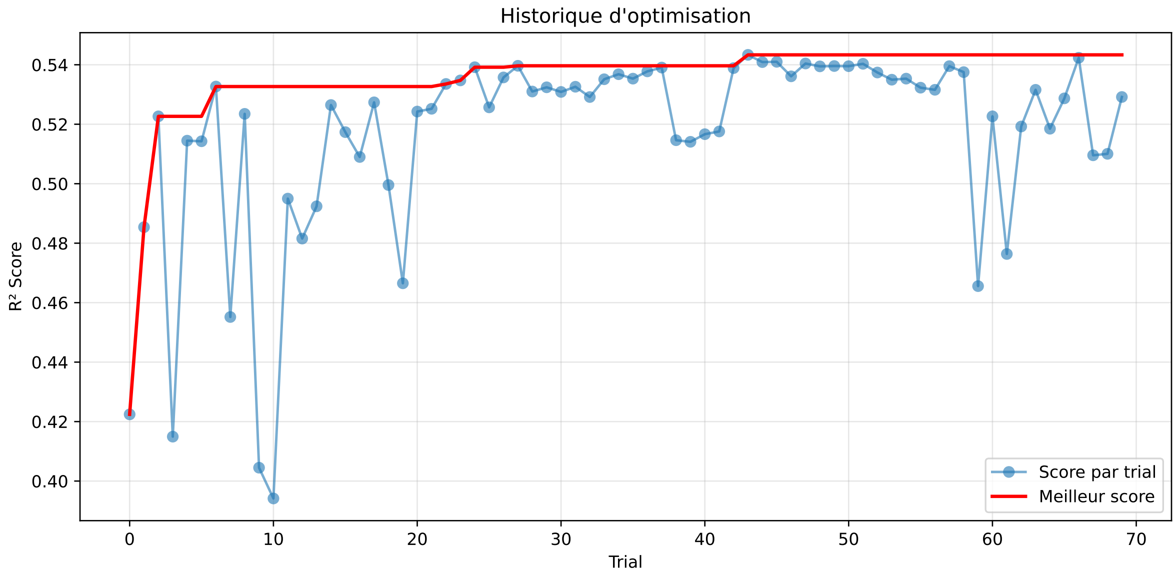


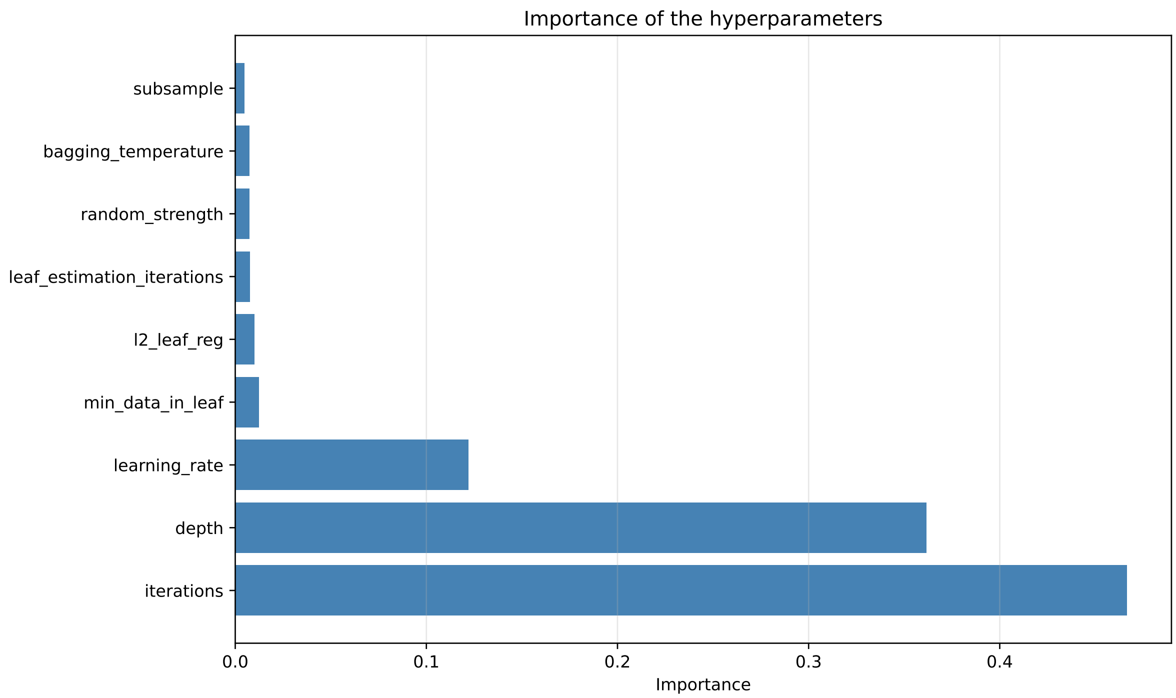
2.5.5 Gridsearch — Group: 270

Len(group) : 4086, **Len(training)** : 2861, **Len(test)** : 1225

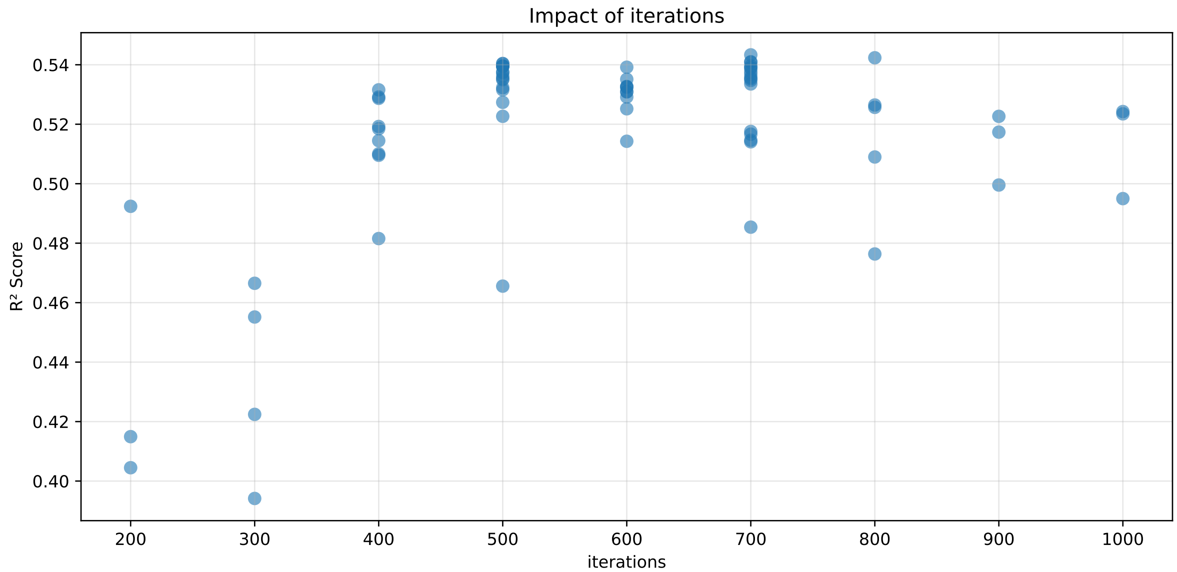
Distribution y_train vs y_test: y_train: mean=2.228, std=0.729, min=0.959, max=7.822 y_test: mean=2.232, std=0.719, min=0.973, max=6.804

==================== TOP Importance FEATURES ==================== feature importance r_umidity 46.160810 r_NH3 17.701615 r_THI 16.357906 r_CO2 12.894088 r_temperature 6.885581

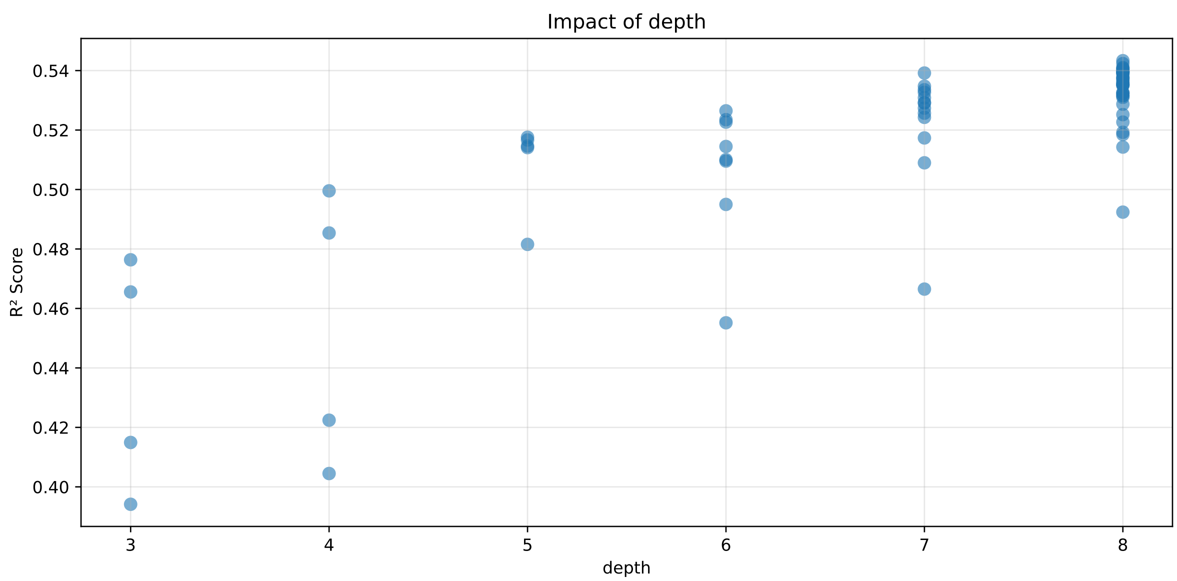




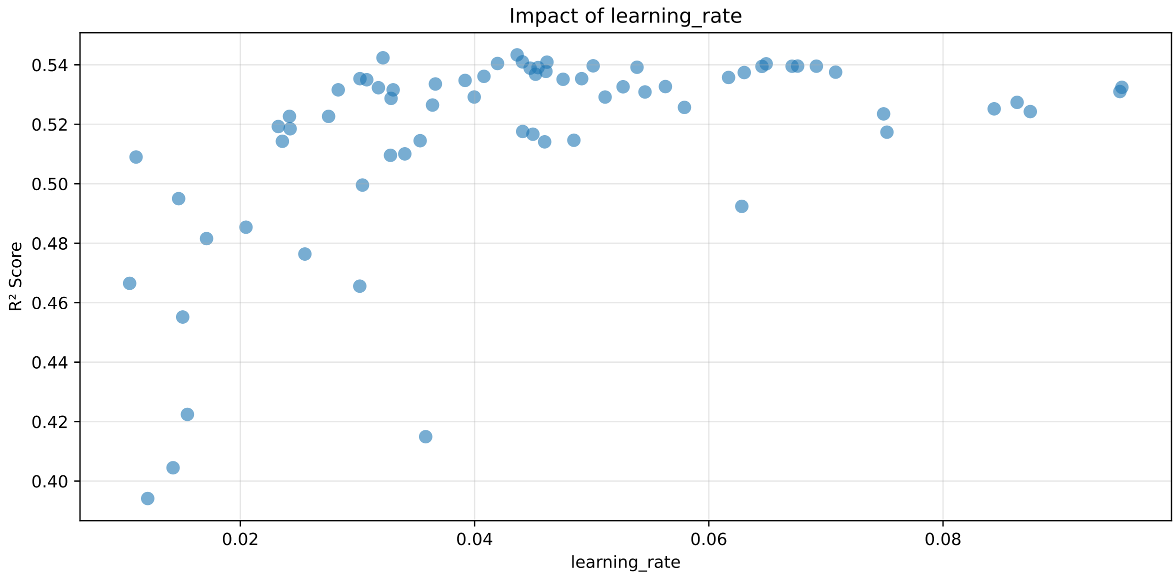
Paramètres avec >5% d'importance : ['iterations', 'depth', 'learning_rate']

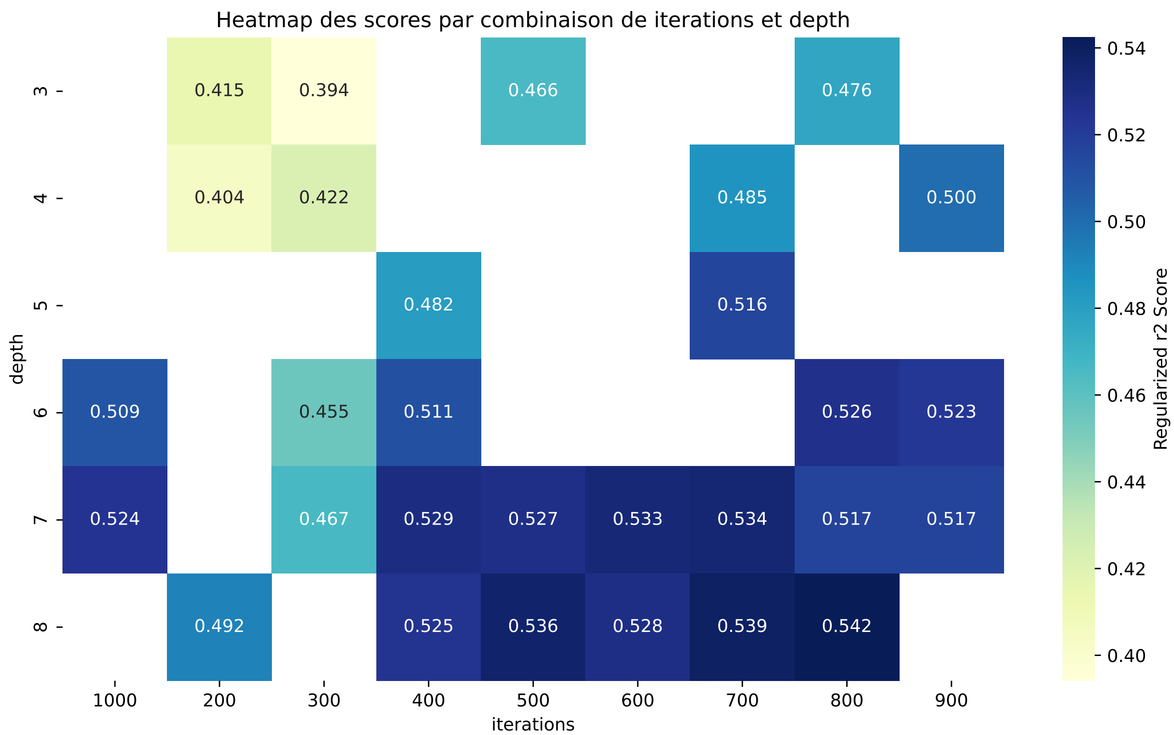


Impact of iterations
R² Score
0.40
0.42
0.44
0.46
0.48
0.50
0.52
0.54
200
300
400
500
600
700
800
900
1000
iterations



Impact of depth
R² Score
0.40
0.42
0.44
0.46
0.48
0.50
0.52
0.54
3
4
5
6
7
8
depth


Impact of learning_rate
R² Score
0.54
0.52
0.50
0.48
0.46
0.44
0.42
0.40
0.02
0.04
0.06
0.08
learning_rate


Heatmap des scores par combinaison de iterations et depth
depth
3
4
5
6
7
8
0.415
0.394
0.466
0.476
0.404
0.422
0.485
0.500
0.482
0.516
0.509
0.455
0.511
0.526
0.523
0.524
0.467
0.529
0.527
0.533
0.534
0.517
0.517
0.492
0.525
0.536
0.528
0.539
0.542
1000
200
300
400
500
600
700
800
900
iterations
Regularized r2 Score
0.54
0.52
0.50
0.48
0.46
0.44
0.42
0.40

rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.4686	0.4224	0.4731	0.0301	0.2839	0.4813	0.0589	1.1393	1.1093
2	0.465	0.4149	0.4647	0.0309	0.2646	0.4735	0.0585	1.1411	1.1207
3	0.4386	0.3941	0.4425	0.0288	0.2608	0.4461	0.0519	1.1317	1.1128
4	0.4544	0.4045	0.4524	0.0295	0.2546	0.4586	0.0541	1.1339	1.1235
5	0.5255	0.4552	0.5135	0.0285	0.2443	0.521	0.0659	1.1447	1.1544
6	0.5586	0.4665	0.5236	0.0276	0.1903	0.5356	0.0691	1.1481	1.1973
7	0.5652	0.4656	0.5187	0.0303	0.1665	0.531	0.0654	1.1405	1.2141
8	0.5964	0.4816	0.5368	0.0289	0.1369	0.5547	0.0732	1.152	1.2386
9	0.6071	0.4854	0.5424	0.0258	0.1202	0.5591	0.0737	1.1519	1.2508
10	0.6229	0.495	0.5535	0.0262	0.1113	0.5741	0.0791	1.1598	1.2584
---	---	---	---	---	---	---	---	---	---
70	0.9653	0.5173	0.5819	0.0205	-0.8264	0.619	0.1016	1.1964	1.8658

Hyperparameters:

Rank 1: {'reg___iterations': 300, 'reg___depth': 4, 'reg___learning_rate': 0.0154801097193033, 'reg___l2_leaf_reg': 25.060424044082165, 'reg___bagging_temperature': 1.452371708292299, 'reg___random_strength': 4.813275956141924, 'reg___subsample': 0.6220860125110919, 'reg___min_data_in_leaf': 50, 'reg___leaf_estimation_iterations': 4}

Rank 2: {'reg___iterations': 200, 'reg___depth': 3, 'reg___learning_rate': 0.03582449083005715, 'reg___l2_leaf_reg': 30.335433727798993, 'reg___bagging_temperature': 4.764612585468852, 'reg___random_strength': 4.850266429028675, 'reg___subsample': 0.8086092419765907, 'reg___min_data_in_leaf': 23, 'reg___leaf_estimation_iterations': 3}

Rank 3: {'reg___iterations': 300, 'reg___depth': 3, 'reg___learning_rate': 0.012090145450609411, 'reg___l2_leaf_reg': 41.64665230824922, 'reg___bagging_temperature': 1.8380006175736785, 'reg___random_strength': 1.753520573008861, 'reg___subsample': 0.6636830711803711, 'reg___min_data_in_leaf': 12, 'reg___leaf_estimation_iterations': 8}

Rank 4: {'reg___iterations': 200, 'reg___depth': 4, 'reg___learning_rate': 0.014254431051156308, 'reg___l2_leaf_reg': 12.341046903633996, 'reg___bagging_temperature': 1.794805437955362, 'reg___random_strength': 2.243578787253271, 'reg___subsample': 0.6344564551518406, 'reg___min_data_in_leaf': 36, 'reg___leaf_estimation_iterations': 9}

Rank 5: {'reg___iterations': 300, 'reg___depth': 6, 'reg___learning_rate': 0.015076796139906442, 'reg___l2_leaf_reg': 25.341782954454388, 'reg___bagging_temperature': 3.2737630246331935, 'reg___random_strength': 4.227408205020117, 'reg___subsample': 0.6344152971546062, 'reg___min_data_in_leaf': 29, 'reg___leaf_estimation_iterations': 4}

Rank 6: {'reg___iterations': 300, 'reg___depth': 7, 'reg___learning_rate': 0.010546173830694824, 'reg___l2_leaf_reg': 30.33699678592842, 'reg___bagging_temperature': 0.32699791225374364, 'reg___random_strength': 1.397051780122411, 'reg___subsample': 0.7633662420887168, 'reg___min_data_in_leaf': 46, 'reg___leaf_estimation_iterations': 10}

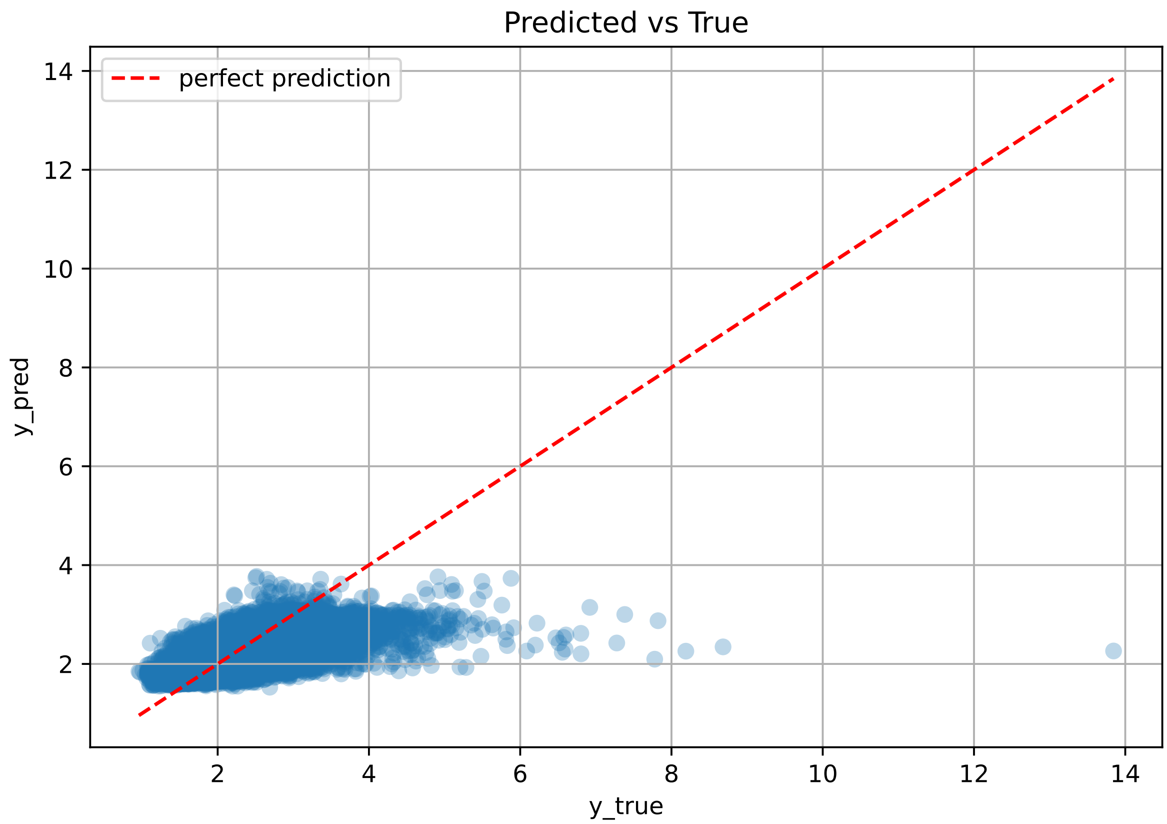
Rank 7: {’reg___iterations’: 500, ’reg___depth’: 3, ’reg___learning_rate’: 0.030196444841202652, ’reg___l2_leaf_reg’: 10.889238092380797, ’reg___bagging_temperature’: 3.695440533163716, ’reg___random_strength’: 4.2900067394346095, ’reg___subsample’: 0.5941502224330992, ’reg___min_data_in_leaf’: 28, ’reg___leaf_estimation_iterations’: 3}

Rank 8: {’reg___iterations’: 400, ’reg___depth’: 5, ’reg___learning_rate’: 0.01711302989503689, ’reg___l2_leaf_reg’: 23.315855690008302, ’reg___bagging_temperature’: 2.6617573233257135, ’reg___random_strength’: 0.9383315887971622, ’reg___subsample’: 0.6311323409246732, ’reg___min_data_in_leaf’: 25, ’reg___leaf_estimation_iterations’: 9}

Rank 9: {’reg___iterations’: 700, ’reg___depth’: 4, ’reg___learning_rate’: 0.020480740328079623, ’reg___l2_leaf_reg’: 18.768745802766325, ’reg___bagging_temperature’: 2.011457658278288, ’reg___random_strength’: 4.537066474495291, ’reg___subsample’: 0.7310554060058028, ’reg___min_data_in_leaf’: 42, ’reg___leaf_estimation_iterations’: 5}

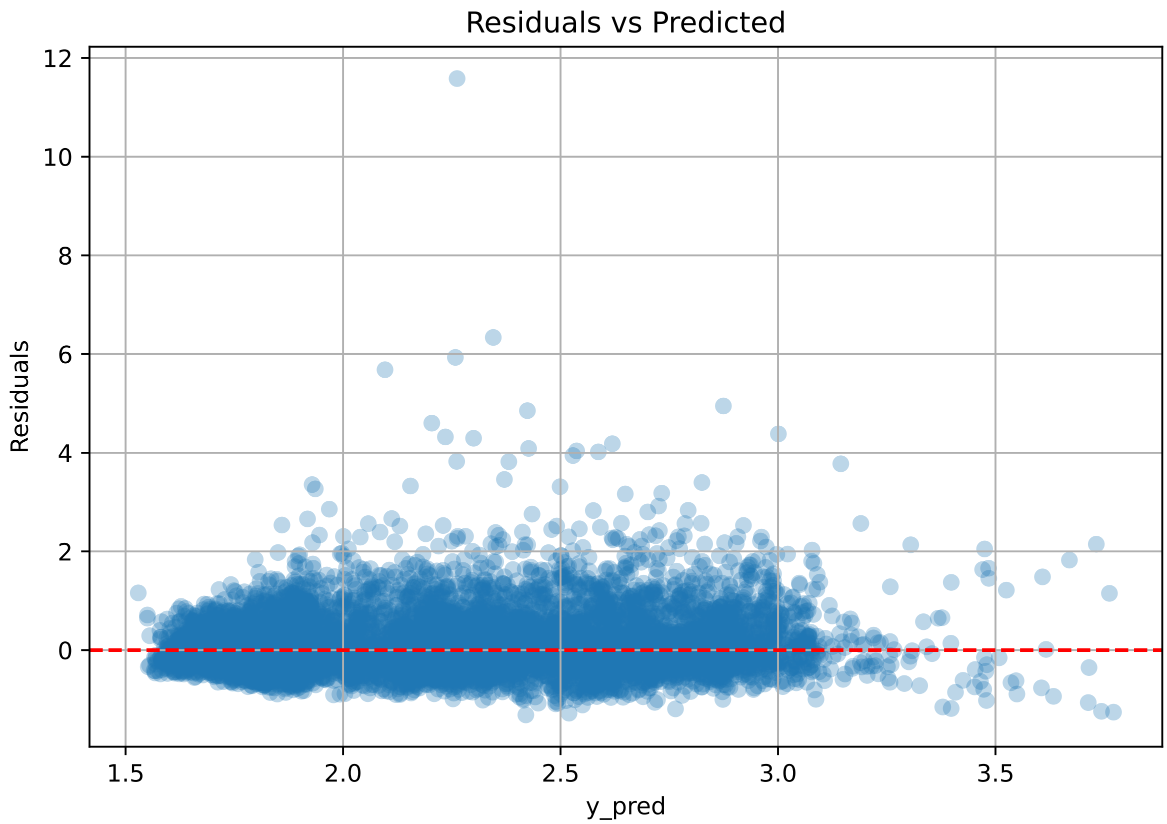
Rank 10: {’reg___iterations’: 1000, ’reg___depth’: 6, ’reg___learning_rate’: 0.014729861446829997, ’reg___l2_leaf_reg’: 36.613699474326864, ’reg___bagging_temperature’: 1.00124672551039, ’reg___random_strength’: 4.573579946737772, ’reg___subsample’: 0.7667945799888376, ’reg___min_data_in_leaf’: 48, ’reg___leaf_estimation_iterations’: 2}

2.5.6 Scatter



Predicted vs True
perfect prediction
y_pred
y_true
14
12
10
8
6
4
2
2
4
6
8
10
12
14

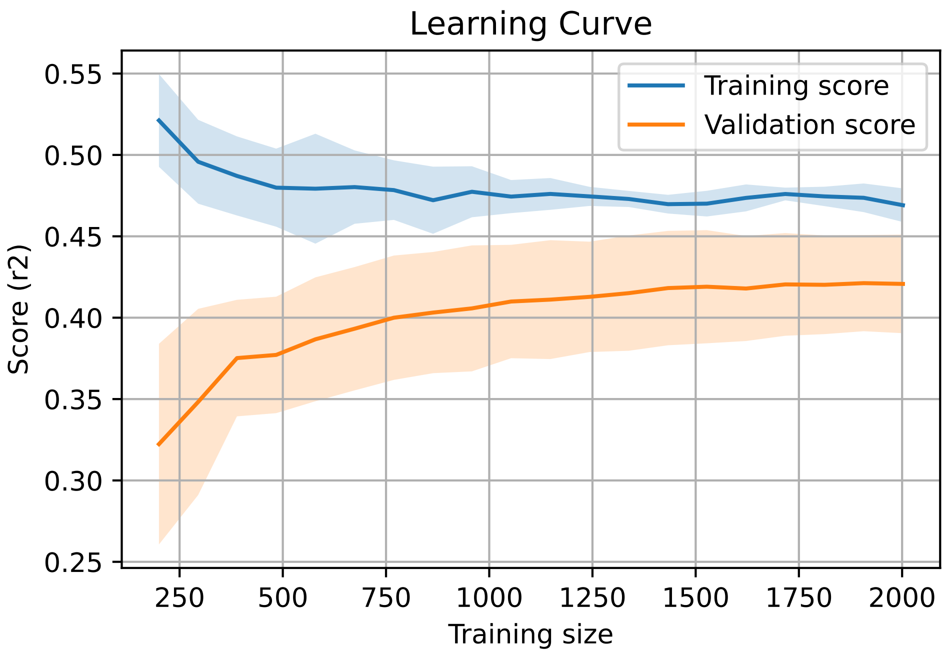
2.5.7 Scatter residuals



2.5.8 Learning curve — Group: 270

Len(group) : 4086, **Len(training)** : 2861,

Len(training_real) : 2003, **Len(test_of_training)** : 858, **Len(test)** : 1225



2.6 Sumup-table

model	group	Len training	Len test	Len total	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	155	2858	1224	4082	0.422	0.432	0.051	0.45	0.45	0.017	1.04	0.977	nan
Linear	270	2861	1225	4086	0.327	0.312	0.035	0.363	0.364	0.052	1.165	1.047	nan
Linear	MEAN	5719	2449	8168	0.375	0.372	0.043	0.406	0.407	0.034	1.103	1.012	nan
LGBM	155	2858	1224	4082	0.526	0.503	0.059	0.54	0.563	0.06	1.12	1.046	0.5027
LGBM	270	2861	1225	4086	0.367	0.327	0.026	0.371	0.397	0.07	1.214	1.124	0.2056
LGBM	MEAN	5719	2449	8168	0.447	0.415	0.043	0.456	0.48	0.065	1.167	1.085	0.354
CatBoost	155	2858	1224	4082	0.555	0.54	0.058	0.576	0.581	0.04	1.074	1.028	0.5404
CatBoost	270	2861	1225	4086	0.469	0.422	0.03	0.473	0.481	0.059	1.139	1.109	0.2839
CatBoost	MEAN	5719	2449	8168	0.512	0.481	0.044	0.525	0.531	0.05	1.107	1.069	0.412

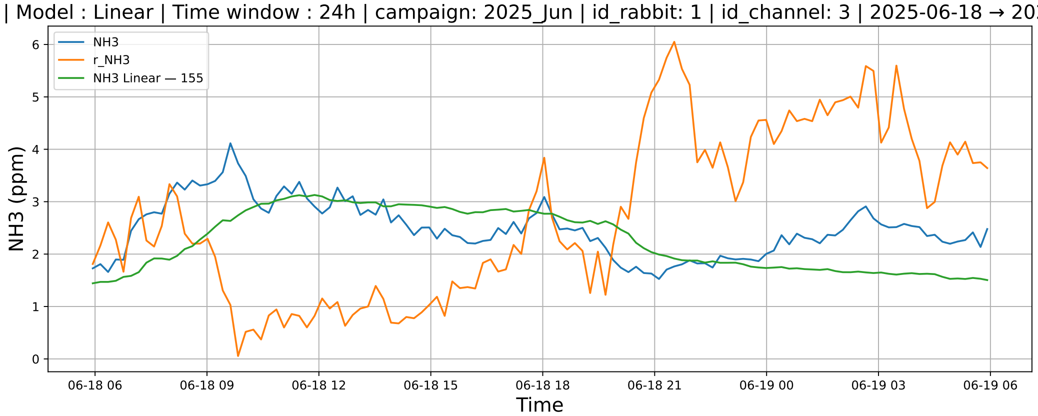
3 Plot

3.1 Standalone plots

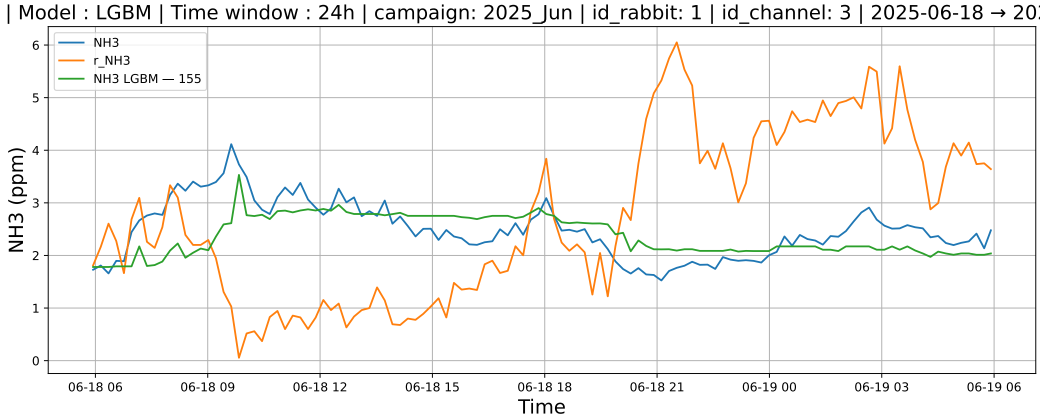
3.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

3.1.1.1 Time window : 24

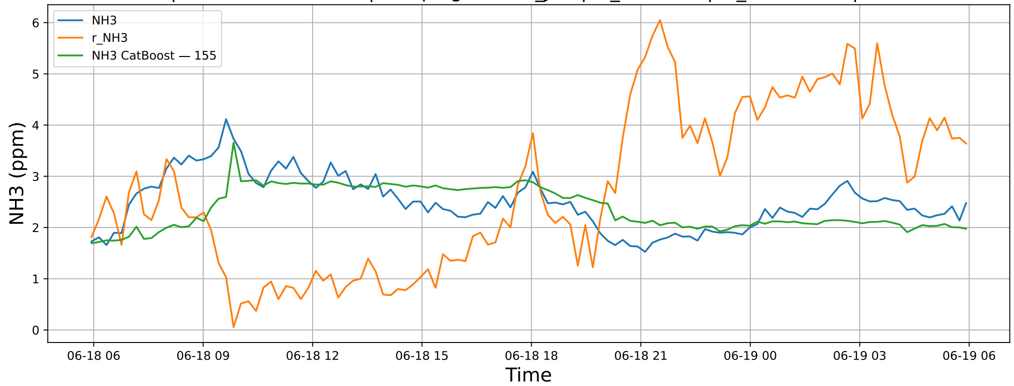
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

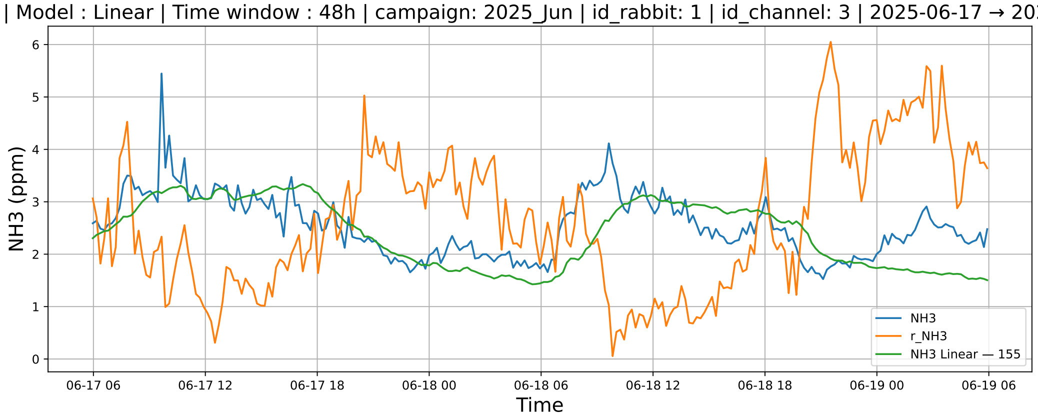


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

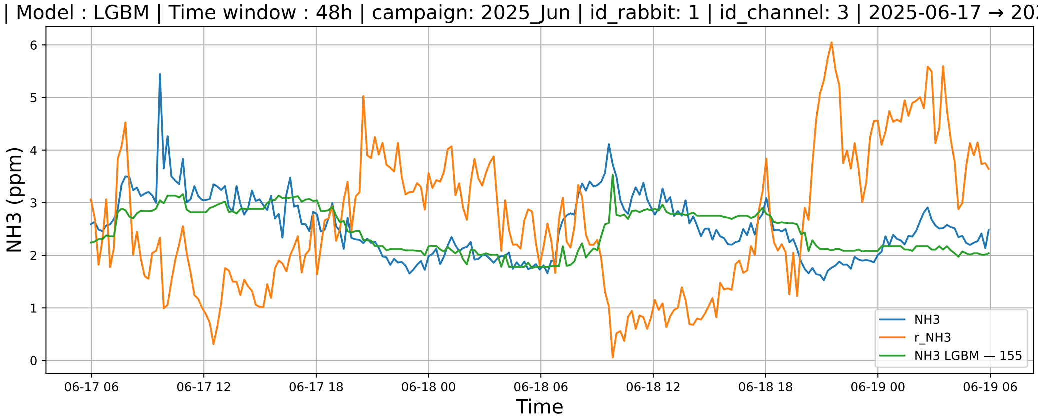


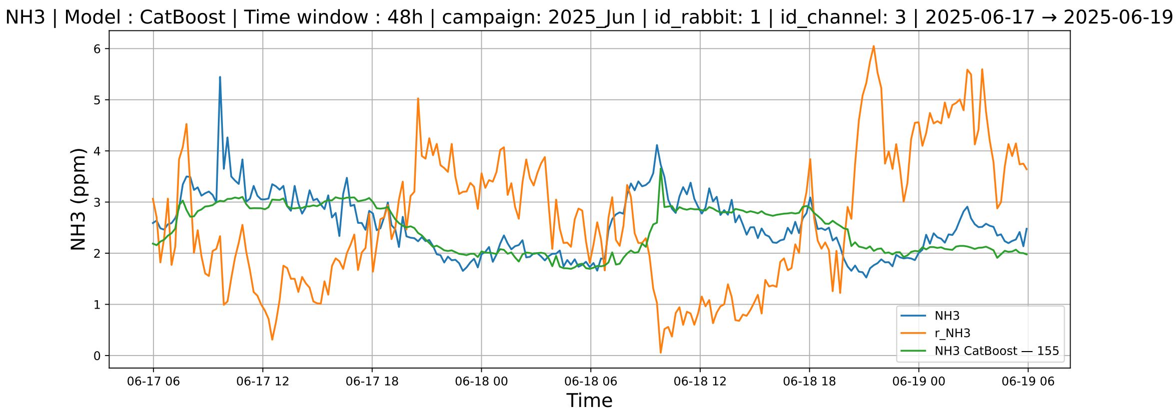
3.1.1.2 Time window : 48

NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

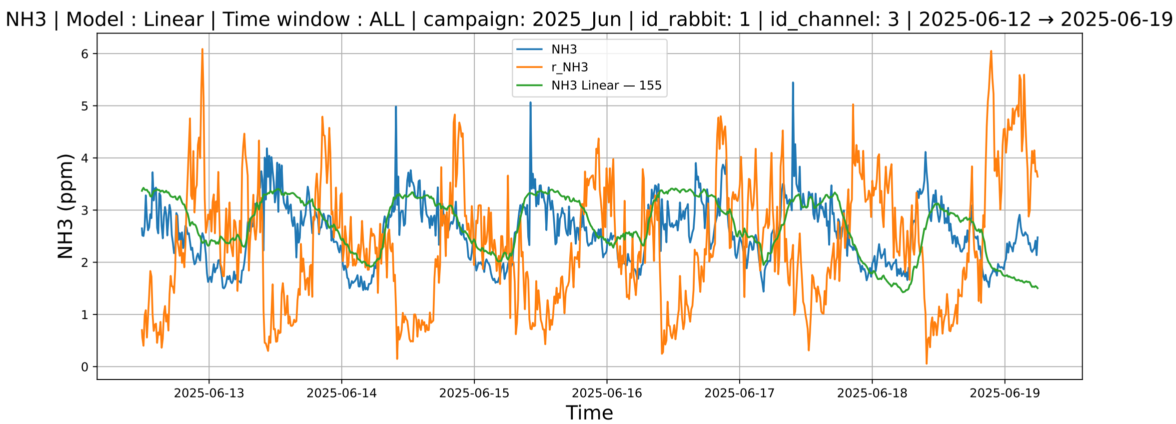


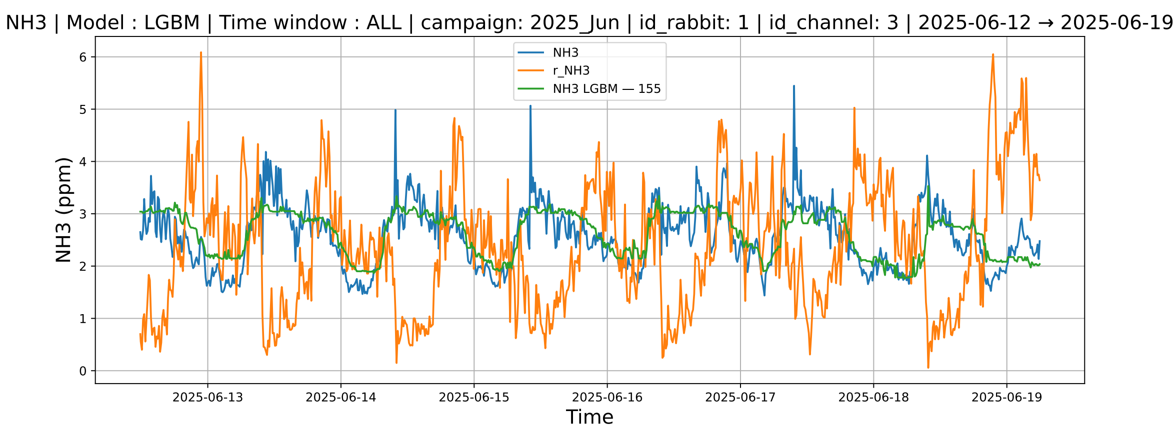
NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



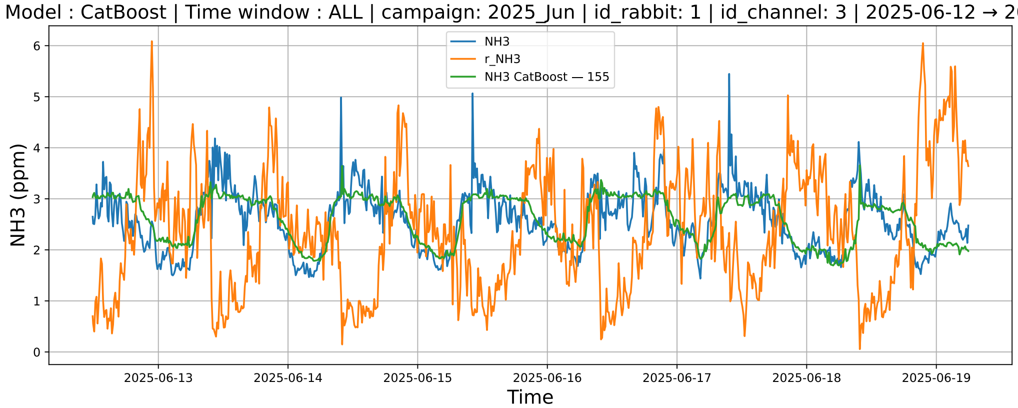


3.1.1.3 Time window : ALL





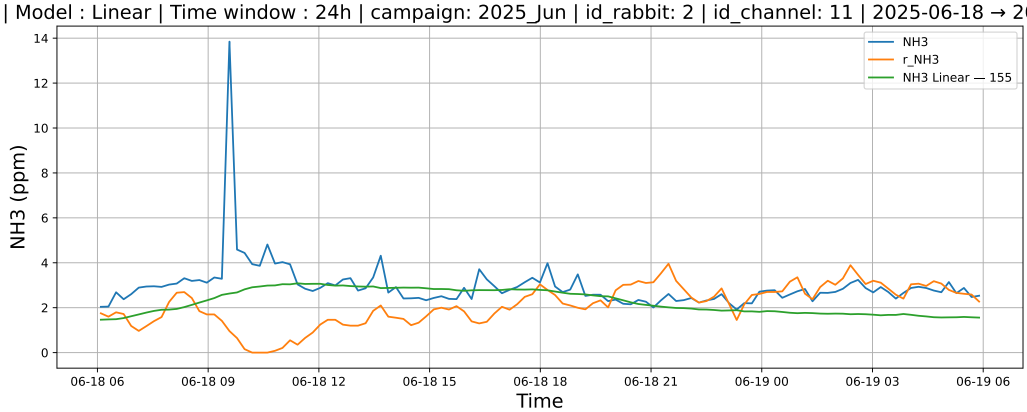
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



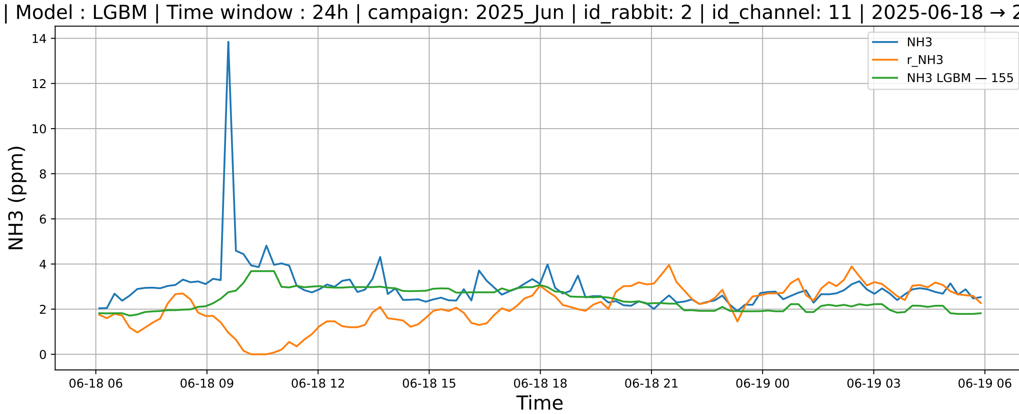
3.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

3.1.2.1 Time window : 24

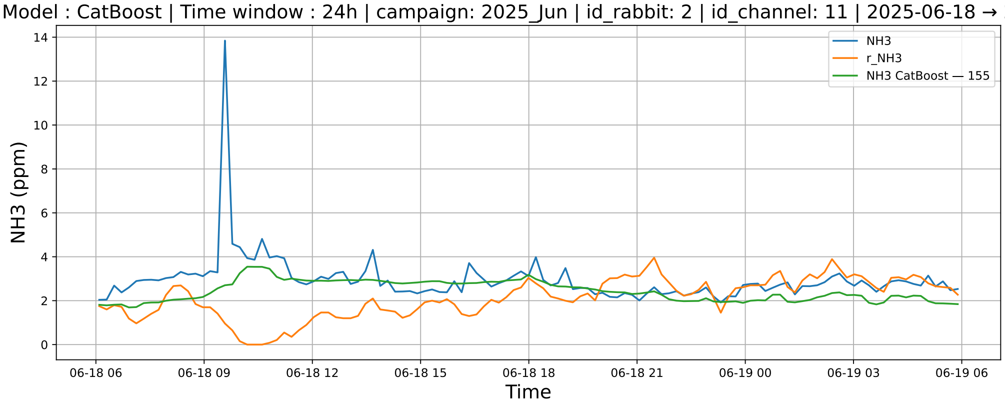
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

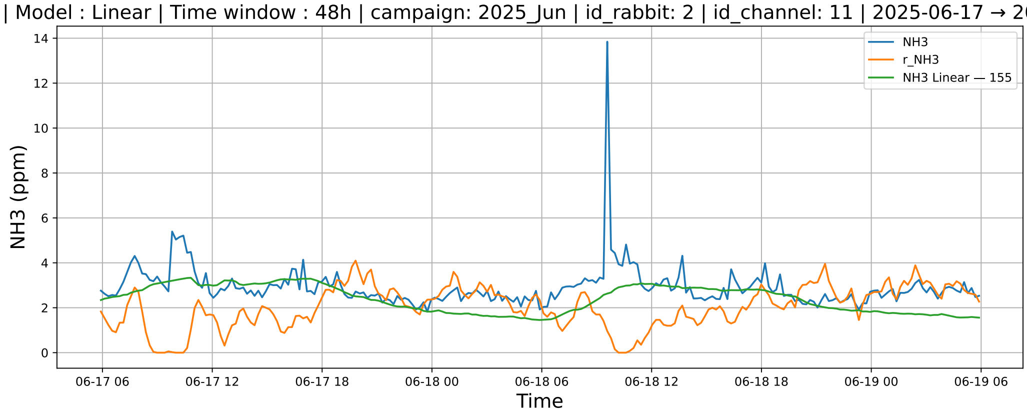


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

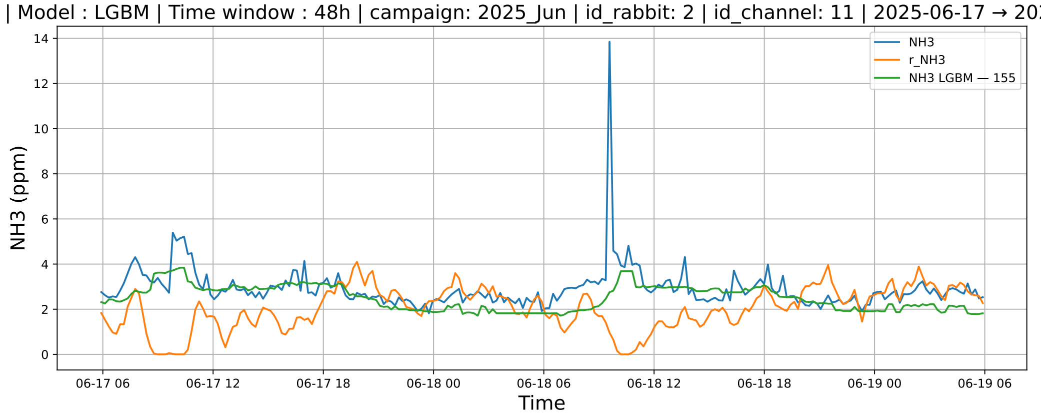


3.1.2.2 Time window : 48

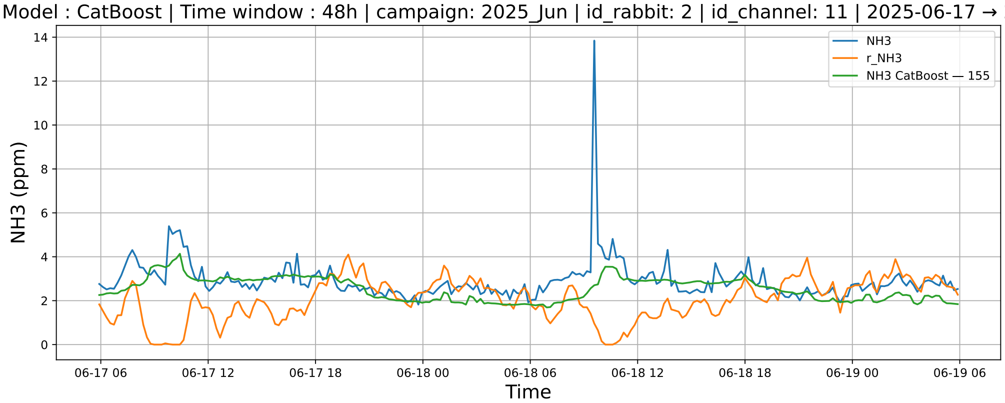
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

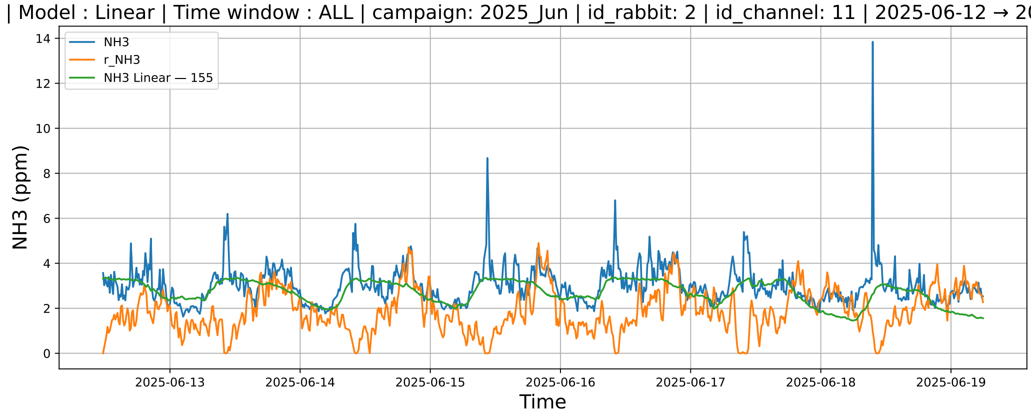


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

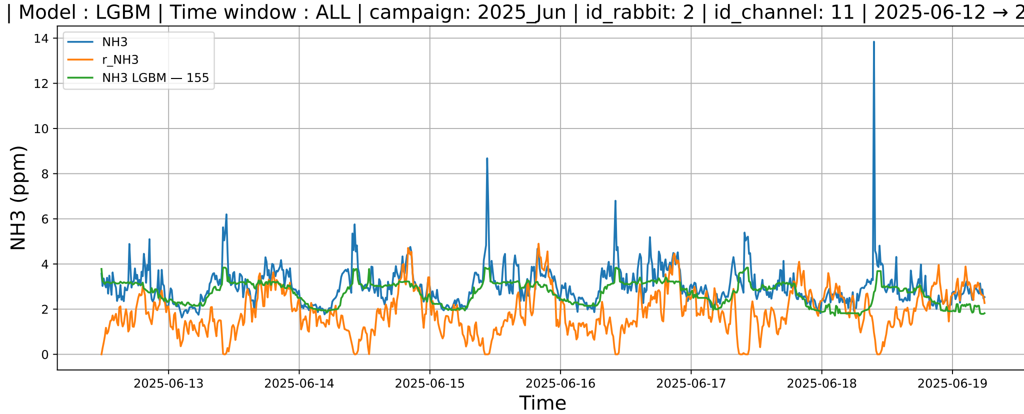


3.1.2.3 Time window : ALL

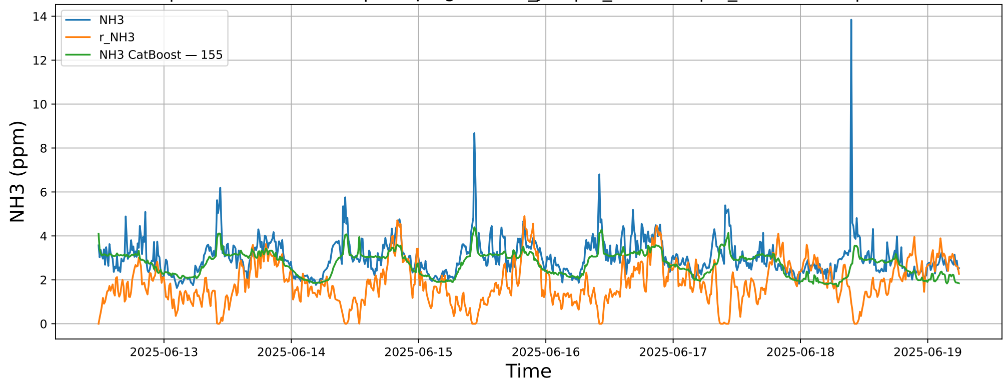
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



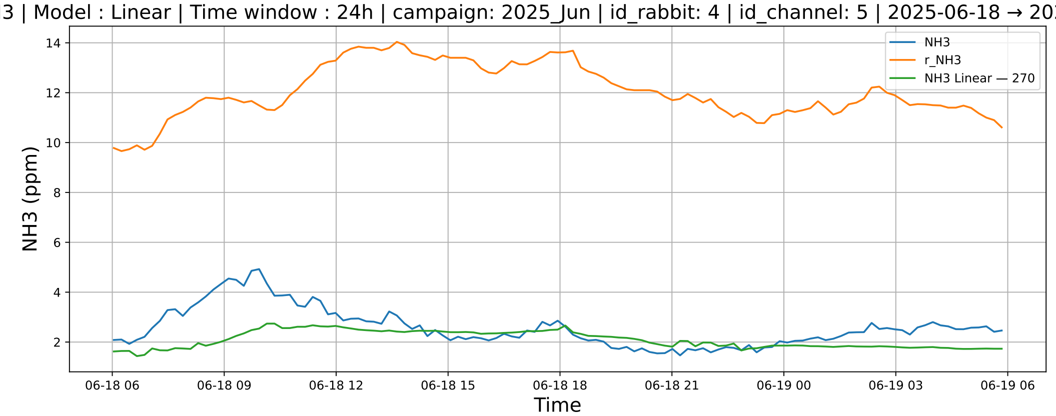
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



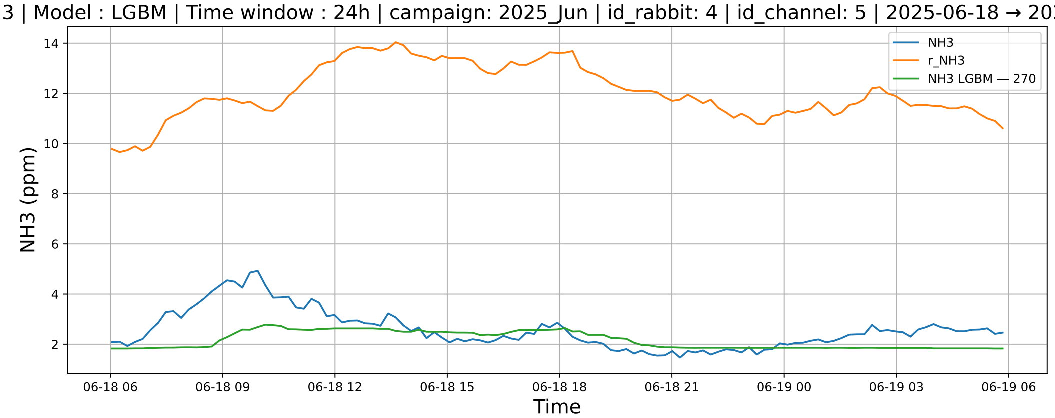
3.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

3.1.3.1 Time window : 24

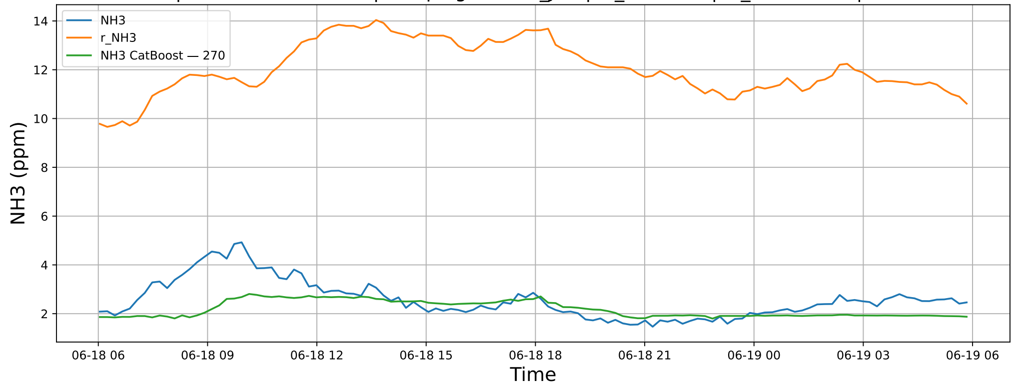
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

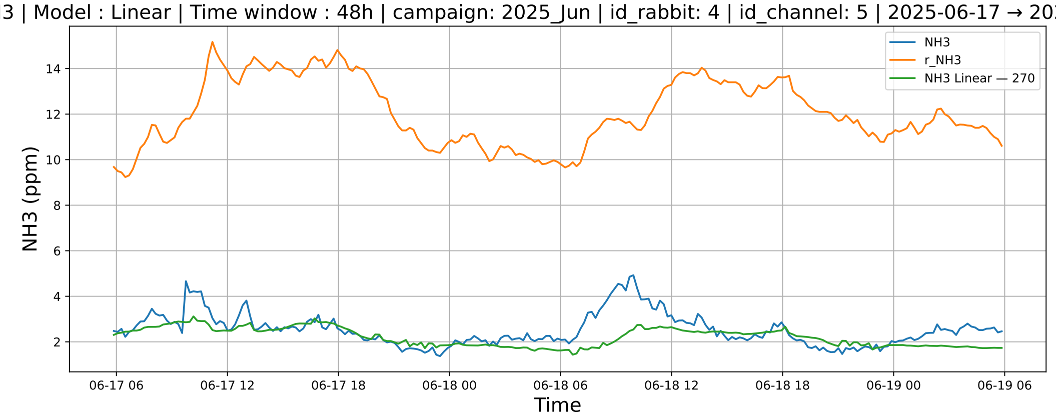


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

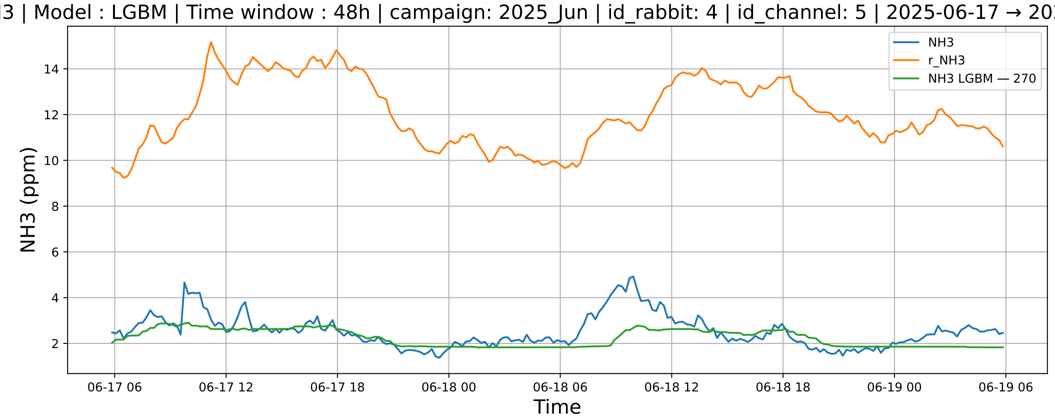


3.1.3.2 Time window : 48

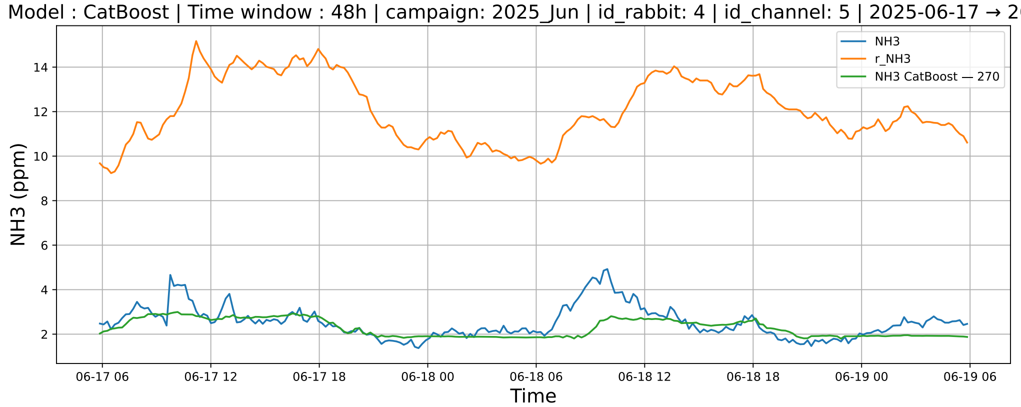
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

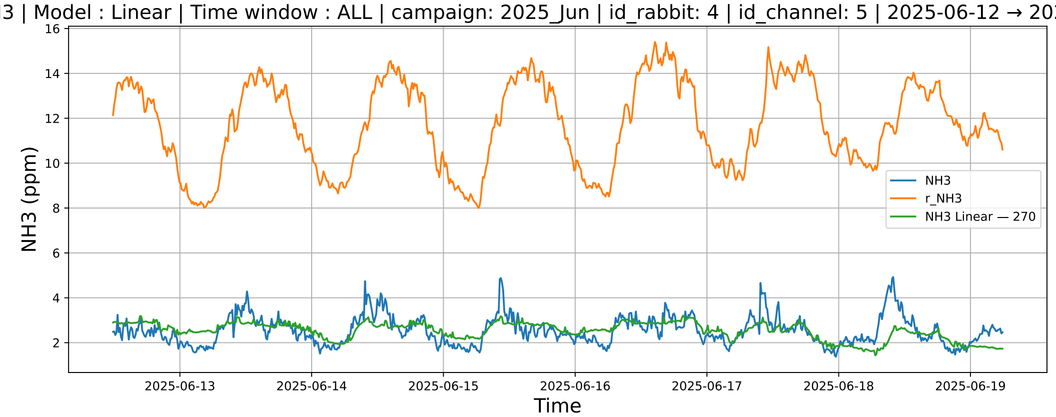


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

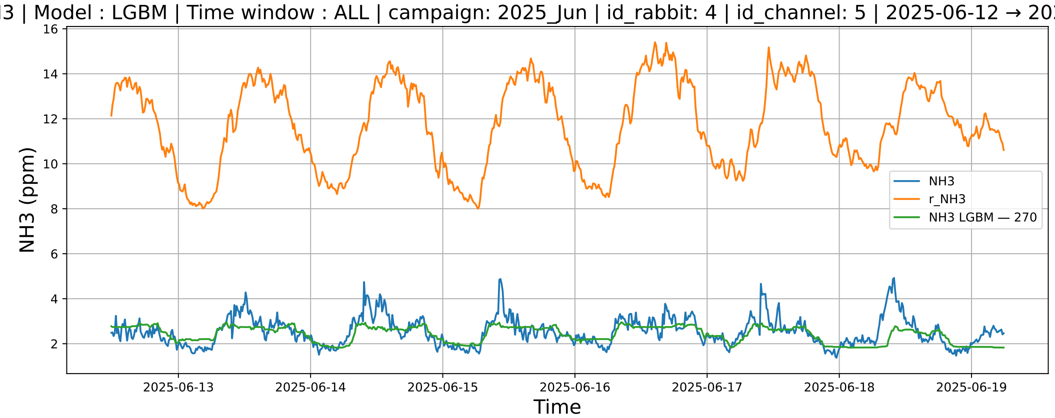


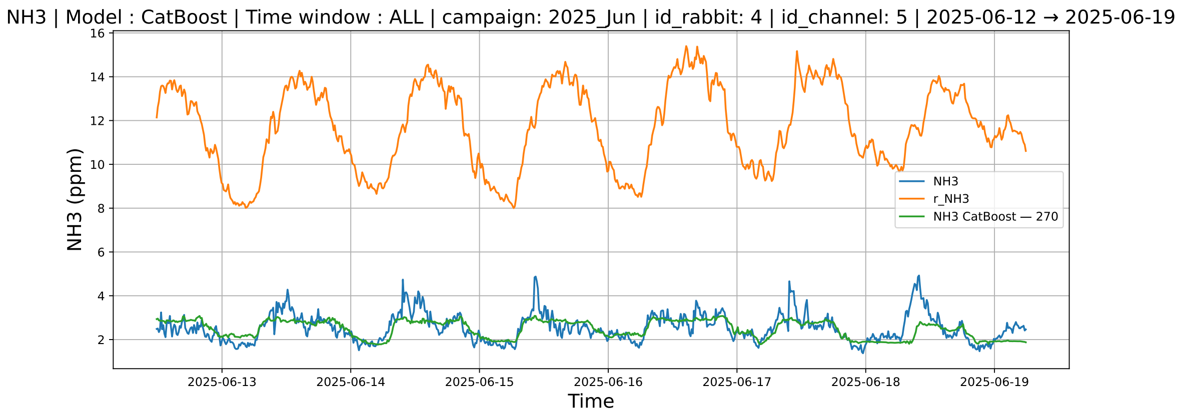
3.1.3.3 Time window : ALL

NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



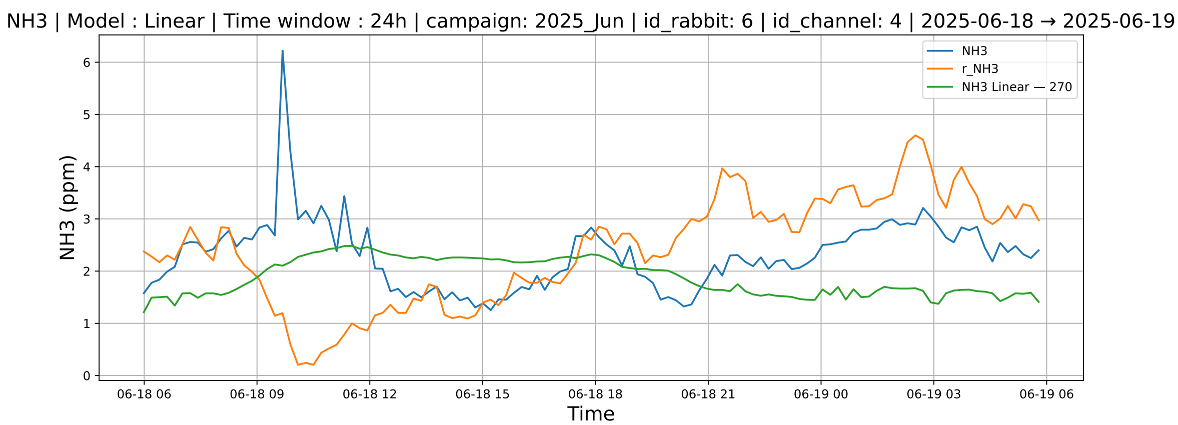
NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19

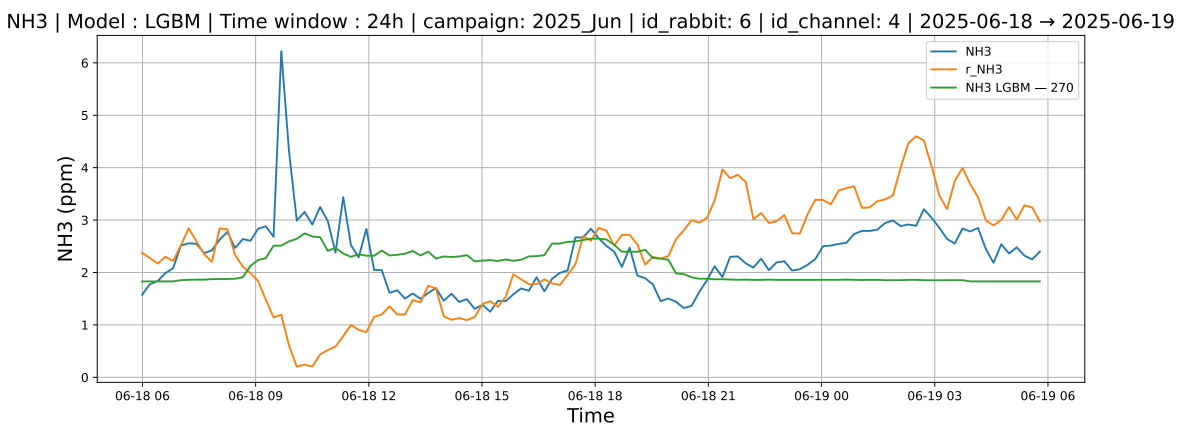




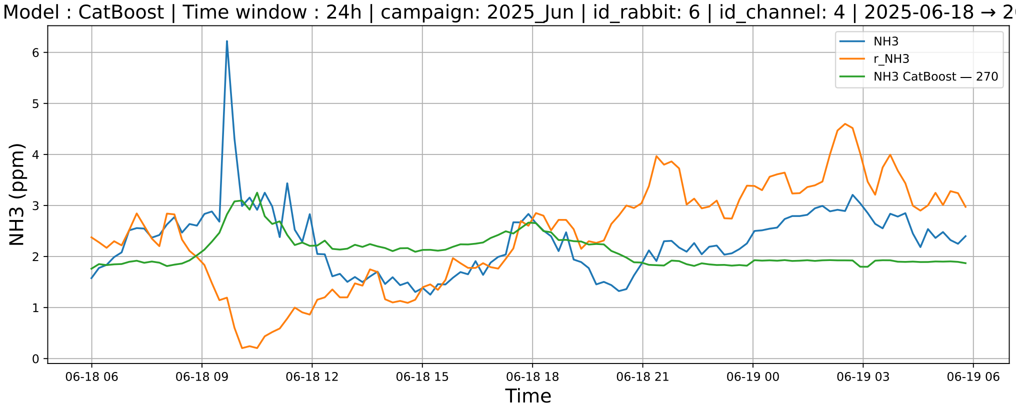
3.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

3.1.4.1 Time window : 24



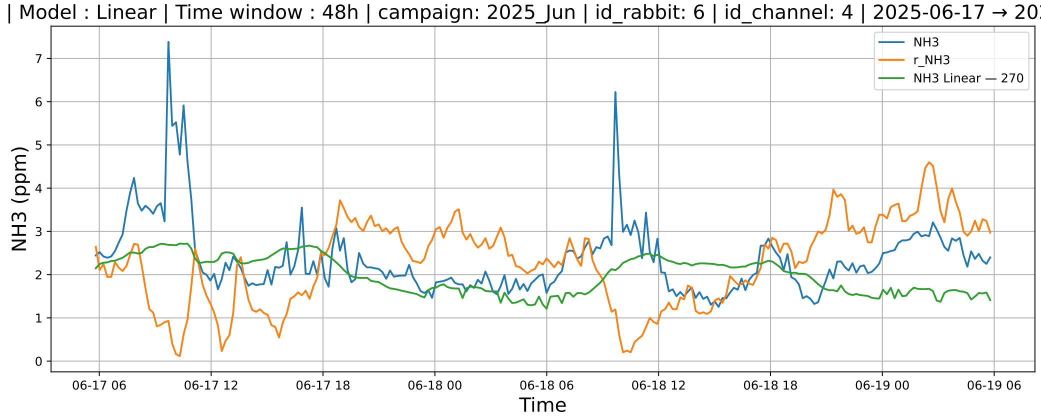


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

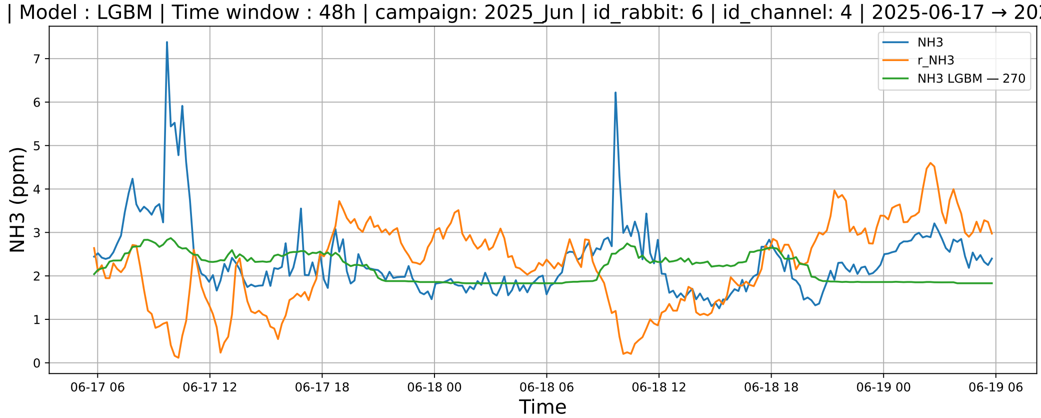


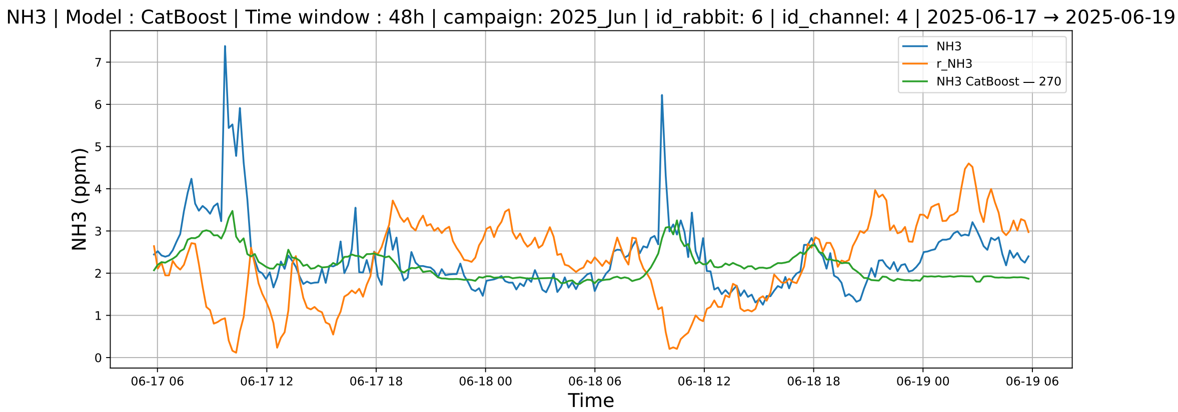
3.1.4.2 Time window : 48

NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

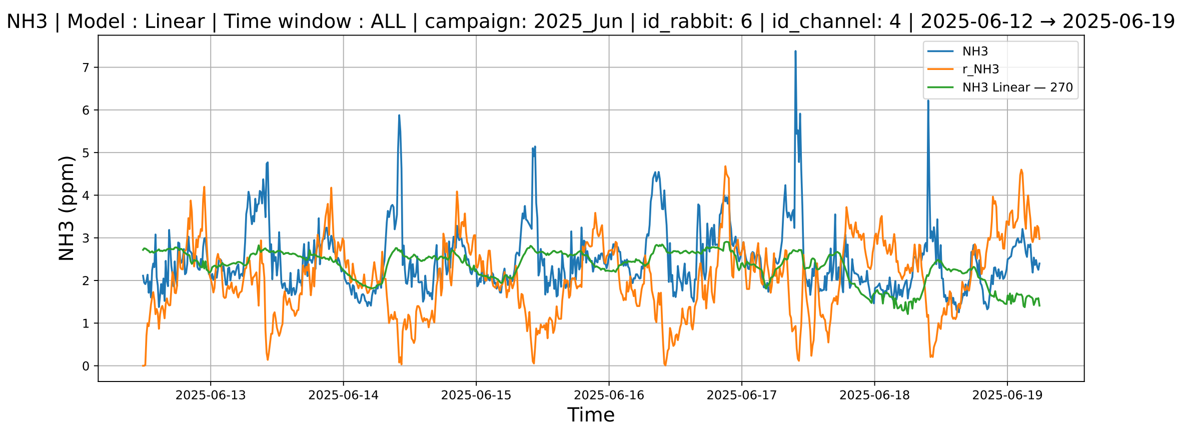


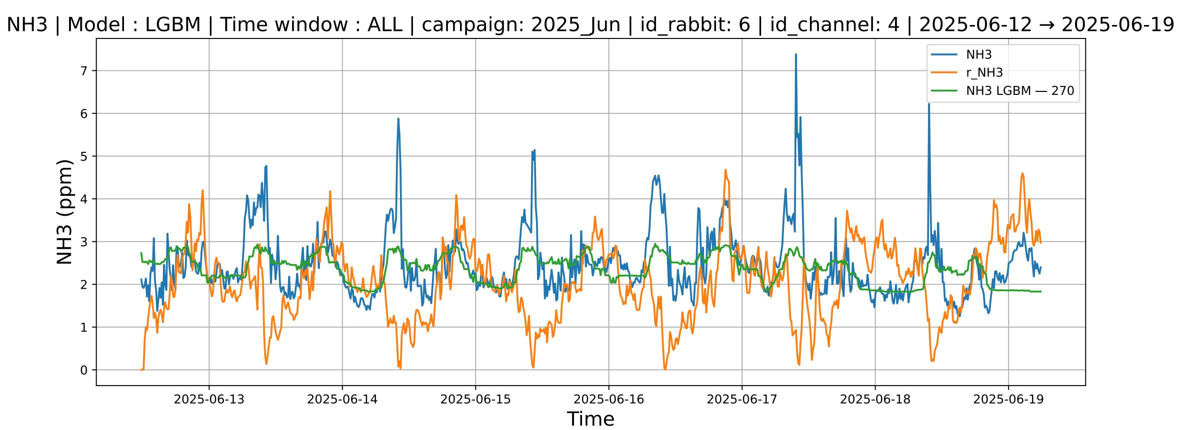
NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



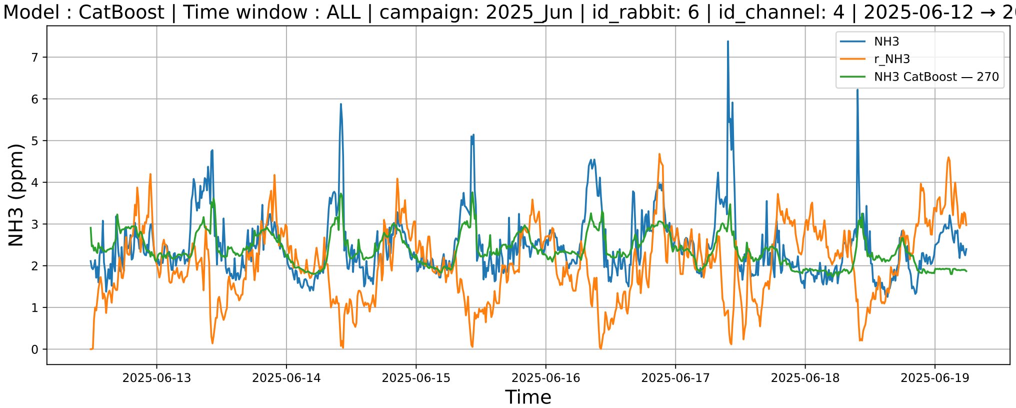


3.1.4.3 Time window : ALL





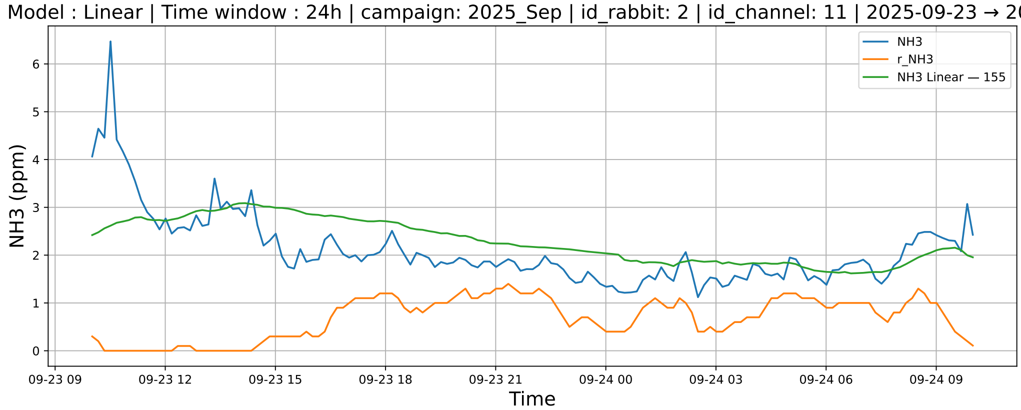
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



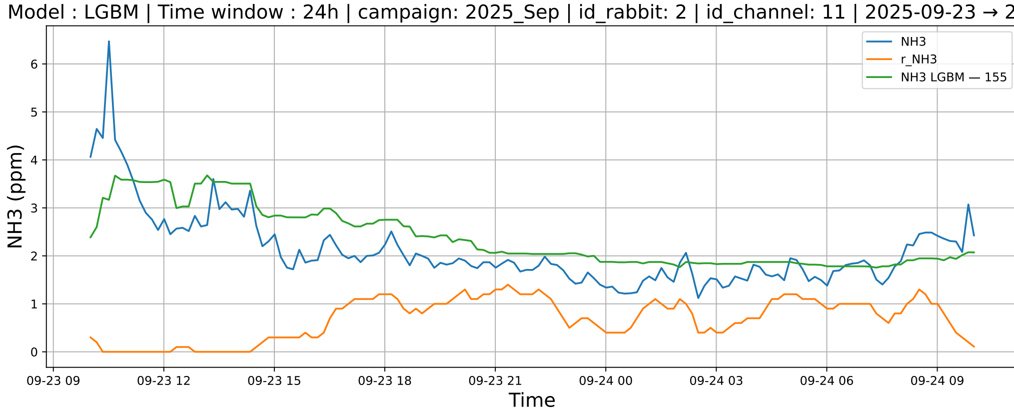
3.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

3.1.5.1 Time window : 24

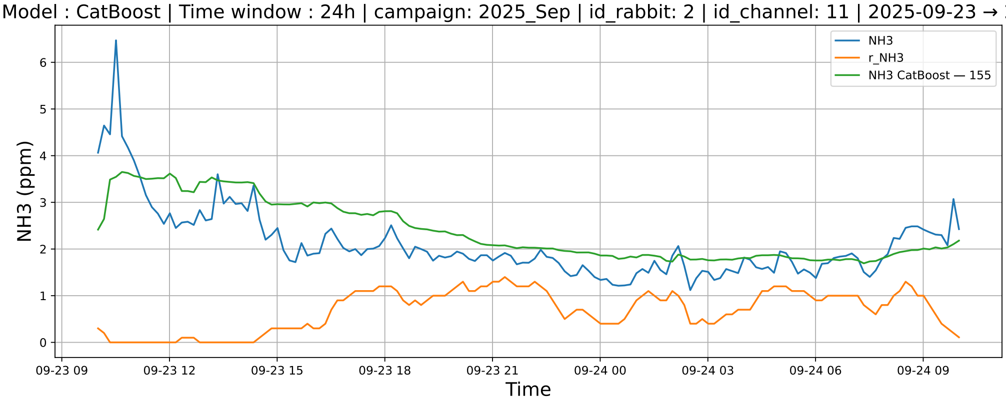
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

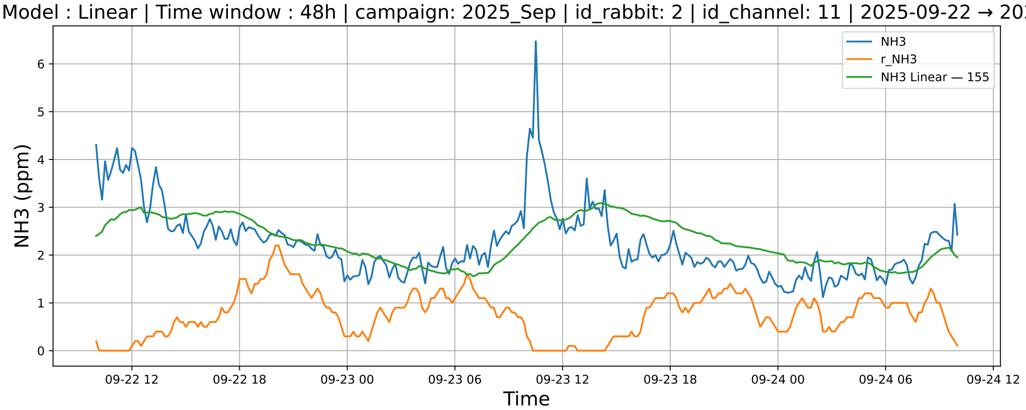


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

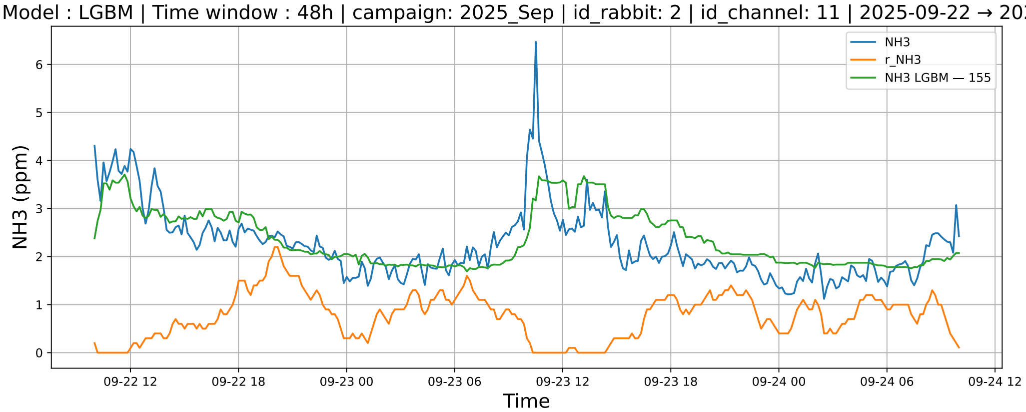


3.1.5.2 Time window : 48

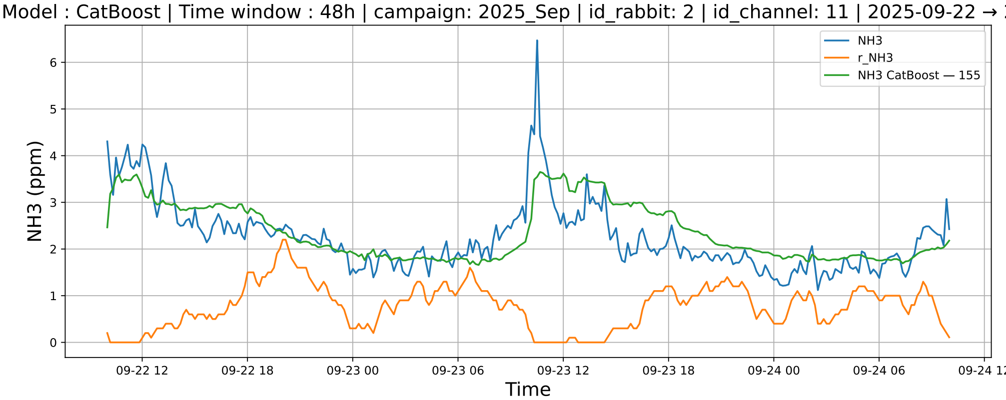
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

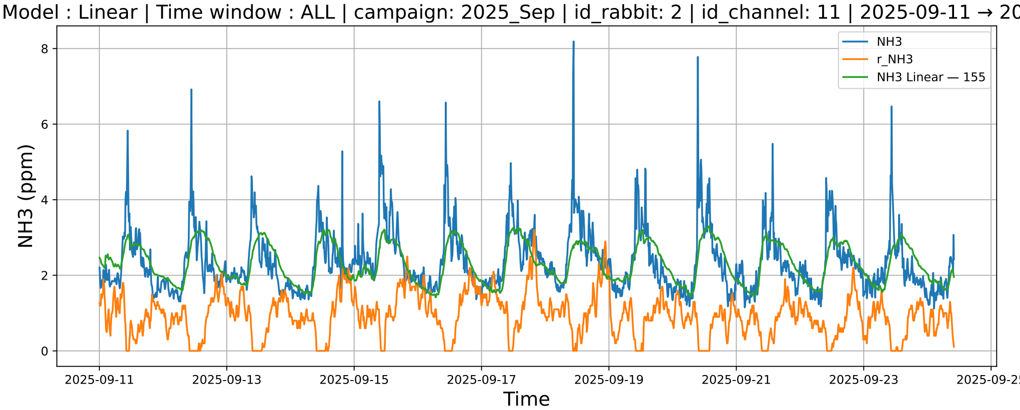


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

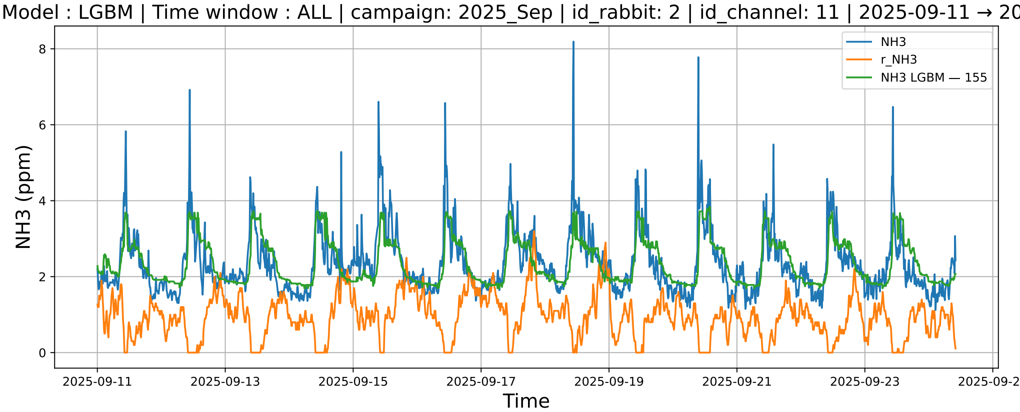


3.1.5.3 Time window : ALL

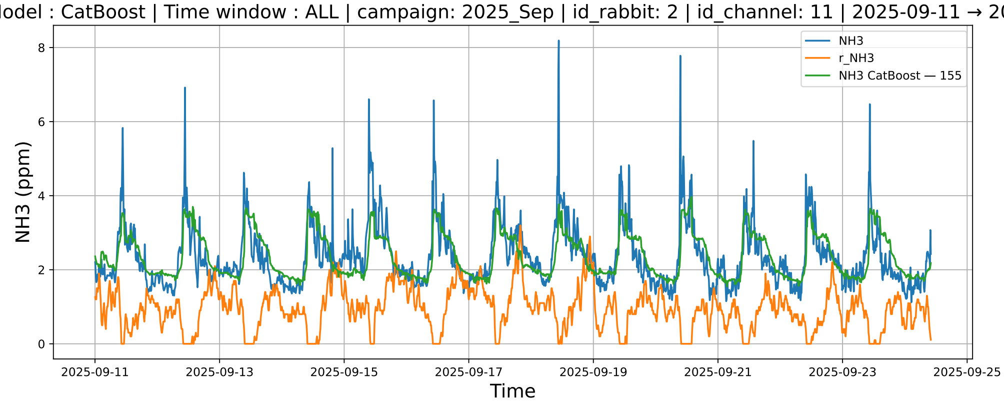
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



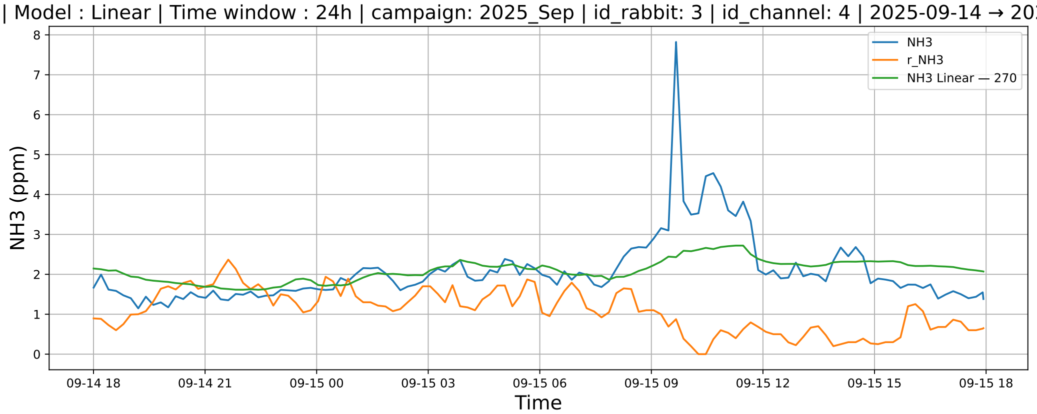
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



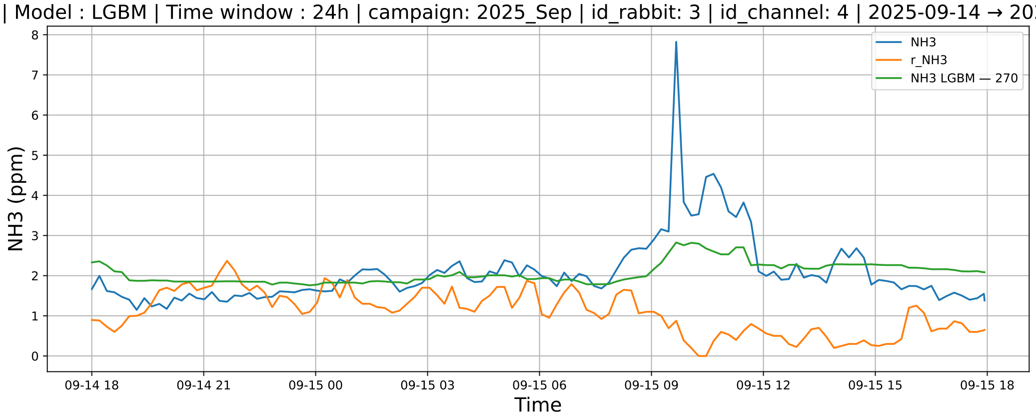
3.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

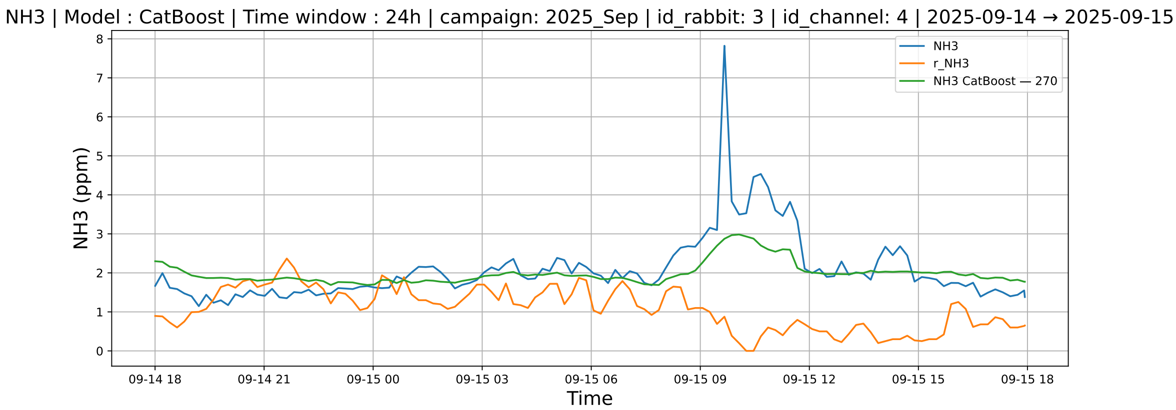
3.1.6.1 Time window : 24

NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

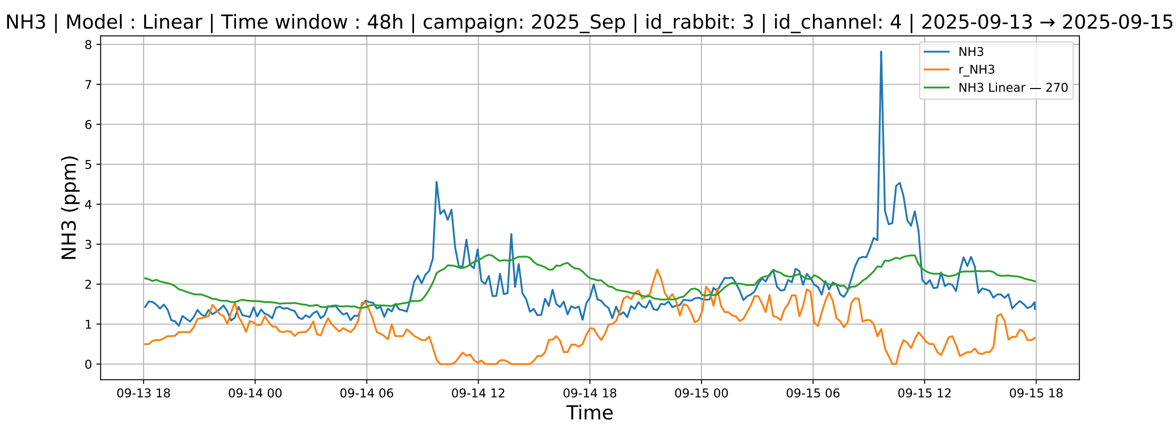


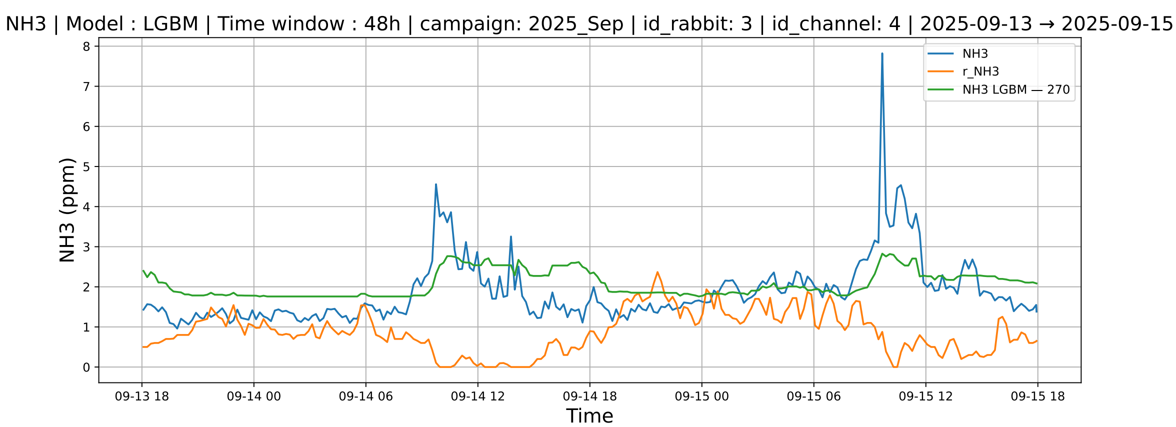
NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



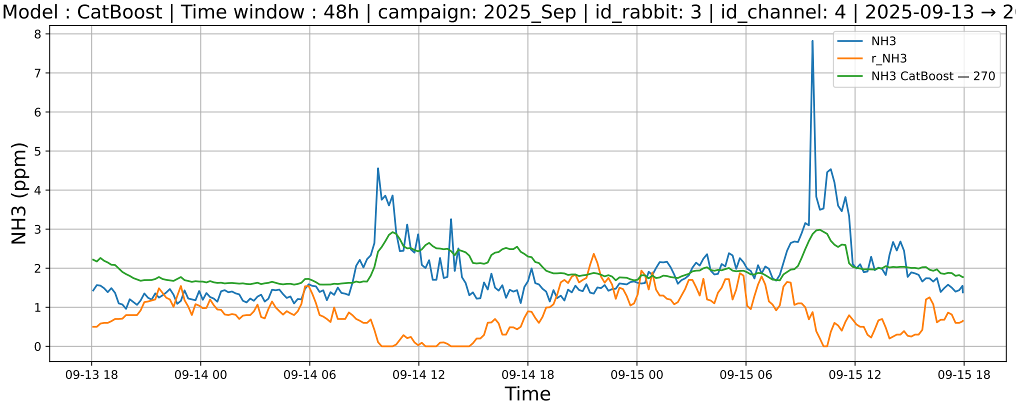


3.1.6.2 Time window : 48



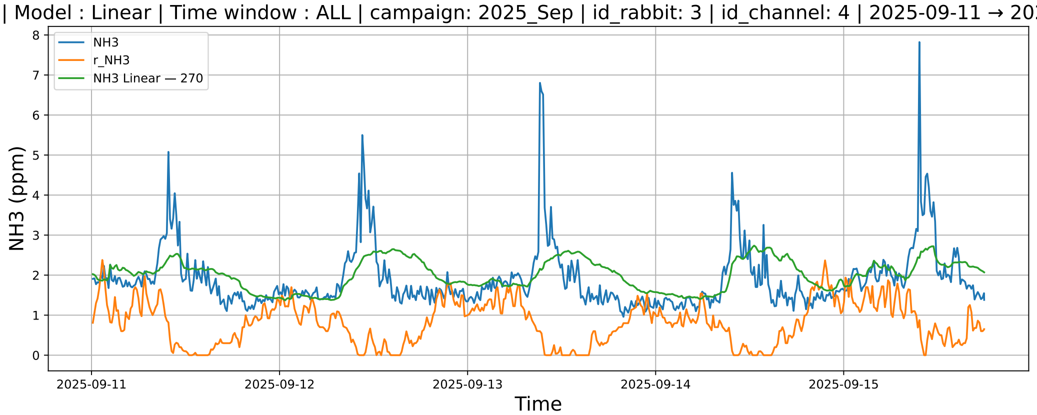


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

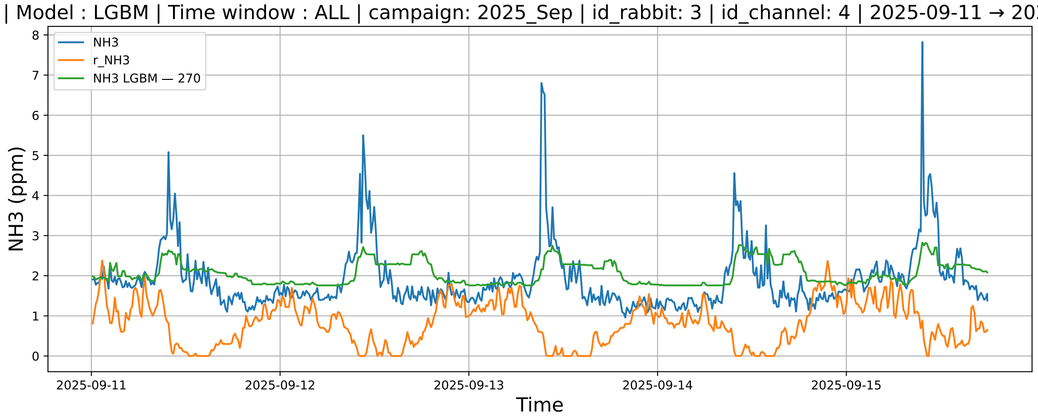


3.1.6.3 Time window : ALL

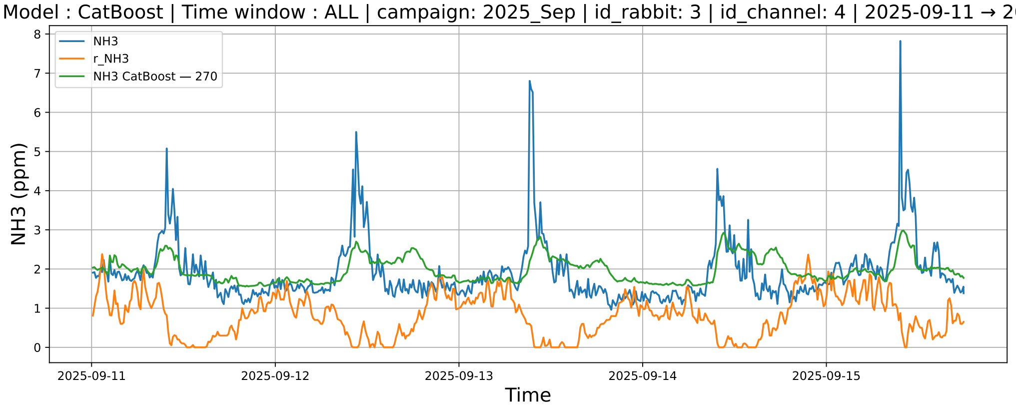
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



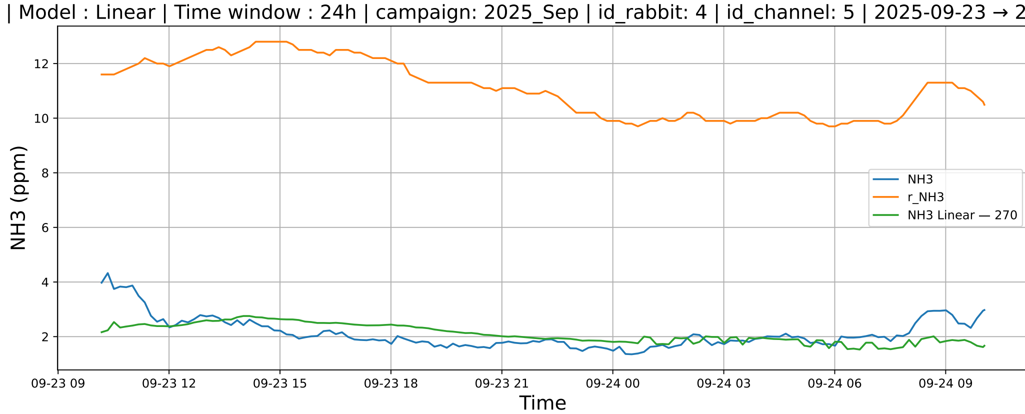
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



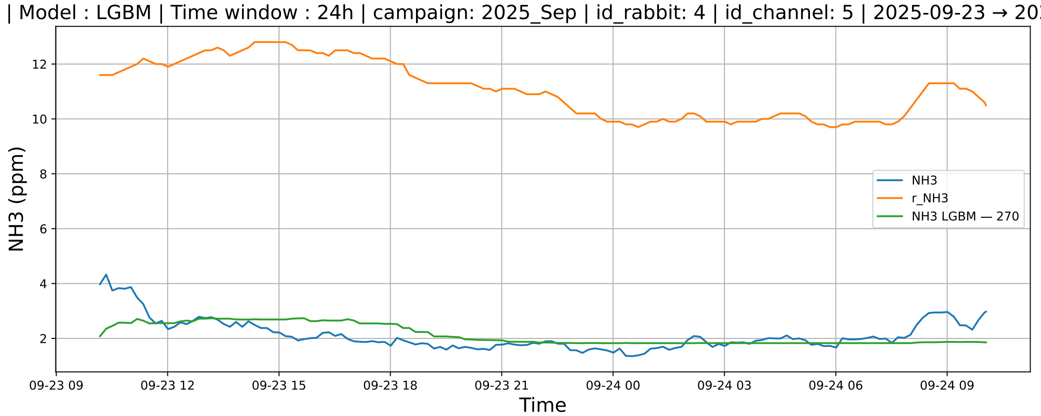
3.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

3.1.7.1 Time window : 24

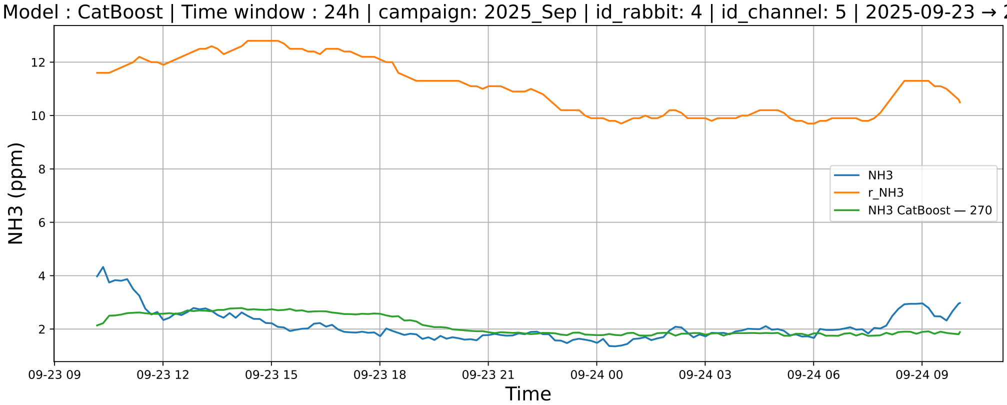
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

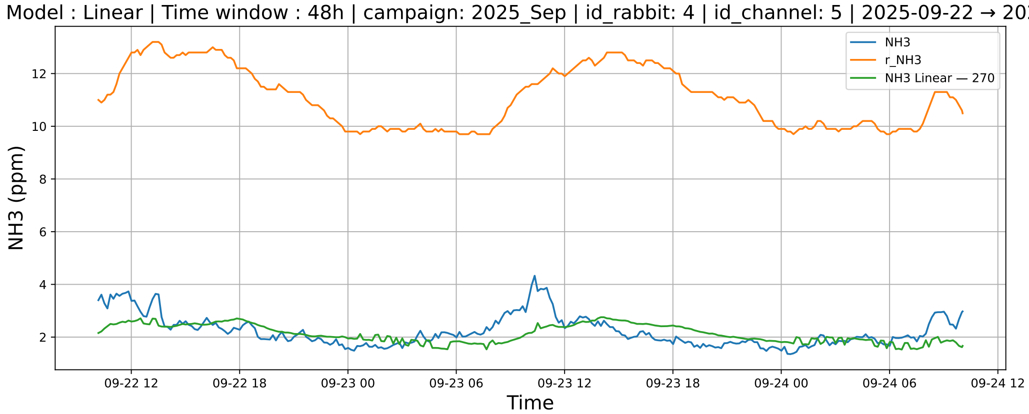


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

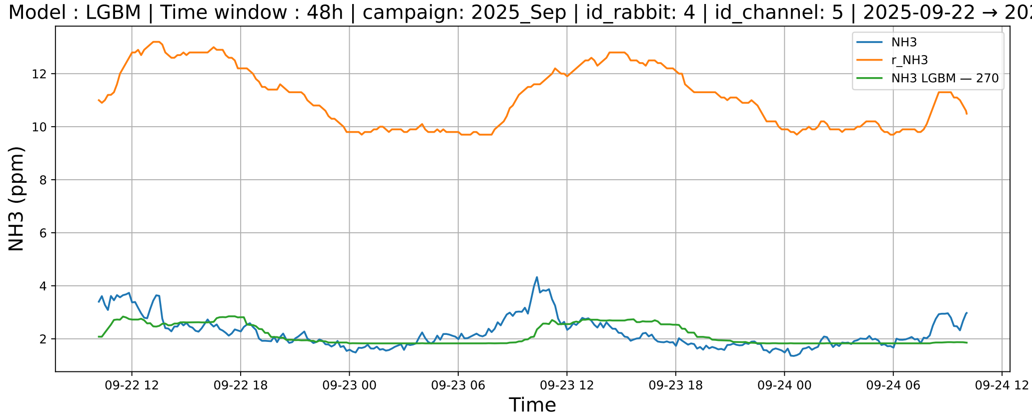


3.1.7.2 Time window : 48

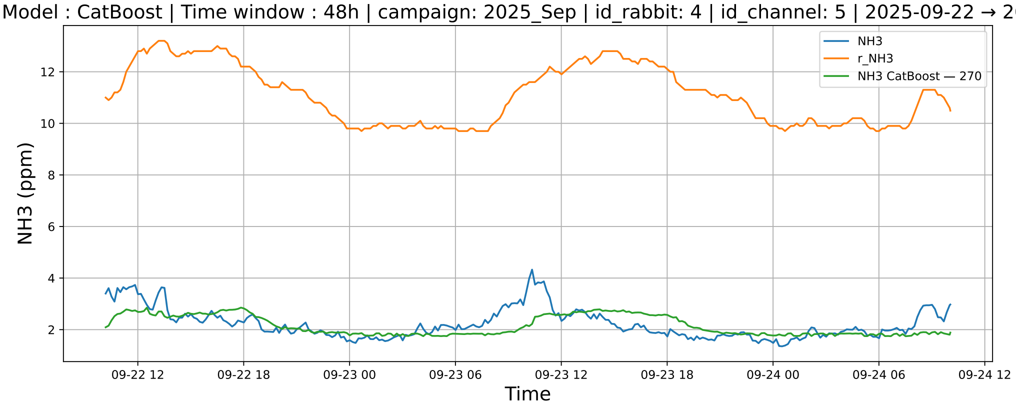
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

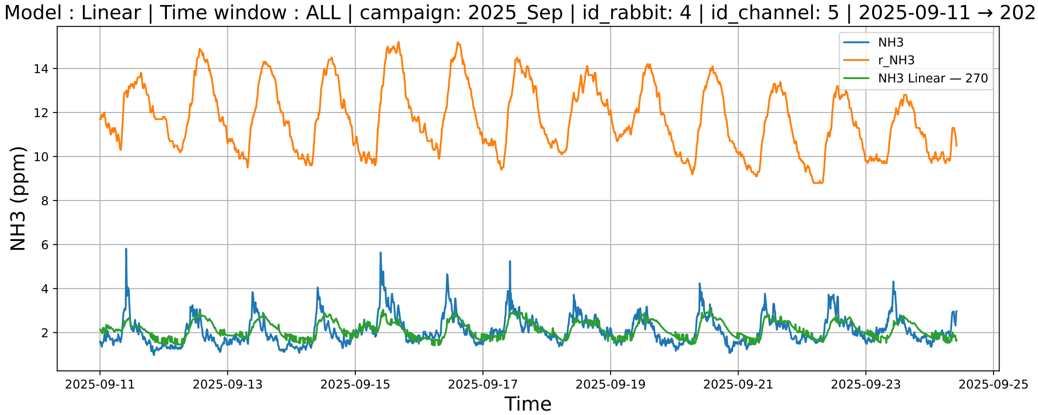


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

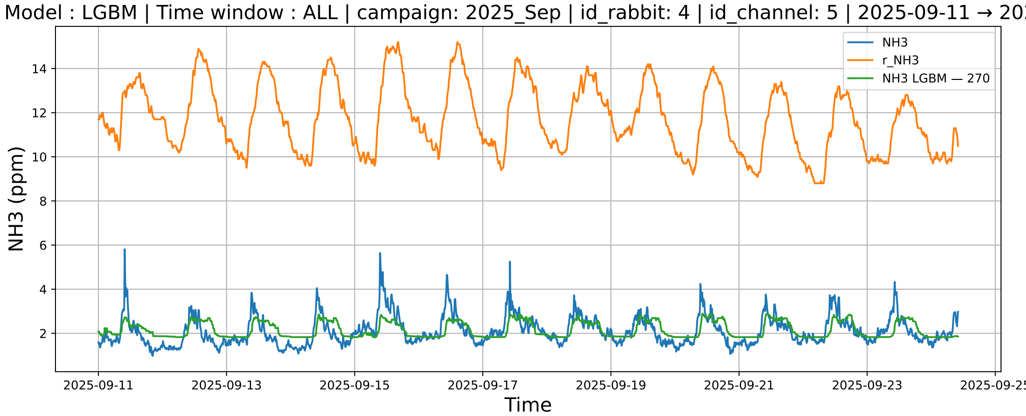


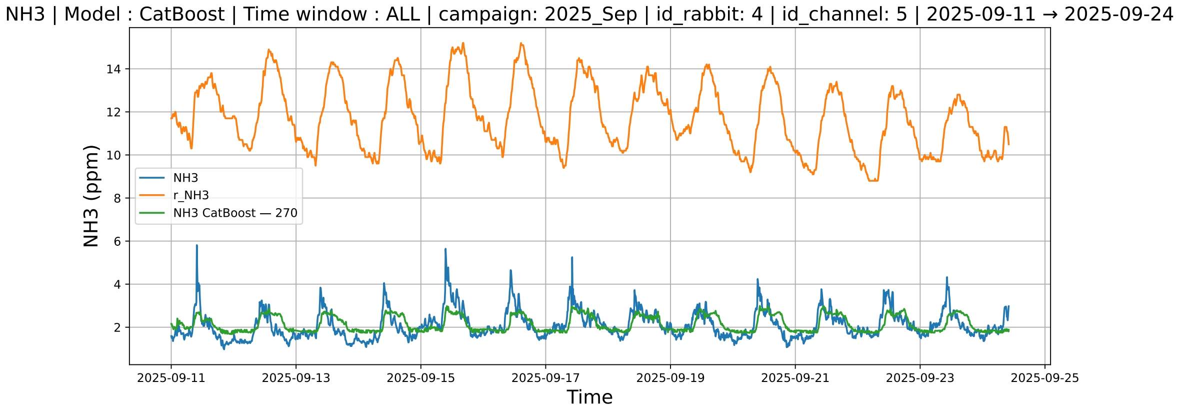
3.1.7.3 Time window : ALL

NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



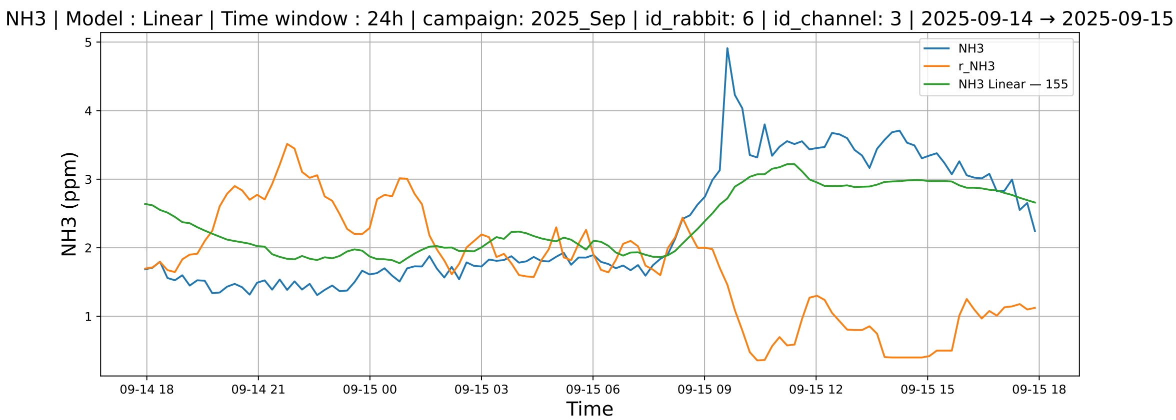
NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24

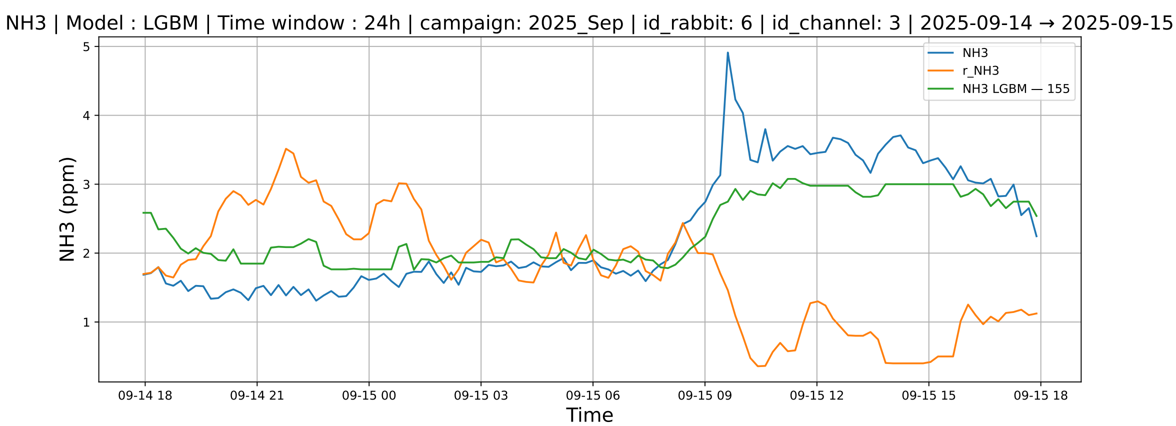




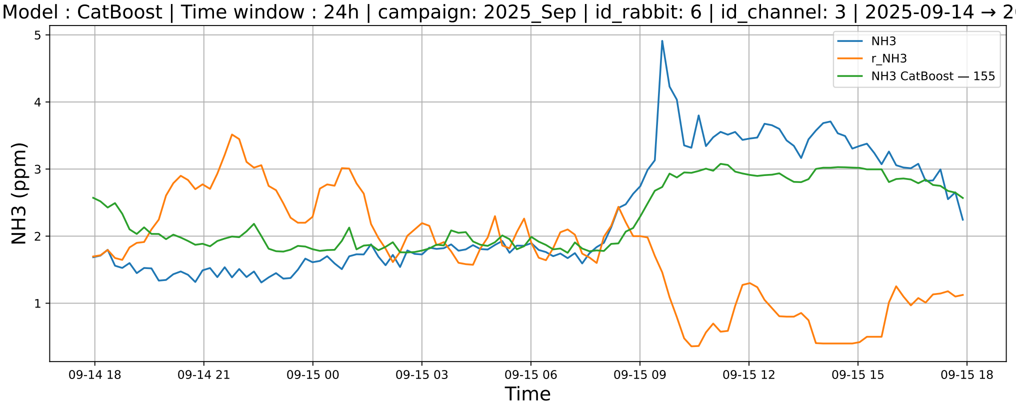
3.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

3.1.8.1 Time window : 24



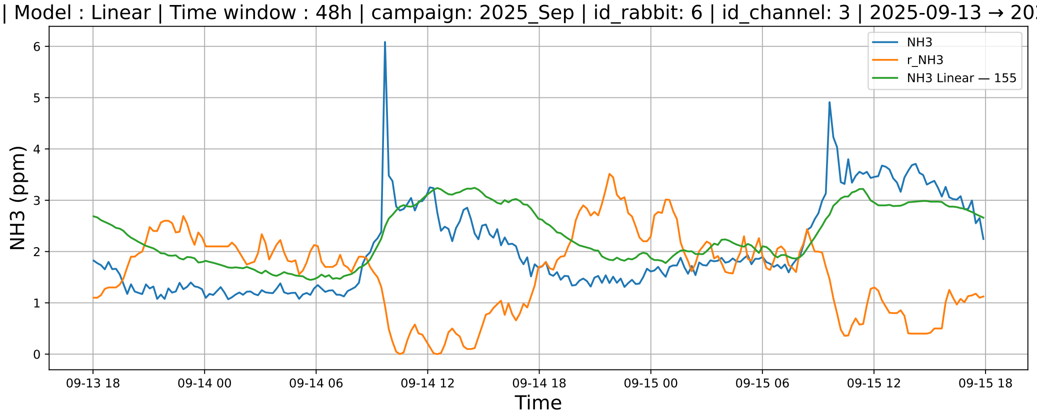


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

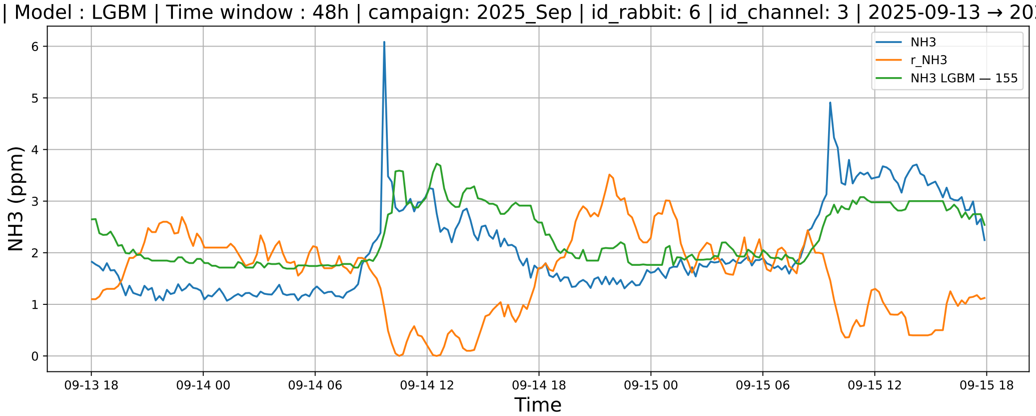


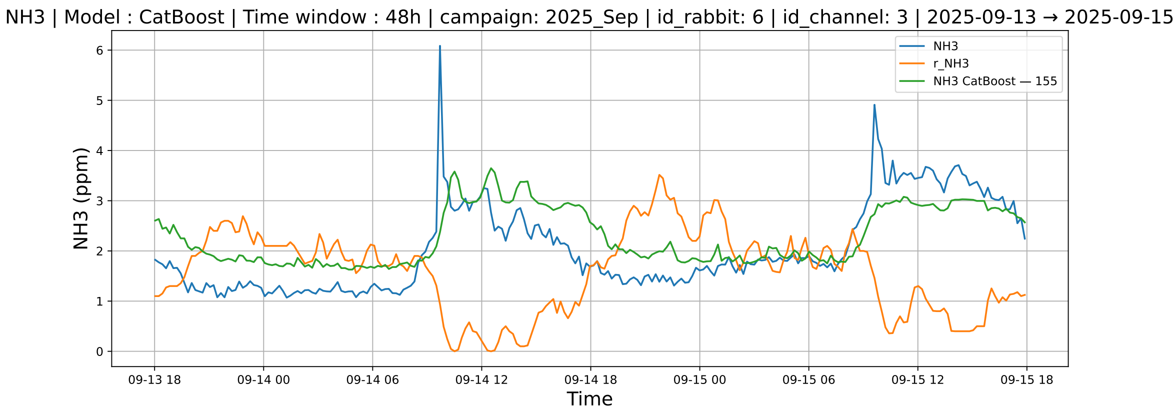
3.1.8.2 Time window : 48

NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

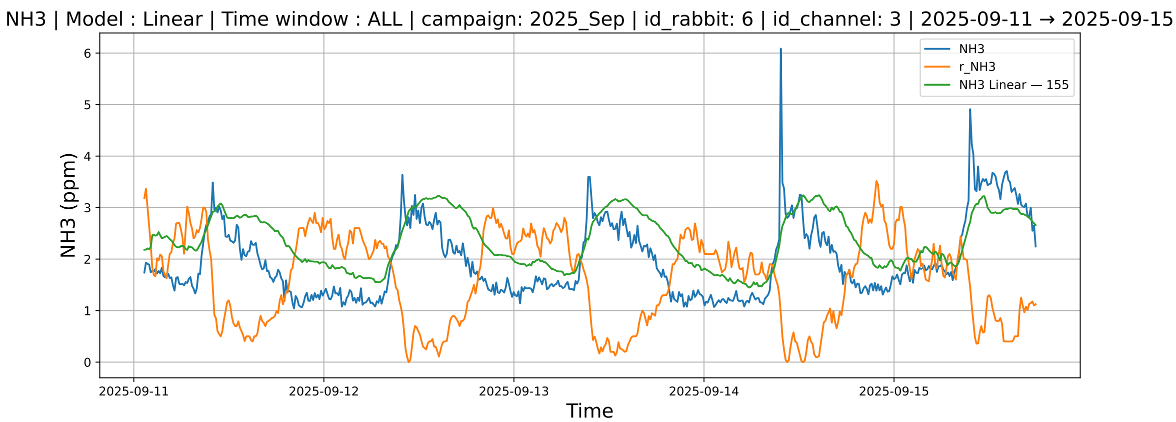


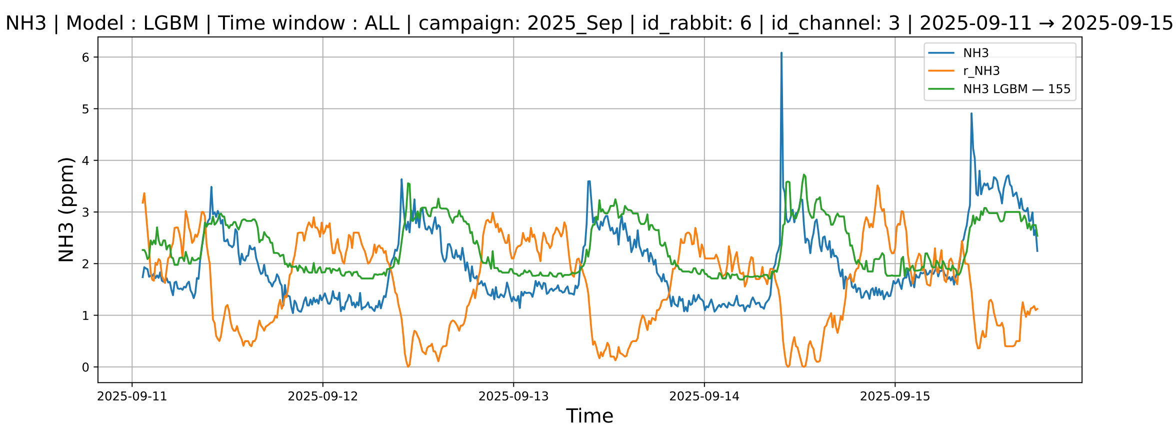
NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

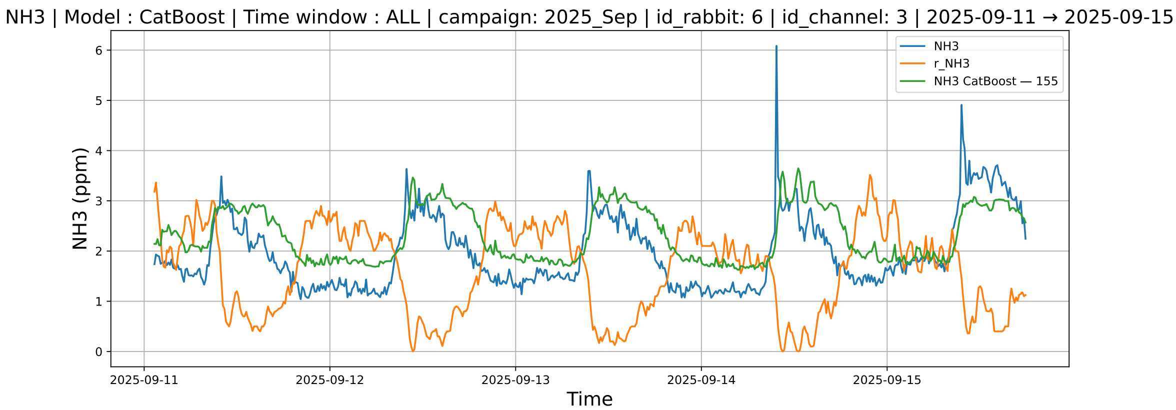




3.1.8.3 Time window : ALL



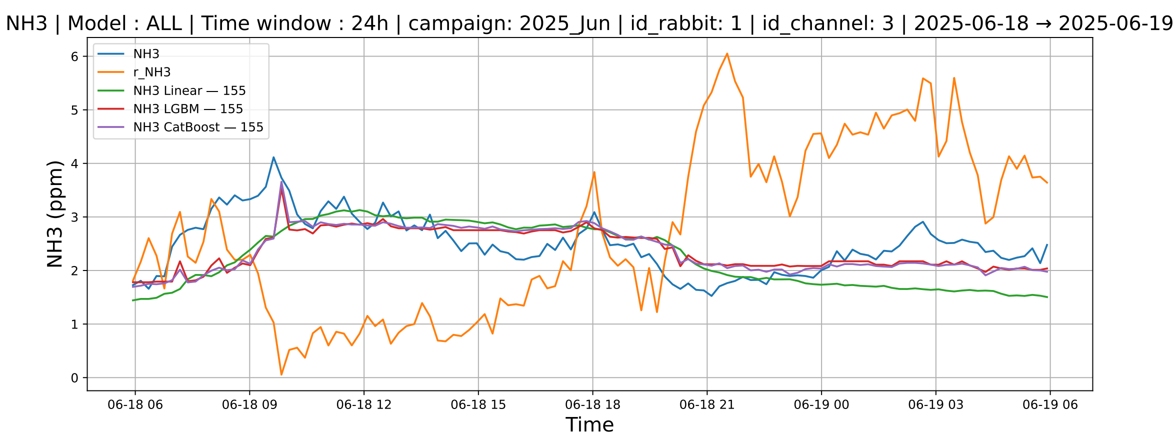




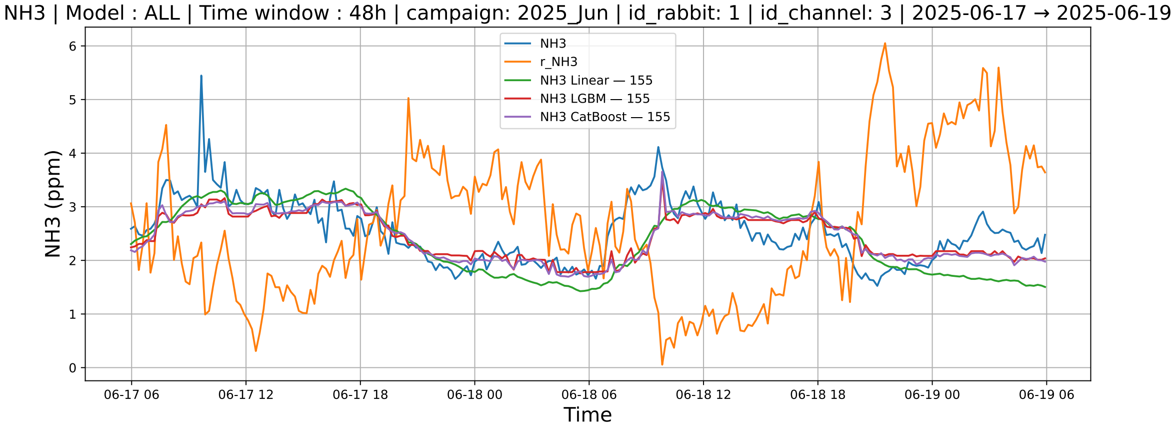
3.2 Compare plots

3.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

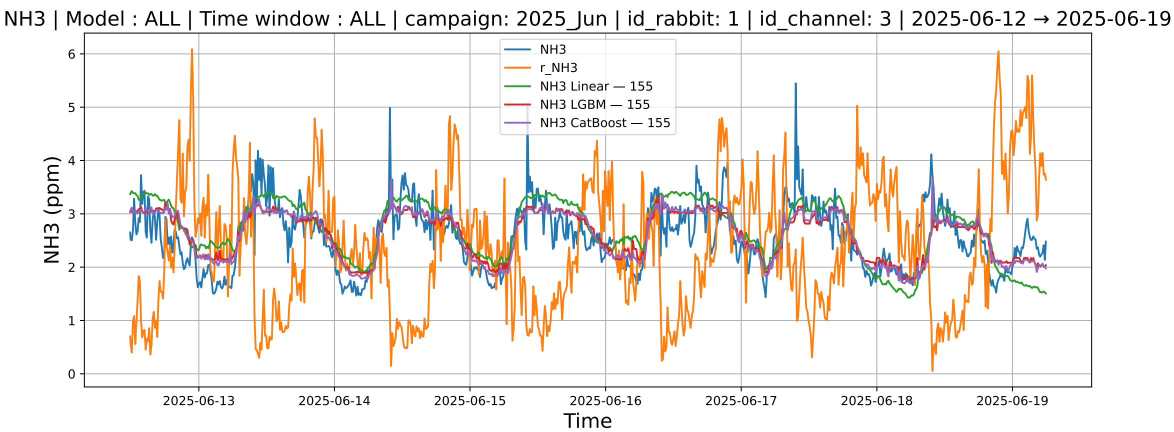
3.2.1.1 Time window : 24



3.2.1.2 Time window : 48

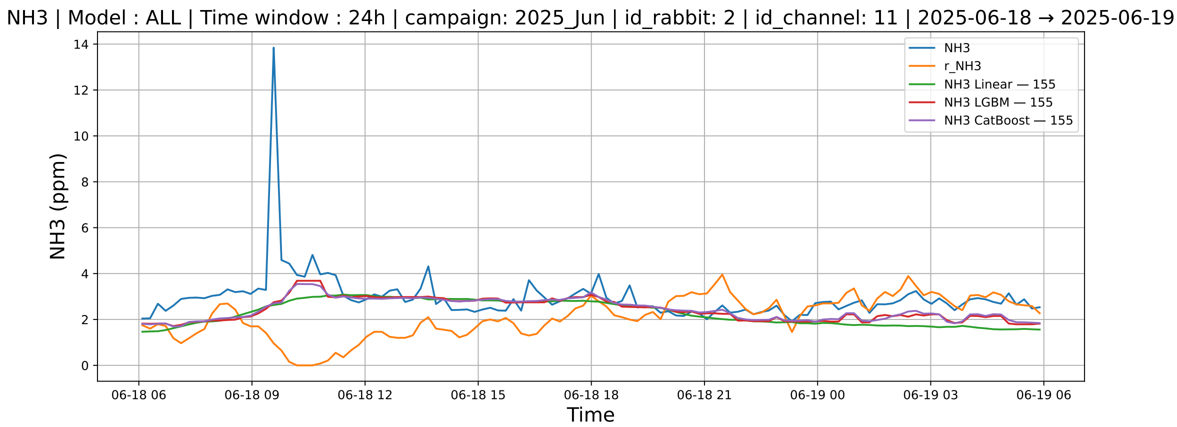


3.2.1.3 Time window : ALL

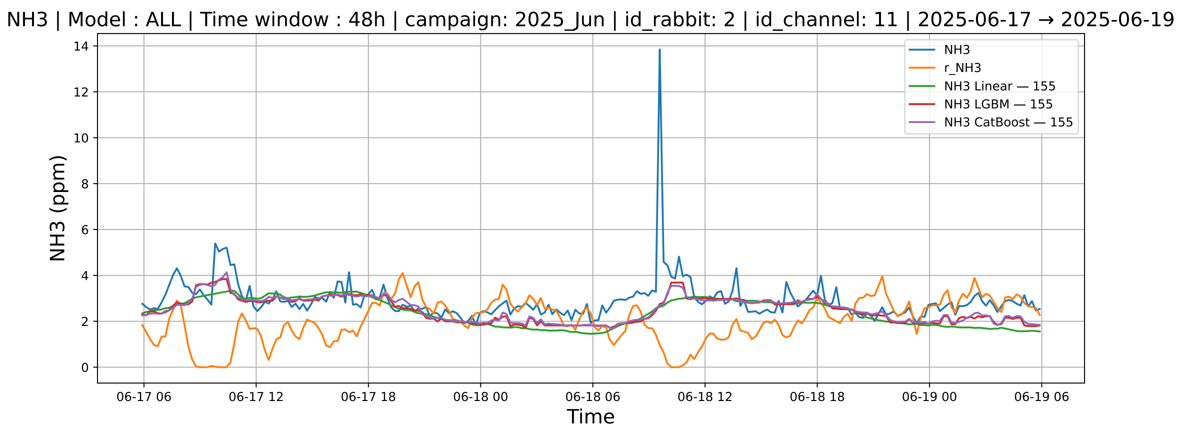


3.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

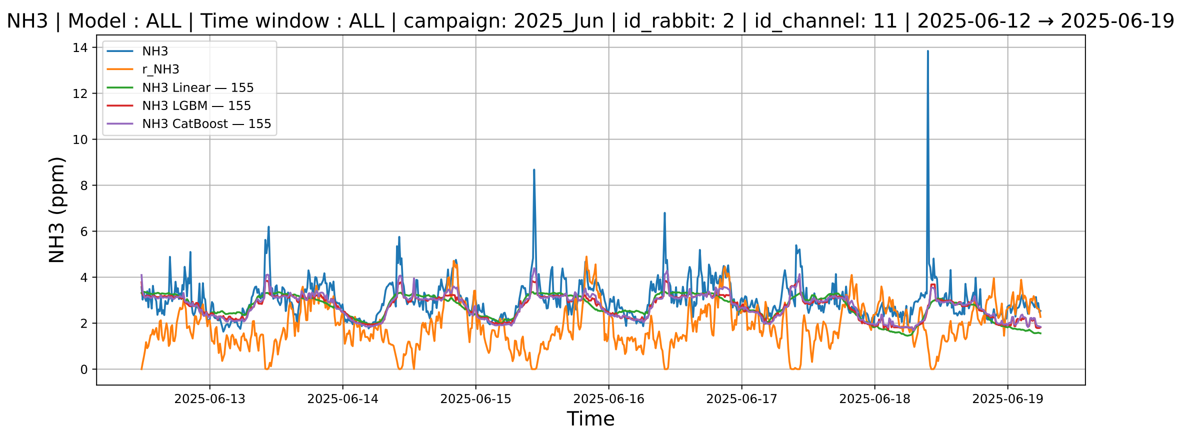
3.2.2.1 Time window : 24



3.2.2.2 Time window : 48



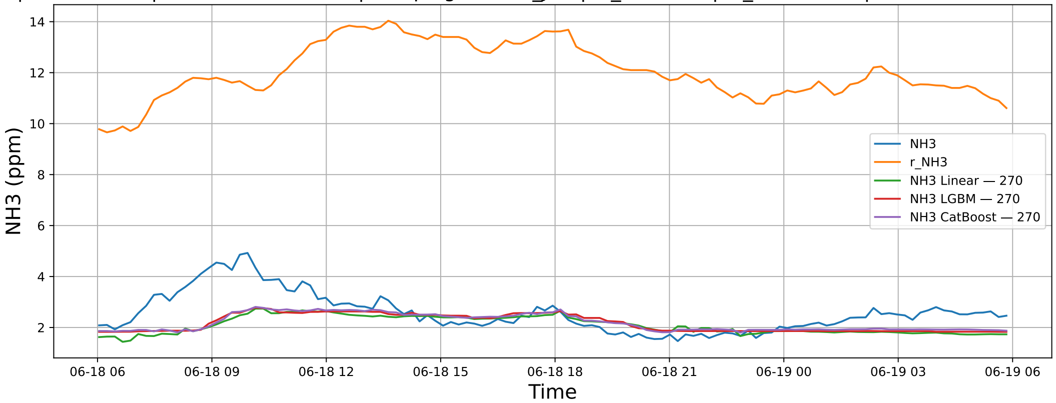
3.2.2.3 Time window : ALL



3.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

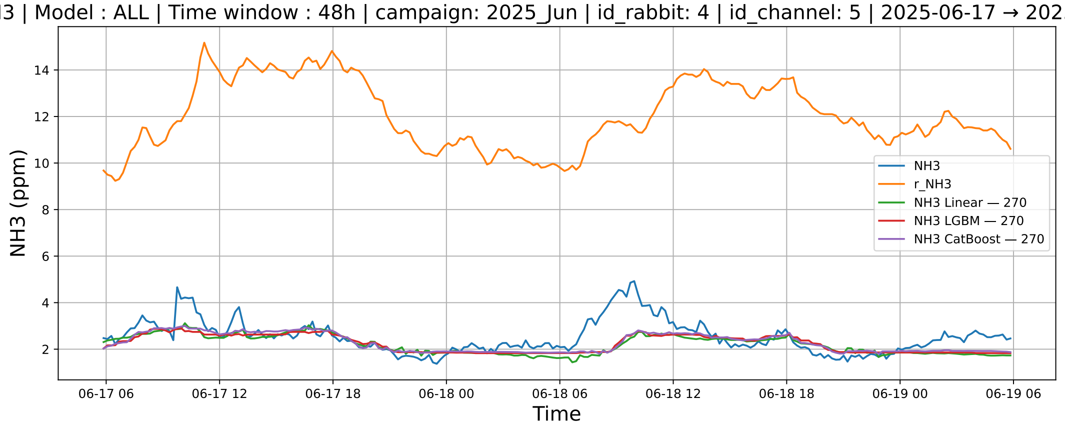
3.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



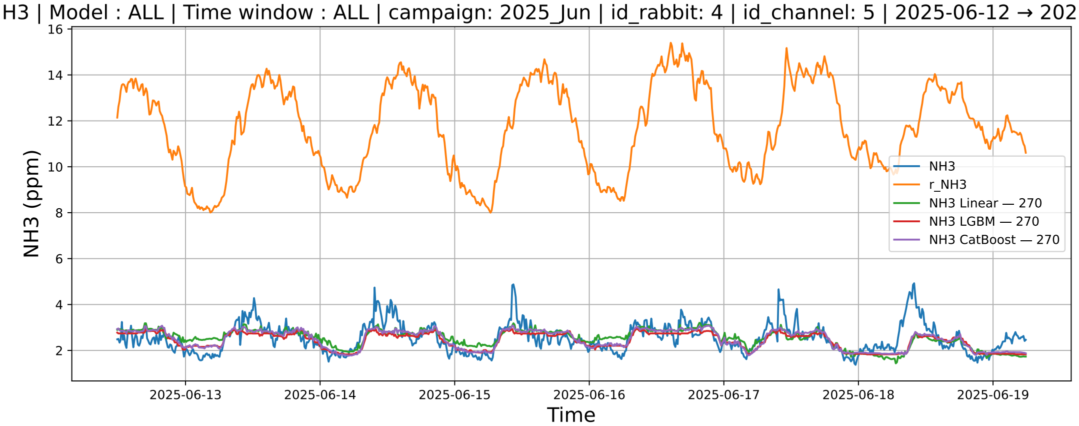
3.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



3.2.3.3 Time window : ALL

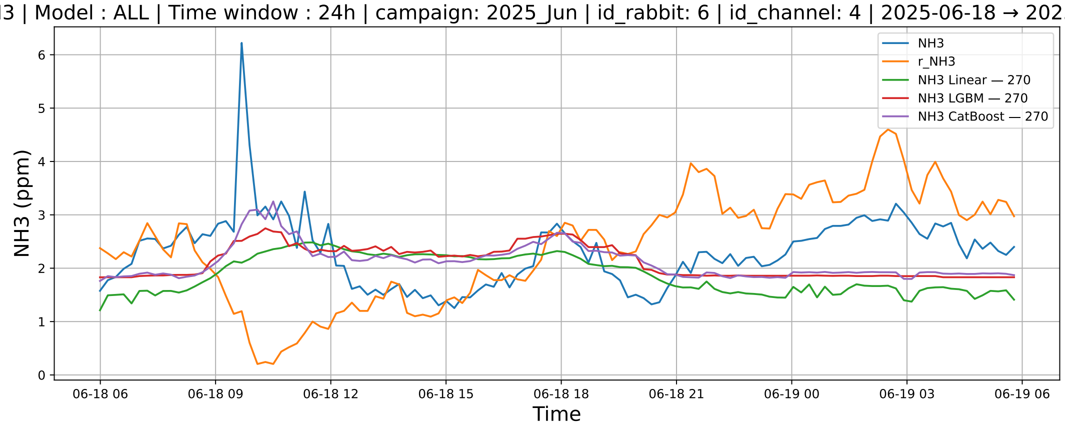
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



3.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

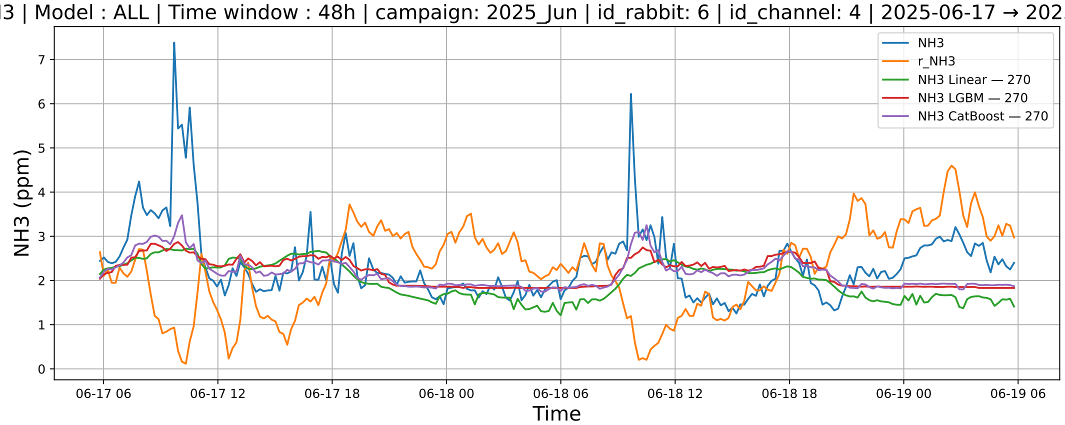
3.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



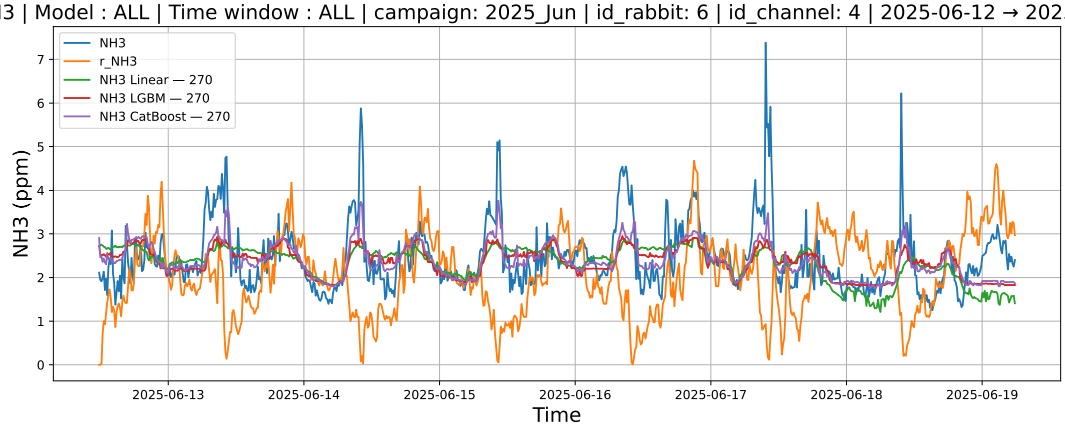
3.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



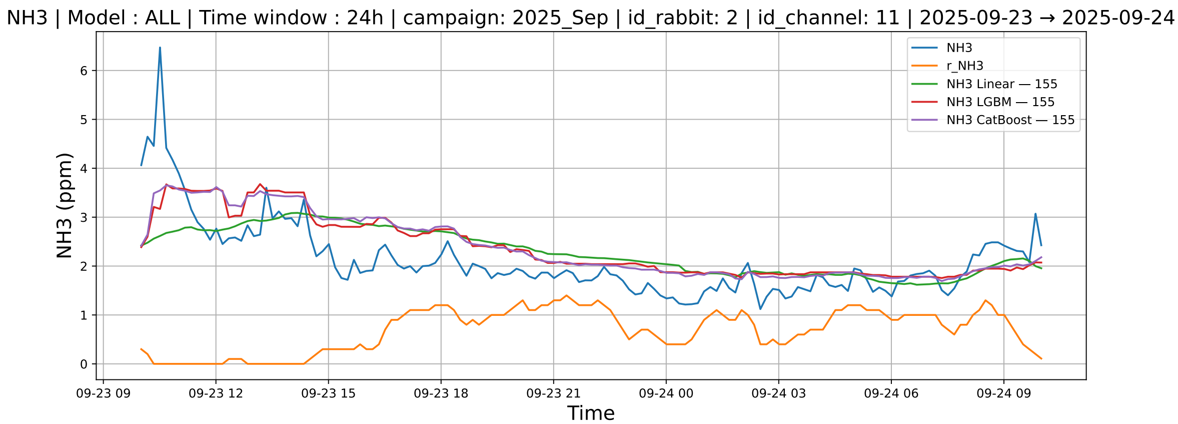
3.2.4.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19

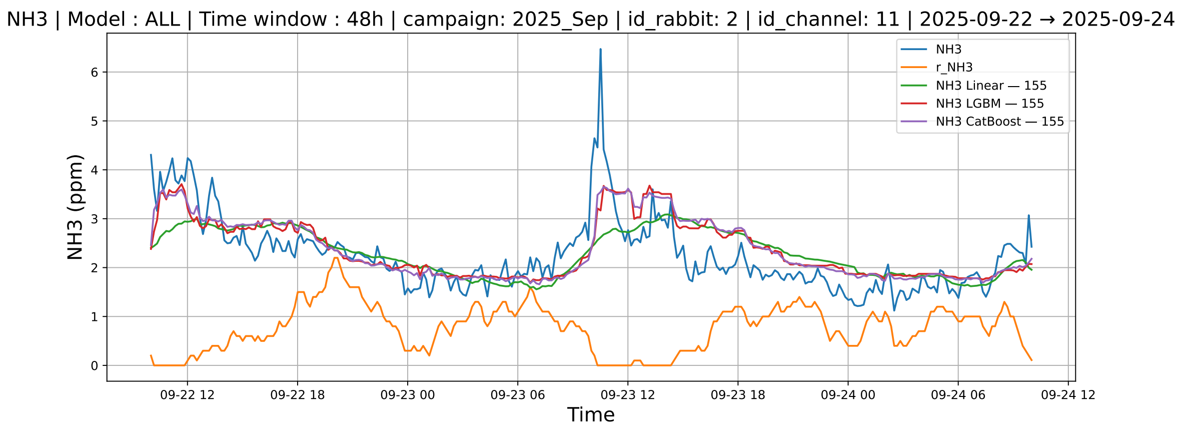


3.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

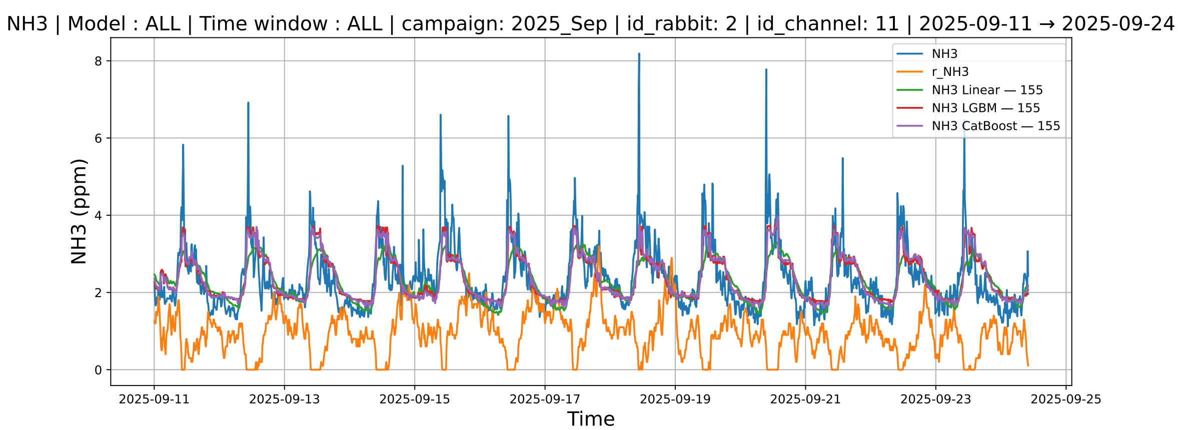
3.2.5.1 Time window : 24



3.2.5.2 Time window : 48



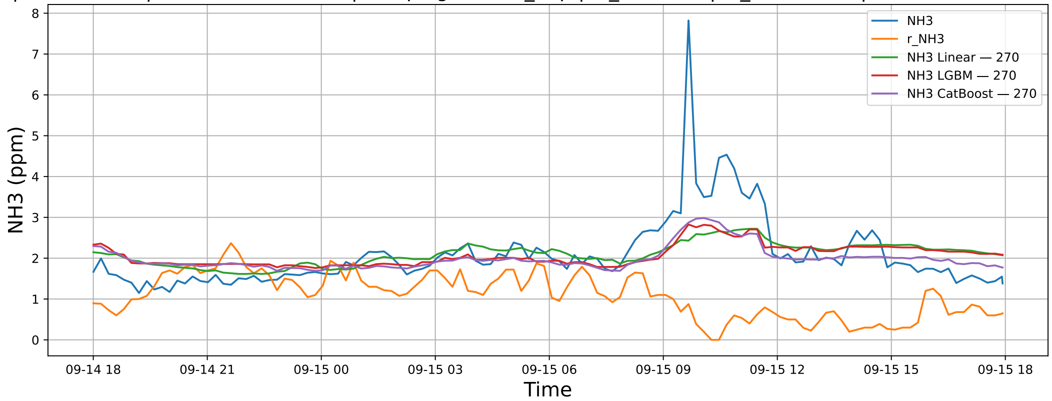
3.2.5.3 Time window : ALL



3.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

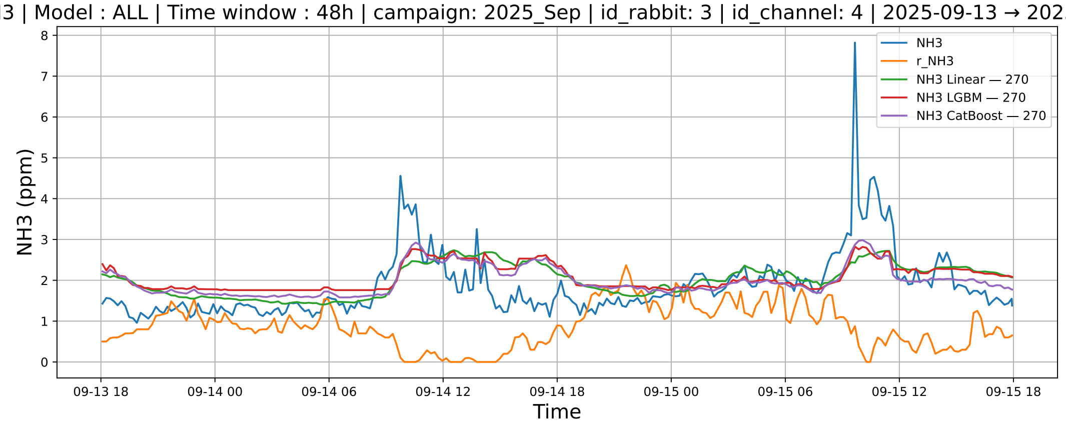
3.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

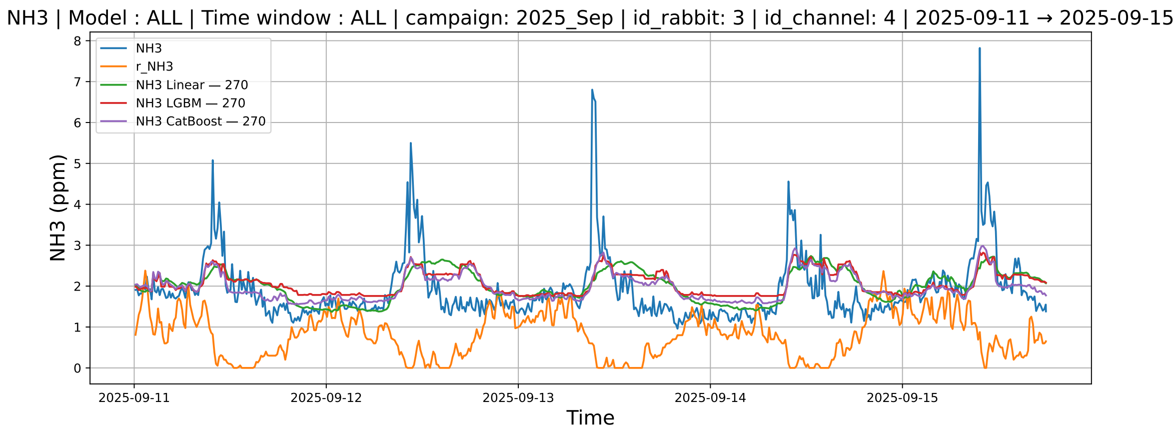


3.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

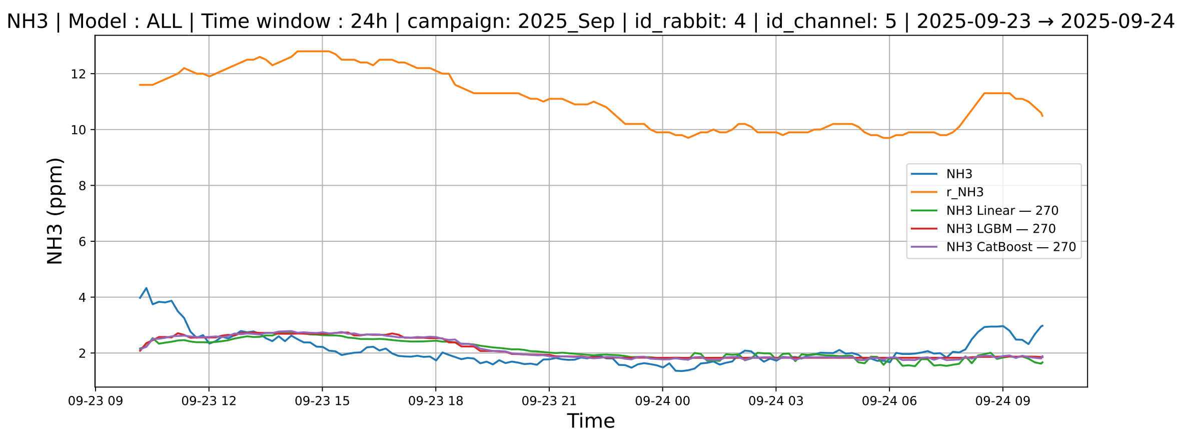


3.2.6.3 Time window : ALL

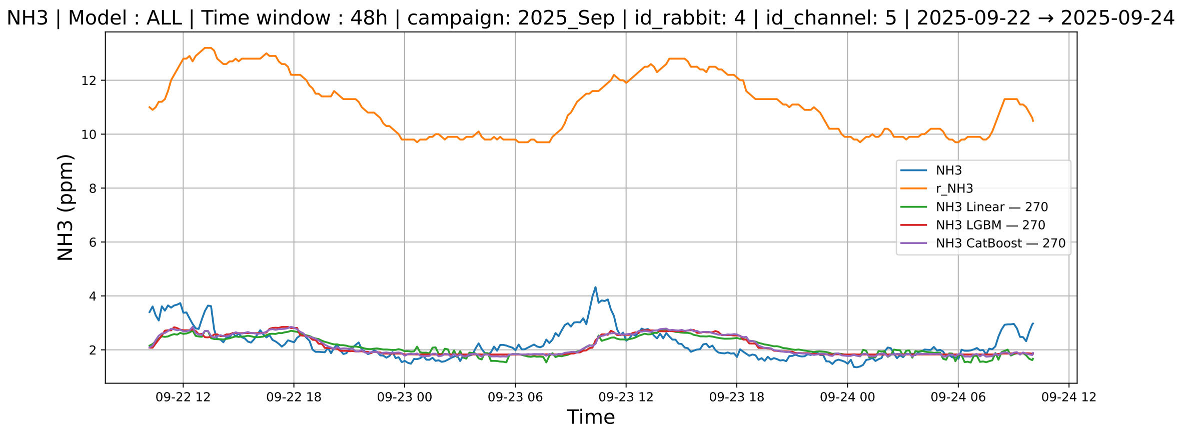


3.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

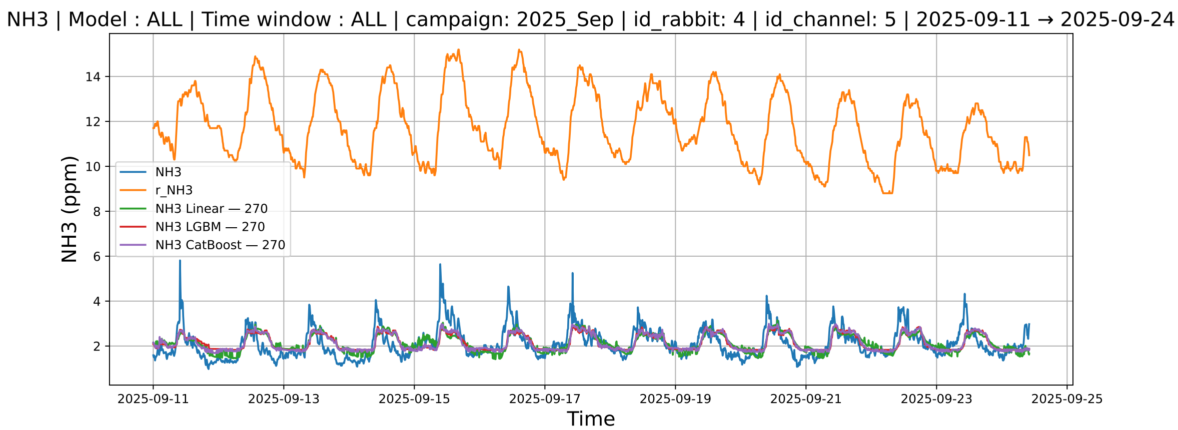
3.2.7.1 Time window : 24



3.2.7.2 Time window : 48



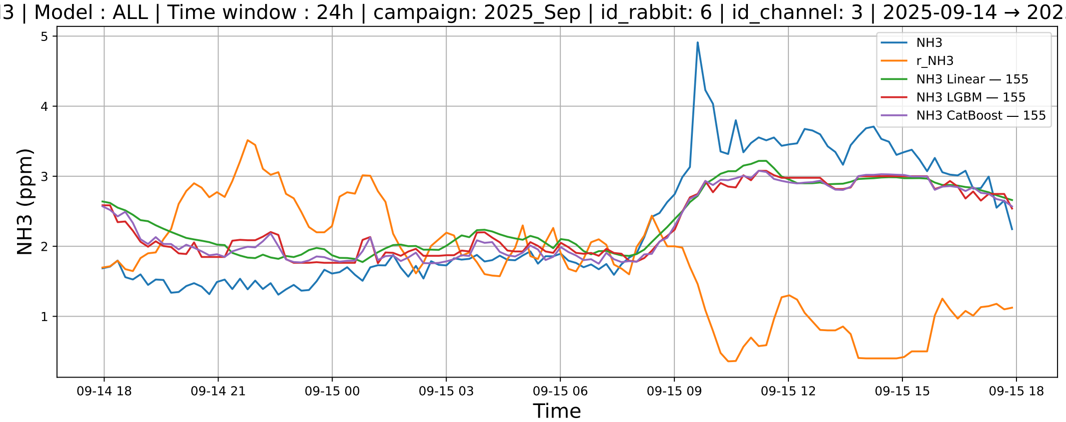
3.2.7.3 Time window : ALL



3.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

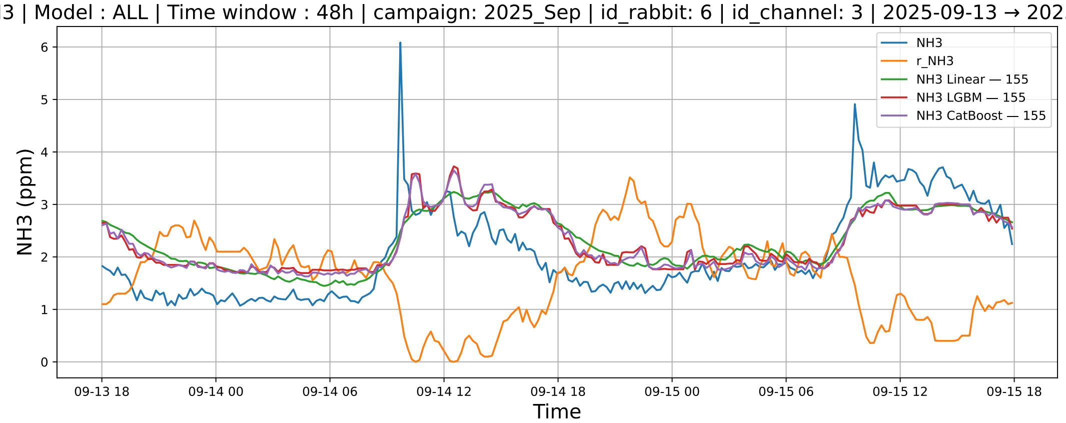
3.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



3.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



3.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15